Vectors

Questions with answers and full explanations

Question 1

VEC-T1-C-001 · Topic 1 · Easy · Concept

Which one of the following quantities is a vector?

(a) Volume of the fuel tank

(b) Displacement of a ship from its port

(c) Speed of the ship

(d) Mass of the cargo on board

Answer: (b) Displacement of a ship from its port

Explanation

Rule: A vector has magnitude and direction and adds by the triangle law. Displacement, position, velocity, acceleration and force are vectors.

Reasoning: Displacement is the straight arrow from the starting point to the final point, so it has size and direction.

Reasoning: Speed, mass and volume are given by one number with a unit only, so they are scalars.

Why the others fail: Speed is only the size of the velocity; mass and volume have no direction at all.

Concept behind the question. Section 1: a vector needs magnitude and direction (and the triangle law); displacement is the standard example, while speed, mass and volume are scalars.

Question 2

VEC-T1-C-002 · Topic 1 · Easy · Concept

Which one of the following is a scalar quantity?

(a) Acceleration of a falling stone

(b) Position of a buoy from the lighthouse

(c) Density of sea water

(d) Force exerted by a tug rope

Answer: (c) Density of sea water

Explanation

Rule: A scalar has magnitude only (one number with a unit). Density, mass, time, temperature, volume, speed and distance are scalars.

Reasoning: Density = mass/volume is a ratio of two scalars, so it has no direction.

Reasoning: Acceleration, force and position are vectors: each needs a direction to be fully stated.

Why the others fail: The stone's acceleration, the rope force and the buoy's position all have a direction, so they are vectors.

Concept behind the question. Section 1 table: density is one of the scalars met so far, whereas position, acceleration and force are vectors.

Question 3

VEC-T1-C-003 · Topic 1 · Easy · Concept

The negative of a non-zero vector $\mathbf{A}$, written $-\mathbf{A}$, is the vector

(a) with the same magnitude as A and the opposite direction

(b) with a negative magnitude and the same direction as A

(c) with half the magnitude of A and the opposite direction

(d) with the same magnitude as A and the same direction

Answer: (a) with the same magnitude as A and the opposite direction

Explanation

Rule: $-\mathrm{A} = (-1)\mathrm{A}$. Multiplying by -1 keeps the magnitude and reverses the direction.

Reasoning: $|-1| \times \mathrm{A} = \mathrm{A}$, so the length is unchanged; the negative sign only flips the arrow.

Why the others fail: A magnitude is never negative; there is no halving in (-1); and ignoring the sign would give A itself.

Concept behind the question. Section 3: the negative of a vector is the special case $\lambda = -1$; the sign belongs to the direction, never to the magnitude.

Question 4

VEC-T1-C-004 · Topic 1 · Easy · Concept

A non-zero vector **A** is multiplied by the number 0. The result 0 **A** is

(a) a vector of zero magnitude pointing along A

(b) a vector of magnitude 1 along A

(c) a vector of the same magnitude as A

(d) the null vector: magnitude zero, no definite direction

Answer: (d) the null vector: magnitude zero, no definite direction

Explanation

Rule: $0\mathrm{A} = 0$, the null (zero) vector. Its magnitude is 0 and its direction cannot be specified.

Reasoning: The magnitude of $\lambda\mathrm{A}$ is $|\lambda|\mathrm{A} = 0 \times \mathrm{A} = 0$. An arrow of zero length points nowhere in particular.

Why the others fail: A zero vector has no default direction; a unit vector has magnitude 1; and the size is certainly not kept.

Concept behind the question. Section 3: multiplying by 0 produces the null vector, which is the only vector without a direction.

Question 5

VEC-T1-C-005 · Topic 1 · Easy · Concept

A displacement arrow is drawn on a chart. It is then redrawn from a different starting point, parallel to the first and of the same length. The two arrows represent

(a) equal vectors

(b) vectors of equal magnitude that are not equal

(c) unequal vectors, because their starting points differ

(d) equal vectors only if both start at the origin

Answer: (a) equal vectors

Explanation

Rule: Vectors are equal if they have the same magnitude and the same direction. In this course vectors are free: sliding one parallel to itself does not change it.

Reasoning: The new arrow has the same length and the same direction as the old one; only its position on the chart differs.

Why the others fail: The starting point is not part of a free vector, so a different start does not make them unequal, and no origin is needed.

Concept behind the question. Section 2: equality of free vectors depends only on magnitude and direction, which is why a vector can be slid parallel to itself.

Question 6

VEC-T1-C-006 · Topic 1 · Easy · Concept

The magnitude of a vector **A** , written | **A** | or A, can be

(a) only positive, even for the null vector

(b) any real number, positive or negative

(c) zero or positive, but never negative

(d) negative when A points towards the negative x-direction

Answer: (c) zero or positive, but never negative

Explanation

Rule: The magnitude of a vector is always positive or zero.

Reasoning: It is a length (with a unit). It is zero only for the null vector and positive for every other vector.

Why the others fail: A minus sign belongs to a component or to the direction, never to the magnitude; and the null vector has magnitude exactly 0.

Concept behind the question. Section 1: representation and magnitude: |A| = A is a length, so it cannot be negative, but it can be zero for the null vector.

Question 7

VEC-T1-C-007 · Topic 1 · Medium · Concept

Which one of the following statements about vector magnitudes and equality is INCORRECT?

(a) The magnitude of a displacement is never greater than the path length.

(b) Two vectors with equal magnitudes are always equal vectors.

(c) The null vector has zero magnitude.

(d) The magnitude of a vector is never negative.

Answer: (b) Two vectors with equal magnitudes are always equal vectors.

Explanation

Rule: Equal vectors need the same magnitude AND the same direction.

Reasoning: Two arrows of equal length pointing in different directions are not equal, so the second statement, option (b), is false.

Reasoning: The other three are true: a magnitude is a length; the straight arrow between two points cannot be longer than any path joining them; and 0A = 0 has zero length.

Why the others fail: They are correct statements, so they cannot be the INCORRECT one asked for.

Concept behind the question. Sections 1-3: equal magnitude is necessary but not sufficient for equality of vectors; the direction must match too.

Question 8

VEC-T1-C-008 · Topic 1 · Medium · Concept

A ferry has a constant velocity **v** directed due west. It is multiplied by a time interval t (positive, in seconds). The product $\mathbf{v}t$ is

(a) a vector: the displacement, directed due east

(b) a vector in m/s, directed due west

(c) a scalar: the distance travelled

(d) a vector: the displacement, directed due west

Answer: (d) a vector: the displacement, directed due west

Explanation

Rule: Multiplying a vector by a positive number keeps its direction. If the number has a unit, the product is a new physical quantity: velocity × time = displacement.

Reasoning: t is positive, so $\mathbf{v}t$ points due west, along v.

Reasoning: Unit: (m/s) $\times$ s = m, which is the unit of displacement, and the quantity still has a direction.

Why the others fail: The product is still a vector; a positive multiplier cannot reverse it; and the unit has become metres, not m/s.

Concept behind the question. Section 3: a scalar that carries a unit (here time) turns a velocity into a different vector, the displacement, with the same direction.

Question 9

VEC-T1-C-009 · Topic 1 · Medium · Concept

A vector $\mathbf{B} = \lambda\mathbf{A}$ points opposite to $\mathbf{A}$ and is shorter than $\mathbf{A}$. The number λ must satisfy

(a) $\lambda < -1$

(b) $-1 < \lambda < 0$

(c) $\lambda > 1$

(d) $0 < \lambda < 1$

Answer: (b) $-1 < \lambda < 0$

Explanation

Rule: λA has magnitude $|\lambda|$A, and it points opposite to A when λ is negative.

Reasoning: Opposite direction means $\lambda < 0$. Shorter than A means $|\lambda| < 1$.

Reasoning: Together, λ is negative with $|\lambda|$ less than 1, i.e. $-1 < \lambda < 0$.

Why the others fail: $\lambda < -1$ gives a longer reversed vector; $0 < \lambda < 1$ gives a shorter vector in the same direction; $\lambda > 1$ gives a longer vector in the same direction.

Concept behind the question. Section 3: the sign of λ fixes the direction (same or reversed) and $|\lambda|$ compared with 1 fixes whether the vector is shortened or lengthened.

Question 10

VEC-T1-C-010 · Topic 1 · Medium · Concept

Which one of the following statements about the null (zero) vector is INCORRECT?

(a) It points along the positive x-axis by default.

(b) Its magnitude is zero.

(c) It is obtained by multiplying any vector by 0.

(d) Its direction cannot be specified.

Answer: (a) It points along the positive x-axis by default.

Explanation

Rule: The null vector 0 = 0A has zero magnitude and no definite direction.

Reasoning: With zero length the arrow has no direction at all, so there is no ”default” direction such as the positive x-axis.

Reasoning: The other three statements agree with the definition 0A = 0.

Why the others fail: They are true statements, so they cannot be the INCORRECT one asked for.

Concept behind the question. Section 3 (Mistakes): a zero vector has no default direction; that is the one property students most often get wrong.

Question 11

VEC-T1-C-011 · Topic 1 · Medium · Concept

A ship moves from P to Q. Two navigators describe the motion using different origins. Which quantity will both of them find to be the same?

(a) The final position vector of the ship

(b) The magnitude of the final position vector

(c) The initial position vector of the ship

(d) The displacement vector from P to Q

Answer: (d) The displacement vector from P to Q

Explanation

Rule: A position vector is the arrow from the chosen origin to the object; the displacement is the arrow from the initial to the final position ($\Delta r = r' - r$).

Reasoning: Changing the origin changes every position vector (and the distance of the ship from the origin), but the arrow PQ between the two end points does not change.

Why the others fail: The initial and final position vectors and the magnitude of the final position vector are all measured from the origin, so they change with it.

Concept behind the question. Section 1: displacement depends only on the end points, so it is the same for every observer; position vectors depend on the origin.

Question 12

VEC-T1-C-012 · Topic 1 · Medium · Concept

For a number λ and a vector $\mathbf{A}$, the product $\lambda\mathbf{A}$ is the null vector. Then

(a) A must be the null vector

(b) λ must be 0

(c) $\lambda = 0$ or A is the null vector

(d) λ and A must have opposite directions

Answer: (c) $\lambda = 0$ or A is the null vector

Explanation

Rule: $|\lambda A| = |\lambda|A$, and a vector is null only when its magnitude is zero.

Reasoning: $|\lambda|A = 0$ means $|\lambda| = 0$ or $A = 0$, so $\lambda = 0$ or A is the null vector.

Why the others fail: Either factor can be responsible, so "λ must be 0" and "A must be null" are each too strong; and the null vector has no direction, so opposite directions mean nothing here.

Concept behind the question. Section 3: a product is null exactly when the number is zero or the vector is already null, just like ordinary multiplication.

Question 13

VEC-T1-C-013 · Topic 1 · Hard · Concept

$\mathbf{B} = \lambda\mathbf{A}$ where $\mathbf{A}$ is a non-zero vector, and $|\mathbf{B}| = |\mathbf{A}|$. Which statement is correct?

(a) $\lambda = +1$ or $\lambda = -1$, so B equals A or $-$A

(b) B can point in any direction, since the magnitudes are equal

(c) $\lambda = -1$ only, so B must equal $-$A

(d) $\lambda = +1$ only, so B must equal A

Answer: (a) $\lambda = +1$ or $\lambda = -1$, so B equals A or $-$A

Explanation

Rule: $|\lambda A| = |\lambda|A$, and λA lies along the line of A (same direction if $\lambda > 0$, opposite if $\lambda < 0$).

Reasoning: $|B| = |A|$ gives $|\lambda|A = A$, so $|\lambda| = 1$, i.e. $\lambda = +1$ or $\lambda = -1$.

Reasoning: $\lambda = +1$ gives B = A (equal vectors); $\lambda = -1$ gives B = $-$A (equal size, opposite direction).

Why the others fail: Each single-value claim misses one of the two possibilities; and B = λA cannot point in an arbitrary direction because it is a multiple of A.

Concept behind the question. Section 3: equal magnitude fixes only $|\lambda| = 1$; the sign of λ decides between B = A and B = $-$A.

Question 14

VEC-T1-C-014 · Topic 1 · Hard · Concept

Two wires carrying currents of 3 A and 4 A into a junction meet there, at right angles to each other. The wire leaving the junction carries 7 A. This shows that electric current

(a) is a vector, and the correct sum should be 5 A by the triangle law

(b) is a vector, because it has a direction in each wire

(c) is a scalar, because it has no direction at all

(d) has a direction but adds like a scalar, so it is not a vector

Answer: (d) has a direction but adds like a scalar, so it is not a vector

Explanation

Rule: A quantity is a vector only if it has a direction AND combines by the triangle law (Section 1).

Reasoning: Currents entering the junction simply add arithmetically: 3 A + 4 A = 7 A, whatever the angle between the wires.

Reasoning: A true vector sum at right angles would be $\sqrt{3^2 + 4^2}$ = 5 A, which is not what is observed. So current fails the triangle-law test.

Why the others fail: A direction alone does not make a vector; the triangle law gives a different sum from the one observed; and current does have a direction, which is why it is the named exception.

Concept behind the question. Section 1 (Mistakes): electric current is the standard example of a quantity with a direction that is still a scalar, because it does not obey the triangle law.

Question 15

VEC-T1-N-015 · Topic 1 · Easy · Calculation

A trawler's velocity **v** is 18 km/h due north. Find the magnitude and direction of 2.5 **v** .

(a) 7.2 km/h, due north

(b) 20.5 km/h, due north

(c) 45 km/h, due north

(d) 45 km/h, due south

Answer: (c) 45 km/h, due north

Explanation

Given: |v| = 18 km/h due north; multiplier λ = 2.5 (positive).

To find: magnitude and direction of λv.

Formula: $|\lambda v| = |\lambda| \times |v|$; direction is the same as v if $\lambda > 0$.

Working: $|\lambda v| = 2.5 \times 18$ km/h = 45 km/h.

Working: $\lambda > 0$, so the direction is unchanged: due north.

Result: 45 km/h, due north. $\rightarrow$ option (c)

Why the others fail: 20.5 adds instead of multiplying; 7.2 divides; "due south" would need a negative multiplier.

Concept behind the question. Section 3: a positive number stretches or shrinks the arrow by that factor without turning it.

Question 16

VEC-T1-N-016 · Topic 1 · Easy · Calculation

An acceleration vector **a** has magnitude 8 m/s^2. What is the magnitude of -3**a** ?

(a) -24 m/s^2

(b) 2.67 m/s^2

(c) 5 m/s^2

(d) 24 m/s^2

Answer: (d) 24 m/s^2

Explanation

Given: |a| = 8 m/s^2; multiplier $\lambda = -3$.

Formula: $|\lambda \mathrm{a}| = |\lambda| \times |\mathrm{a}|$.

Working: $|\lambda| = |-3| = 3$.

Working: $|\lambda \mathrm{a}| = 3 \times 8 = 24$ m/s^2.

Result: 24 m/s^2 (the vector points opposite to a, but its magnitude is positive). $\rightarrow$ option (d)

Why the others fail: -24 gives the magnitude a minus sign; 2.67 divides; 5 adds λ to the magnitude.

Concept behind the question. Section 3: a negative multiplier only reverses the direction; the magnitude is $|\lambda|$ times the original and is never negative.

Question 17

VEC-T1-N-017 · Topic 1 · Easy · Calculation

A train moves with a constant velocity of 90 km/h due west for 20 minutes. Find its displacement.

(a) 30 km due east

(b) 30 km due west

(c) 1800 km due west

(d) 4.5 km due west

Answer: (b) 30 km due west

Explanation

Given: v = 90 km/h due west; t = 20 min.

Formula: displacement = velocity $\times$ time (a positive number with a unit, so the direction stays the same as v).

Working: t = 20/60 h = 1/3 h.

Working: displacement = 90 $\times$ (1/3) = 30 km, directed due west.

Result: 30 km due west. $\rightarrow$ option (b)

Why the others fail: 1800 km uses 20 as hours; ”due east” reverses the vector; 4.5 divides.

Concept behind the question. Section 3: velocity $\times$ time = displacement; the time is a positive scalar, so the displacement points the same way as the velocity.

Question 18

VEC-T1-N-018 · Topic 1 · Easy · Calculation

A car has a constant acceleration of 2.5 m/s^2 due south for 8 s. What is the change in its velocity?

(a) 10.5 m/s due south

(b) 0.31 m/s due south

(c) 20 m/s due south

(d) 20 m/s due north

Answer: (c) 20 m/s due south

Explanation

Given: a = 2.5 m/s^2 due south; t = 8 s.

Formula: change in velocity = acceleration $\times$ time (multiplying a vector by a positive number that has a unit).

Working: magnitude = 2.5 m/s^2 $\times$ 8 s = 20 m/s.

Working: the number (time) is positive, so the direction is the same as the acceleration: due south.

Result: 20 m/s due south. $\rightarrow$ option (c)

Why the others fail: ”due north” flips the direction; 0.31 divides; 10.5 adds.

Concept behind the question. Section 3: acceleration × time is a velocity (m/s^2 × s = m/s); a positive scalar multiplier never changes the direction.

Question 19

VEC-T1-N-019 · Topic 1 · Easy · Calculation

A tug's displacement is 90 m due west in 15 s. Find its average velocity (multiply the displacement by 1/t).

(a) 6 m/s due west

(b) 1350 m/s due west

(c) 6 m/s due east

(d) 75 m/s due west

Answer: (a) 6 m/s due west

Explanation

Given: displacement = 90 m due west; t = 15 s.

Formula: average velocity = (1/t) × displacement; 1/t is a positive number with unit s^{-1}.

Working: magnitude = 90 m ÷ 15 s = 6 m/s.

Working: the multiplier is positive, so the direction stays due west.

Result: 6 m/s due west. → option (a)

Why the others fail: 1350 multiplies by t; "due east" reverses the direction; 75 subtracts.

Concept behind the question. Section 3: dividing a vector by a positive scalar is multiplying by its reciprocal; the result is a new vector (velocity) in the same direction.

Question 20

VEC-T1-N-020 · Topic 1 · Easy · Calculation

A tug sails 12 km east and then 5 km west along the same straight channel. Find its displacement from the start.

(a) 13 km east

(b) 7 km east

(c) 17 km east

(d) 7 km west

Answer: (b) 7 km east

Explanation

Given: 12 km east, then 5 km west on the same line. Take east as positive.

Formula: displacement = final position − initial position (only end points matter).

Working: final position = +12 − 5 = +7 km.

Result: 7 km east of the start (the path length is 17 km). → option (b)

Why the others fail: 17 km is the path length; "west" has the wrong sign; 13 km would need the legs at right angles.

Concept behind the question. Section 1: displacement depends only on the end points; back-and-forth motion adds to the path length but partly cancels in the displacement.

Question 21

VEC-T1-N-021 · Topic 1 · Easy · Calculation

Water from a 40 L tank and a 25 L tank is poured into one drum, through pipes at right angles to each other. What is the total volume of water in the drum?

(a) 15 L

(b) 32.5 L

(c) 47.2 L

(d) 65 L

Answer: (d) 65 L

Explanation

Given: volumes 40 L and 25 L; the pipe directions are irrelevant.

Formula: scalars of the same unit add by ordinary algebra: V = V_1 + V_2.

Working: V = 40 L + 25 L = 65 L.

Result: 65 L. → option (d)

Why the others fail: 47.2 L treats volume as a vector; 15 L subtracts; 32.5 L averages.

Concept behind the question. Section 1: volume is a scalar, so quantities add by plain arithmetic and the angle between the pipes plays no role.

Question 22

VEC-T1-N-022 · Topic 1 · Easy · Calculation

Two forces act along the same direction: $\mathbf{F}_1$ of 15 N and $\mathbf{F}_2$ of 6 N. If $\mathbf{F}_2 = \lambda\mathbf{F}_1$, find λ .

(a) -0.4

(b) 0.4

(c) 2.5

(d) 9

Answer: (b) 0.4

Explanation

Given: F_1 = 15 N, F_2 = 6 N, same direction.

Formula: $F_2 = \lambda F_1$ with $|F_2| = |\lambda|\ |F_1|$; same direction means $\lambda > 0$.

Working: $|\lambda|$ = 6/15 = 0.4.

Result: λ = +0.4. → option (b)

Why the others fail: 2.5 is the reciprocal; -0.4 would reverse the direction; 9 subtracts.

Concept behind the question. Section 3: when two vectors lie along the same line in the same direction, the number connecting them is the positive ratio of their magnitudes.

Question 23

VEC-T1-N-023 · Topic 1 · Medium · Calculation

A displacement **A** is 24 m due east and a displacement **B** is 9 m due west. If **B** = λ**A** , find λ .

(a) -15

(b) -2.67

(c) -0.375

(d) 0.375

Answer: (c) -0.375

Explanation

Given: |A| = 24 m (east), |B| = 9 m (west).

Formula: |B| = $|\lambda||$A|; B is opposite to A, so $\lambda < 0$.

Working: $|\lambda|$ = 9/24 = 0.375.

Working: opposite direction → $\lambda = -0.375$.

Result: $\lambda = -0.375$. → option (c)

Why the others fail: 0.375 has the wrong sign; -2.67 is the reciprocal; -15 subtracts.

Concept behind the question. Section 3: a negative number reverses the direction and its absolute value is the ratio of the magnitudes.

Question 24

VEC-T1-N-024 · Topic 1 · Medium · Calculation

A runner covers one quarter of a circular track of radius 14 m, starting at a point on the track (take π = 22/7). What is the magnitude of the runner's displacement?

(a) 14 m

(b) 19.8 m

(c) 22 m

(d) 28 m

Answer: (b) 19.8 m

Explanation

Given: radius r = 14 m; the runner covers a quarter of the circle, so the start and end points are at 90° at the centre.

Formula: displacement = straight line between start and end; for a right angle at the centre it is the hypotenuse $\sqrt{r^2 + r^2}$ = r$\sqrt{2}$.

Working: $|\Delta r| = 14 \times \sqrt{2} = 14 \times 1.414 = 19.8$ m.

Result: 19.8 m (the path length is 22 m). $\rightarrow$ option (b)

Why the others fail: 22 m is the arc length; 14 m is the radius; 28 m is the diameter, which would need a half circle.

Concept behind the question. Section 1: displacement is the straight arrow between the end points, always shorter than a curved path joining them.

Question 25

VEC-T1-N-025 · Topic 1 · Medium · Calculation

A cargo belt moves a parcel with a constant velocity of 0.5 m/s due east. For what time must it run to give the parcel a displacement of 18 m due east?

(a) 0.028 s

(b) 9 s

(c) 18.5 s

(d) 36 s

Answer: (d) 36 s

Explanation

Given: v = 0.5 m/s east; displacement = 18 m east.

Formula: displacement = velocity $\times$ time, so the multiplying number t = displacement/velocity (both along east).

Working: t = 18 m $\div$ 0.5 m/s = 36 s.

Result: 36 s. $\rightarrow$ option (d)

Why the others fail: 9 multiplies; 0.028 is the reciprocal; 18.5 adds quantities with different units.

Concept behind the question. Section 3: velocity $\times$ time = displacement can be read backwards; since both vectors point east, the scalar time is the ratio of their magnitudes.

Question 26

VEC-T1-N-026 · Topic 1 · Medium · Calculation

A is a force of 8 N. Given $\mathbf{B} = -2\mathbf{A}$ and $\mathbf{C} = 1.5\ \mathbf{B}$, find the magnitude of **C** and its direction relative to **A** .

(a) 24 N, opposite to **A**

(b) 16 N, opposite to **A**

(c) 4 N, opposite to **A**

(d) 24 N, along **A**

Answer: (a) 24 N, opposite to **A**

Explanation

Given: |A| = 8 N; B = (−2)A; C = (1.5)B.

Formula: C = (1.5)(−2)A = λA with λ equal to the product of the two multipliers.

Working: $\lambda = (1.5) \times (-2) = -3$.

Working: $|C| = |\lambda| \times |A| = 3 \times 8 = 24$ N.

Working: $\lambda < 0$, so C is opposite to A.

Result: 24 N, opposite to A. → option (a)

Why the others fail: ”along A” ignores the minus sign; 4 N adds the multipliers; 16 N uses −2 only.

Concept behind the question. Section 3: successive multiplications by numbers combine into a single multiplier equal to their product, with sign and size both tracked.

Question 27

VEC-T1-N-027 · Topic 1 · Medium · Calculation

A lifeboat patrols a straight coast: 40 km north, then 25 km south, then another 10 km south. Find the ratio (path length) : (magnitude of displacement).

(a) 15 : 1

(b) 1 : 15

(c) 3 : 1

(d) 1 : 1

Answer: (a) 15 : 1

Explanation

Given: legs +40 km, −25 km, −10 km (north positive).

Formula: path length = sum of the leg lengths; displacement = final − initial = sum of the signed legs.

Working: path length = 40 + 25 + 10 = 75 km.

Working: displacement = 40 − 25 − 10 = 5 km (north), so the magnitude is 5 km.

Working: ratio = 75 : 5 = 15 : 1.

Result: 15 : 1. → option (a)

Why the others fail: 1 : 1 holds only for motion in one direction; 1 : 15 is inverted; 3 : 1 uses displacement 25 km.

Concept behind the question. Section 1: path length counts every leg, displacement counts only the net change of position, so the two differ whenever the motion turns back.

Question 28

VEC-T1-N-028 · Topic 1 · Medium · Calculation

A stone is stuck on the rim of a wheel of radius 0.5 m. The wheel turns about a fixed centre through 1.5 revolutions. Find the magnitude of the stone’s displacement (take $\pi = 3.14$).

(a) 0 m

(b) 0.5 m

(c) 1 m

(d) 4.71 m

Answer: (c) 1 m

Explanation

Given: radius = 0.5 m; turns = 1.5.

Formula: displacement = straight line from the start point to the end point.

Working: 1 full revolution brings the stone back to the start; the extra half revolution takes it to the opposite end of a diameter.

Working: $|\Delta \mathbf{r}|$ = diameter = $2 \times 0.5 = 1$ m.

Result: 1 m (the path length is $1.5 \times 2 \times 3.14 \times 0.5 = 4.71$ m). $\rightarrow$ option (c)

Why the others fail: 4.71 m is the path length; 0.5 m is only the radius; 0 m would be right only after a whole number of turns.

Concept behind the question. Section 1: displacement depends only on where the stone starts and ends; here that is two opposite points of the rim.

Question 29

VEC-T1-N-029 · Topic 1 · Medium · Calculation

A lifeboat sails 8 km east and then 6 km north in a total time of 2 h. Find the ratio (average speed) : (magnitude of average velocity).

(a) 1 : 1

(b) 5 : 7

(c) 7 : 1

(d) 7 : 5

Answer: (d) 7 : 5

Explanation

Given: legs 8 km east and 6 km north; total time 2 h.

Formula: average speed = path length / time; average velocity = displacement / time.

Working: path length = 8 + 6 = 14 km, so average speed = 7 km/h.

Working: displacement = $\sqrt{8^2 + 6^2}$ = 10 km, so |average velocity| = 5 km/h.

Working: ratio = 7 : 5 = 7 : 5.

Result: 7 : 5. $\rightarrow$ option (d)

Why the others fail: 5 : 7 is inverted; 1 : 1 needs a straight one-way path; 7 : 1 uses a displacement of 2 km.

Concept behind the question. Section 1: the same time divides path length (a scalar) and displacement (a vector); the ratio is path length to displacement magnitude.

Question 30

VEC-T1-N-030 · Topic 1 · Medium · Calculation

The position vector $\mathbf{r}$ of a buoy from a lighthouse is 12 km towards the north-east. Find the vector $-\frac{1}{4}\mathbf{r}$.

(a) 3 km towards the north-east

(b) 3 km towards the south-west

(c) 8 km towards the south-west

(d) 48 km towards the south-west

Answer: (b) 3 km towards the south-west

Explanation

Given: $|\mathbf{r}| = 12$ km, north-east; multiplier $\lambda = -1/4$.

Formula: $|\lambda \mathbf{r}| = |\lambda|\,|\mathbf{r}|$; negative λ reverses the direction.

Working: $|\lambda \mathbf{r}| = (1/4) \times 12 = 3$ km.

Working: reversing north-east gives south-west.

Result: 3 km towards the south-west. $\rightarrow$ option (b)

Why the others fail: "north-east" ignores the sign; 48 km multiplies by 4; 8 km subtracts.

Concept behind the question. Section 3: multiplying by $-1/4$ shrinks the arrow to a quarter and turns it through 180°.

Question 31

VEC-T1-N-031 · Topic 1 · Medium · Calculation

A ship's tank of volume 3.5 m^3 is full of oil of density 800 kg/m^3. A leak removes 0.5 m^3 of oil. What is the mass of oil left?

(a) 400 kg

(b) 2400 kg

(c) 2800 kg

(d) 3200 kg

Answer: (b) 2400 kg

Explanation

Given: V = 3.5 m^3; ρ = 800 kg/m^3; leaked volume = 0.5 m^3.

Formula: mass = density $\times$ volume (scalars multiply and add by ordinary algebra, with no direction involved).

Working: volume left = $3.5 - 0.5 = 3$ m^3.

Working: mass left = $800 \times 3 = 2400$ kg.

Result: 2400 kg. $\rightarrow$ option (b)

Why the others fail: 2800 kg is the original mass; 3200 kg adds the leak; 400 kg is the mass that leaked out.

Concept behind the question. Section 1: mass, volume and density are scalars; they combine by ordinary algebra and multiplying different units is fine.

Question 32

VEC-T1-N-032 · Topic 1 · Medium · Calculation

A boat moves along a straight canal with velocity -54 km/h (east is taken as positive). Find the magnitude of its velocity in m/s.

(a) -15 m/s

(b) 15 m/s

(c) 54 m/s

(d) 194.4 m/s

Answer: (b) 15 m/s

Explanation

Given: v = -54 km/h with east positive, so the boat moves west.

Formula: magnitude = |v|, always positive; 1 km/h = 1/3.6 m/s.

Working: |v| = 54 km/h.

Working: $54 \div 3.6 = 15$ m/s.

Result: 15 m/s (directed west). $\rightarrow$ option (b)

Why the others fail: -15 gives a negative magnitude; 194.4 multiplies by 3.6; 54 leaves the unit unconverted.

Concept behind the question. Section 1: the minus sign of a 1-D velocity tells the direction; the magnitude is positive.

Question 33

VEC-T1-N-033 · Topic 1 · Hard · Calculation

A is a force of 8 N due east. Given $\mathbf{B} = -2\mathbf{A}$, $\mathbf{C} = 0.75\,\mathbf{B}$ and $\mathbf{D} = -\frac{1}{3}\mathbf{C}$, find the magnitude and direction of **D**.

(a) 4 N, due east

(b) 12.7 N, due west

(c) 4 N, due west

(d) 36 N, due east

Answer: (a) 4 N, due east

Explanation

Given: |A| = 8 N due east; B = (−2)A; C = (0.75)B; D = (−1/3)C.

Formula: successive multiplications by numbers give one overall multiplier λ = product of the multipliers; |D| = $|\lambda||$A|.

Working: $\lambda = (-2) \times (0.75) \times (-1/3) = (-1.5) \times (-1/3) = +0.5$.

Working: $|D| = 0.5 \times 8 = 4$ N.

Working: $\lambda > 0$ (two negative signs cancel), so D points the same way as A: due east.

Result: 4 N, due east. → option (a)

Why the others fail: ”due west” loses a sign change; 12.7 N comes from adding multipliers; 36 N multiplies by 3.

Concept behind the question. Section 3: two negative multipliers reverse the direction twice, so the final vector points along the original again.

Question 34

VEC-T1-N-034 · Topic 1 · Hard · Calculation

A boat moves at a constant speed of 5.5 m/s round three-quarters of a circular pond of radius 7 m (take π = 22/7). Find the magnitude of its average velocity for this journey.

(a) 1.17 m/s

(b) 1.65 m/s

(c) 2.33 m/s

(d) 5.5 m/s

Answer: (b) 1.65 m/s

Explanation

Given: speed = 5.5 m/s; r = 7 m; arc = 3/4 of the circle.

Formula: time = path length ÷ speed; average velocity = displacement ÷ time.

Working: path length = $(3/4)(2\pi r) = 0.75 \times 2 \times (22/7) \times 7 = 33$ m, so t = 33/5.5 = 6 s.

Working: start and end are 270° apart round the circle, i.e. 90° apart at the centre, so displacement = $r\sqrt{2} = 7 \times 1.414 = 9.9$ m.

Working: |average velocity| = 9.9 ÷ 6 = 1.65 m/s.

Result: 1.65 m/s. → option (b)

Why the others fail: 5.5 m/s is the average speed; 1.17 m/s uses the radius as displacement; 2.33 m/s uses the diameter.

Concept behind the question. Section 1: average speed uses the path length and average velocity uses the displacement over the same time, so for a curved path they differ.

Question 35

VEC-T1-N-035 · Topic 1 · Hard · Calculation

A raft drifts with a constant velocity of 0.8 m/s due east for 12.5 minutes. Its displacement is $\Delta\mathbf{r}$ and a vector $\mathbf{B} = -0.4\,\Delta\mathbf{r}$ is formed. Find the magnitude and direction of **B** .

(a) 1500 m, due west

(b) 240 m, due east

(c) 240 m, due west

(d) 4 m, due west

Answer: (c) 240 m, due west

Explanation

Given: v = 0.8 m/s due east; t = 12.5 min; B = (−0.4)Δr.

Formula: $\Delta r = v \times t$ (positive number with a unit, same direction as v); then $|B| = |-0.4|\ |\Delta r|$, direction reversed.

Working: $t = 12.5 \times 60 = 750$ s.

Working: $|\Delta r| = 0.8 \times 750 = 600$ m, due east.

Working: $|B| = 0.4 \times 600 = 240$ m; the multiplier is negative, so B points due west.

Result: 240 m, due west. → option (c)

Why the others fail: "due east" ignores the sign; 4 m uses 12.5 as seconds; 1500 m divides by 0.4.

Concept behind the question. Section 3: a unit-carrying scalar (time) turns velocity into displacement, and a further negative number reverses and scales it.

Question 36

VEC-T2-C-001 · Topic 2 · Easy · Concept

Which construction correctly gives **A** + **B** by the triangle law?

(a) Draw **B** starting at the head of **A** ; the resultant length is the length of **A** plus the length of **B**

(b) Draw **B** starting at the tail of **A** ; join the head of **A** to the head of **B**

(c) Draw **B** starting at the head of **A** ; join the head of **B** to the tail of **A** , arrow pointing at the tail of **A**

(d) Draw **B** starting at the head of **A** ; join the tail of **A** to the head of **B**

Answer: (d) Draw **B** starting at the head of **A** ; join the tail of **A** to the head of **B**

Explanation

Rule: Triangle law: put the tail of B at the head of A; A + B is the arrow from the tail of A to the head of B.

Reasoning: Option (d) follows this exactly, so it is the construction for **A** + **B** .

Reasoning: The resultant always runs from the first tail to the last head.

Why the others fail: Joining the two heads after a tail-to-tail drawing gives A − B or B − A; reversing the closing arrow gives −(A + B); adding lengths works only when the vectors lie along one line in the same direction.

Concept behind the question. The triangle (head-to-tail) law of Section 4 is the basic rule for adding any two vectors of the same kind; the resultant closes the triangle from first tail to last head.

Question 37

VEC-T2-C-002 · Topic 2 · Easy · Concept

A and **B** are drawn from a common point O as two adjacent sides of a parallelogram. By the parallelogram law, **A** + **B** is represented by

(a) the diagonal joining the head of **A** to the head of **B**

(b) a line of length equal to the sum of the lengths of **A** and **B**

(c) the diagonal starting at O

(d) the side of the parallelogram that is longer

Answer: (c) the diagonal starting at O

Explanation

Rule: Parallelogram law: with A and B tail to tail, complete the parallelogram; the diagonal from the common tail is A + B.

Reasoning: The diagonal from O ends at the opposite corner, which is reached by going along A and then along B.

Why the others fail: The diagonal between the two heads is a difference (B − A or A − B); a single side is only A or B; summing lengths gives the maximum possible value and ignores the angle.

Concept behind the question. The parallelogram law is the triangle law redrawn tail to tail; both place the sum on the diagonal from the common starting point.

Question 38

VEC-T2-C-003 · Topic 2 · Easy · Concept

For any two different vectors, which of the following relations is NOT correct?

(a) $\mathbf{A} - \mathbf{B} = \mathbf{B} - \mathbf{A}$

(b) $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$

(c) $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$

(d) $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$

Answer: (a) $\mathbf{A} - \mathbf{B} = \mathbf{B} - \mathbf{A}$

Explanation

Rule: Addition is commutative and associative; subtraction is addition of the negative; B − A is the opposite of A − B.

Reasoning: B − A = −(A − B), so A − B = B − A only if the vectors are equal, which the stem excludes.

Why the others fail: The other three relations are exactly the commutative law, the associative law and the definition of subtraction, so they hold for all vectors.

Concept behind the question. Order does not matter for addition but it does for subtraction: swapping the order reverses the direction of the difference (Section 4).

Question 39

VEC-T2-C-004 · Topic 2 · Easy · Concept

Four displacement vectors are drawn head to tail, one after another, and the head of the fourth ends exactly on the tail of the first. Their vector sum is

(a) the null vector

(b) a vector equal to the sum of the four magnitudes

(c) the longest of the four vectors

(d) the first vector, since it starts the figure

Answer: (a) the null vector

Explanation

Rule: Polygon law: the sum of vectors joined head to tail is the arrow from the first tail to the last head.

Reasoning: Here the last head coincides with the first tail, so that arrow has zero length: the sum is the null vector.

Why the others fail: Adding magnitudes would give the path length, not the vector sum; neither the longest nor the first side is the closing arrow.

Concept behind the question. When the arrows of a polygon close on themselves the net displacement is zero, as for a boat that returns to its starting point (Section 4).

Question 40

VEC-T2-C-005 · Topic 2 · Easy · Concept

For a non-zero vector $\mathbf{A}$, the sum $\mathbf{A} + (-\mathbf{A})$ is

(a) a vector of zero magnitude pointing along $\mathbf{A}$

(b) a vector of magnitude 2A pointing along $\mathbf{A}$

(c) a vector of zero magnitude, the null vector, with no definite direction

(d) a vector of magnitude 2A pointing opposite to $\mathbf{A}$

Answer: (c) a vector of zero magnitude, the null vector, with no definite direction

Explanation

Rule: A + (−A) = 0, the null vector (Section 4); the null vector has zero length and its direction is undefined.

Reasoning: −A has the same length as A but points the opposite way, so placing it head to tail with A brings the arrow back to the start: zero length.

Why the others fail: The two arrows cancel, so the magnitude is zero, not 2A, and a zero-length arrow has no direction to assign.

Concept behind the question. The null vector is the vector sum of a vector and its negative; physically it is the zero displacement of a boat that returns to its starting point.

Question 41

VEC-T2-C-006 · Topic 2 · Medium · Concept

A and **B** are different non-zero vectors. Compared with the vector $\mathbf{A} - \mathbf{B}$, the vector $\mathbf{B} - \mathbf{A}$ is

(a) different in magnitude but in the same direction

(b) equal in magnitude and in direction

(c) equal in magnitude and perpendicular to it

(d) equal in magnitude and opposite in direction

Answer: (d) equal in magnitude and opposite in direction

Explanation

Rule: B − A = B + (−A) and A − B = A + (−B), and B − A = −(A − B).

Reasoning: Multiplying a vector by −1 keeps its length and reverses its direction, so the two differences are equal in size and opposite in direction.

Why the others fail: The order of subtraction matters only for direction; the size is the same, and the reversal is 180°, not 90°.

Concept behind the question. A − B and B − A represent the same difference diagonal of the parallelogram read in opposite senses; the mistakes list in Section 4 warns not to draw one in the direction of the other.

Question 42

VEC-T2-C-007 · Topic 2 · Medium · Concept

A tug makes three successive displacements **P** , **Q** , **R** in that order. Navigator 1 adds **P** and **Q** first and then adds **R** . Navigator 2 adds **Q** and **R** first and then adds **P** . The final displacement found by the two navigators is

(a) different, because the grouping changes the answer

(b) the same in magnitude but opposite in direction

(c) the same direction but larger for Navigator 2

(d) the same in magnitude and direction

Answer: (d) the same in magnitude and direction

Explanation

Rule: Vector addition is associative: (P + Q) + R = P + (Q + R); either way the sum is the arrow from the first tail to the last head.

Reasoning: Both navigators join the same three arrows head to tail; only the order in which they combine them differs, and the closing arrow from the start point to the end point is unchanged.

Why the others fail: Grouping cannot change the start point or end point of the tug, so neither the size nor the direction of the net displacement can change.

Concept behind the question. Associativity (Section 4) lets us add any number of vectors in any grouping; the net displacement depends only on the start and end positions.

Question 43

VEC-T2-C-008 · Topic 2 · Medium · Concept

Two non-zero vectors have magnitudes A and B. Which statement about the magnitude R of their resultant is correct?

(a) R can be smaller than each of A and B

(b) R can never be zero

(c) R always equals A + B

(d) R is always larger than each of A and B

Answer: (a) R can be smaller than each of A and B

Explanation

Rule: The resultant lies between $|A - B|$ and A + B: the largest is for vectors in the same direction, the smallest for opposite directions.

Reasoning: For two equal vectors in opposite directions R = 0, which is smaller than each of A and B; so R can be smaller than each.

Why the others fail: R = A + B only when the vectors point the same way; R = 0 is possible when A = B and they are opposite.

Concept behind the question. The Section 4 mistakes list warns against adding magnitudes directly; a resultant may be smaller than each of the vectors added.

Question 44

VEC-T2-C-009 · Topic 2 · Medium · Concept

Three non-zero vectors of magnitudes a, b and c add up to the null vector. Which condition must hold?

(a) $a^2 + b^2 = c^2$ for the largest of them

(b) a = b = c

(c) No one of a, b, c is greater than the sum of the other two

(d) a + b + c = 0

Answer: (c) No one of a, b, c is greater than the sum of the other two

Explanation

Rule: Vectors that sum to zero close into a polygon (a triangle for three), so each side is at most the sum of the other two.

Reasoning: If one magnitude exceeded the sum of the other two, the two shorter arrows could never reach back to the start, and the triangle would not close.

Why the others fail: Magnitudes are positive, so a + b + c cannot be 0; Pythagoras holds only for a right-angled triangle; closed triangles need not be equilateral (3, 4, 5 also close).

Concept behind the question. A closed polygon needs every side to be no longer than the sum of the others; this is the triangle condition behind the polygon law (Section 4).

Question 45

VEC-T2-C-010 · Topic 2 · Medium · Concept

Which statement about the triangle law and the parallelogram law of vector addition is correct?

(a) The diagonal from the common tail gives $\mathbf{A} - \mathbf{B}$

(b) They give the same resultant; the diagonal from the common tail equals the closing side of the triangle

(c) The parallelogram law works only for vectors of equal magnitude

(d) They give the same magnitude but opposite directions

Answer: (b) They give the same resultant; the diagonal from the common tail equals the closing side of the triangle

Explanation

Rule: The parallelogram law and the triangle law give the same resultant (Section 4).

Reasoning: Sliding B to the head of A turns half of the parallelogram into the triangle, and its closing side lies along the diagonal from the common tail.

Why the others fail: Both laws give one and the same vector; the diagonal between the heads is the difference; the parallelogram law holds for any two vectors.

Concept behind the question. Because a free vector may be slid parallel to itself, the parallelogram and the triangle are two drawings of one rule (Sections 2 and 4).

Question 46

VEC-T2-C-011 · Topic 2 · Hard · Concept

Three vectors lying in one plane have the magnitudes listed in the options. For which set can the three vectors NOT add up to the null vector for any choice of directions?

(a) 4, 4 and 8 units

(b) 3, 4 and 5 units

(c) 2, 3 and 6 units

(d) 5, 5 and 5 units

Answer: (c) 2, 3 and 6 units

Explanation

Rule: Three vectors can sum to zero only if the largest magnitude is not more than the sum of the other two.

Reasoning: For 2, 3 and 6 the largest is 6 but $2 + 3 = 5 < 6$, so the triangle can never close.

Reasoning: For 4, 4 and 8 the largest equals the sum of the others ($4 + 4 = 8$), so the vectors close when all lie on one line (two 4s in one direction against the 8).

Why the others fail: 3, 4, 5 and 5, 5, 5 form actual triangles, and 4, 4, 8 closes on a line, so all three of these sets can sum to zero.

Concept behind the question. The triangle condition has an equality case: when one magnitude equals the sum of the other two the vectors are collinear and still sum to zero (Section 4, polygon law).

Question 47

VEC-T2-C-012 · Topic 2 · Hard · Concept

Four non-zero vectors **A** , **B** , **C** , **D** are drawn head to tail in that order and form a closed quadrilateral. **A** + **B** is not the null vector. Which of the following statements is INCORRECT?

(a) $\mathbf{A} + \mathbf{B} + \mathbf{C} + \mathbf{D} = 0$

(b) $\mathbf{D} = -(\mathbf{A} + \mathbf{B} + \mathbf{C})$

(c) The magnitude of **A** + **B** equals the magnitude of **C** + **D**

(d) $\mathbf{A} + \mathbf{B} = \mathbf{C} + \mathbf{D}$

Answer: (d) $\mathbf{A} + \mathbf{B} = \mathbf{C} + \mathbf{D}$

Explanation

Rule: A closed polygon has zero vector sum, so $A + B + C + D = 0$.

Reasoning: Moving $C + D$ to the other side gives $A + B = -(C + D)$: equal in magnitude but opposite in direction.

Reasoning: Since $A + B$ is not the null vector, $A + B = C + D$ would force both to be zero, which is false; so that statement is the incorrect one.

Why the others fail: $A + B + C + D = 0$, $D = -(A + B + C)$ and $|A + B| = |C + D|$ all follow directly from the zero sum, so they are correct statements.

Concept behind the question. The two halves of a closed polygon are equal in size and opposite in direction: this sign is the usual trap (Section 4, polygon law).

Question 48

VEC-T2-N-013 · Topic 2 · Easy · Calculation

A runner jogs 70 m east along a straight track and then 30 m west. Find the displacement of the runner.

(a) 40 m east

(b) 30 m west

(c) 40 m west

(d) 100 m east

Answer: (a) 40 m east

Explanation

Given: 70 m east, then 30 m west (a straight line).

To find: net displacement.

Formula: Take east as positive and add the signed displacements: d = d_1 + d_2.

Working: d = (+70) + (−30) = +40 m

Result: 40 m east → option (a)

Why the others fail: 100 m is the path length; 40 m west has the wrong direction; 30 m west is only the second leg.

Concept behind the question. Collinear vectors in opposite directions subtract; displacement depends only on the end points, not the path (Sections 1 and 4).

Question 49

VEC-T2-N-014 · Topic 2 · Easy · Calculation

A ship sails 9 km east and then 12 km north. What is the magnitude of its resultant displacement?

(a) 3 km

(b) 15 km

(c) 21 km

(d) 225 km

Answer: (b) 15 km

Explanation

Given: 9 km east, then 12 km north (the two legs are at right angles).

To find: magnitude of the resultant displacement R.

Formula: Triangle law with a right angle: R = $\sqrt{a^2 + b^2}$.

Working: R = $\sqrt{9^2 + 12^2}$ = $\sqrt{81 + 144}$ = $\sqrt{225}$ = 15 km

Result: 15 km → option (b)

Why the others fail: 21 km adds the legs (valid only on one line); 3 km subtracts them; 225 stops before taking the square root.

Concept behind the question. For perpendicular vectors the closing side of the triangle is the hypotenuse, always shorter than the sum of the two legs (Section 4).

Question 50

VEC-T2-N-015 · Topic 2 · Easy · Calculation

Two displacement vectors, each of magnitude 5 m, are at right angles to each other. What is the magnitude of their sum?

(a) 3.54 m

(b) 7.07 m

(c) 10 m

(d) 50 m

Answer: (b) 7.07 m

Explanation

Given: two vectors of 5 m each, perpendicular.

To find: magnitude R of the sum.

Formula: R = $\sqrt{a^2 + b^2}$ for a right angle; for equal sizes R = $\sqrt{2} \times$ a.

Working: R = $\sqrt{5^2 + 5^2} = \sqrt{50}$ = 7.07 m

Result: 7.07 m $\rightarrow$ option (b)

Why the others fail: 10 m is the sum of the magnitudes (parallel vectors); 50 has no square root; 3.54 m divides instead of multiplies by $\sqrt{2}$.

Concept behind the question. Two equal perpendicular vectors form the sides of a square whose diagonal, the resultant, is $\sqrt{2}$ times a side (Section 4).

Question 51

VEC-T2-N-016 · Topic 2 · Easy · Calculation

Two forces of 8 N and 5 N act at a point, and their directions can be chosen freely. What are the largest and the smallest possible magnitudes of their resultant?

(a) 13 N and 0 N

(b) 9.4 N and 3 N

(c) 8 N and 5 N

(d) 13 N and 3 N

Answer: (d) 13 N and 3 N

Explanation

Given: forces of 8 N and 5 N, directions free.

To find: largest and smallest magnitude of the resultant.

Formula: Largest when parallel (same direction): R = A + B. Smallest when antiparallel (opposite directions): R = |A − B|.

Working: R_{max} = 8 + 5 = 13 N; R_{min} = 8 − 5 = 3 N

Result: 13 N and 3 N $\rightarrow$ option (d)

Why the others fail: Zero needs equal magnitudes; 9.4 N is the right-angle value, not the maximum; 8 N and 5 N are the inputs, not the extremes.

Concept behind the question. The resultant of two vectors is greatest along one line in the same sense and smallest along one line in opposite senses (Section 4).

Question 52

VEC-T2-N-017 · Topic 2 · Easy · Calculation

A drone flies 120 m east, then 120 m west, then 50 m north. Find its displacement from the start.

(a) 290 m north

(b) 50 m south

(c) 50 m north

(d) 130 m

Answer: (c) 50 m north

Explanation

Given: legs 120 m east, 120 m west, 50 m north.

To find: displacement.

Formula: Add the legs head to tail; opposite equal legs cancel.

Working: East-west: +120 − 120 = 0 m. North-south: +50 m. So the displacement is 50 m north.

Result: 50 m north $\rightarrow$ option (c)

Why the others fail: 290 m is the path length; 130 m forgets that the west leg cancels the east leg; south reverses the sign.

Concept behind the question. Equal and opposite legs cancel exactly (A + (−A) = 0), so only the leftover leg contributes to displacement (Section 4).

Question 53

VEC-T2-N-018 · Topic 2 · Easy · Calculation

Train A moves at 10 m/s east and train B at 6 m/s east on parallel tracks. Find $\vec{v}_A - \vec{v}_B$.

(a) 4 m/s east

(b) 16 m/s east

(c) 16 m/s west

(d) 4 m/s west

Answer: (a) 4 m/s east

Explanation

Given: v_A = 10 m/s east, v_B = 6 m/s east (east positive).

To find: $v_A - v_B$.

Formula: A − B = A + (−B): reverse B, then add.

Working: $v_A - v_B$ = (+10) + (−6) = +4 m/s, i.e. 4 m/s east

Result: 4 m/s east → option (a)

Why the others fail: 4 m/s west is $v_B - v_A$; 16 m/s adds the vectors instead of subtracting, in either sense.

Concept behind the question. Subtracting a vector means adding its negative; for collinear vectors this is ordinary signed subtraction (Section 4).

Question 54

VEC-T2-N-019 · Topic 2 · Easy · Calculation

A car moving at 20 m/s north turns a corner and then moves at 15 m/s east. Find the magnitude of the change in its velocity (final velocity − initial velocity).

(a) 5 m/s

(b) 25 m/s

(c) 35 m/s

(d) 625 m/s

Answer: (b) 25 m/s

Explanation

Given: initial velocity 20 m/s north, final velocity 15 m/s east (at right angles).

To find: $|v_{final} - v_{initial}|$.

Formula: Δv = v_f + (−v_i). The vectors v_f and $-v_i$ are at right angles, so |Δv| = $\sqrt{v_f^2 + v_i^2}$.

Working: |Δv| = $\sqrt{15^2 + 20^2}$ = $\sqrt{625}$ = 25 m/s

Result: 25 m/s → option (b)

Why the others fail: 35 and 5 m/s treat the velocities as collinear; 625 is the value before the square root.

Concept behind the question. Reversing the initial velocity and adding it head to tail gives the change in velocity; for perpendicular directions the sizes combine by Pythagoras (Section 4).

Question 55

VEC-T2-N-020 · Topic 2 · Medium · Calculation

A lifeboat leaves its base, goes 5 km east, then 12 km north, and then returns straight to the base. How long is the return leg?

(a) 7 km

(b) 13 km

(c) 17 km

(d) 169 km

Answer: (b) 13 km

Explanation

Given: legs 5 km east and 12 km north, then a straight return to the start.

To find: length of the return leg.

Formula: Closed polygon: the three legs add to zero, so the return leg is minus the resultant of the first two; its length is $\sqrt{a^2 + b^2}$.

Working: length = $\sqrt{5^2 + 12^2}$ = $\sqrt{169}$ = 13 km

Result: 13 km $\rightarrow$ option (b)

Why the others fail: 17 km is the sum of the legs (they are not along one line); 7 km subtracts them; 169 has no square root.

Concept behind the question. If the arrows close on themselves their sum is zero, so the last leg exactly cancels the displacement of the first two (polygon law, Section 4).

Question 56

VEC-T2-N-021 · Topic 2 · Medium · Calculation

Two displacements of magnitudes 9 m and 4 m are added, with the angle between them unknown. Which of these can be the magnitude of the resultant?

(a) 0 m

(b) 3 m

(c) 8 m

(d) 14 m

Answer: (c) 8 m

Explanation

Given: magnitudes 9 m and 4 m, angle unknown.

To find: which value is possible for the resultant.

Formula: $|A - B| \leq R \leq A + B$ (smallest for opposite, largest for same direction).

Working: R_{min} = 9 − 4 = 5 m; R_{max} = 9 + 4 = 13 m. So 5 m $\leq$ R $\leq$ 13 m.

Result: 8 m lies in the range 5 m to 13 m. $\rightarrow$ option (c)

Why the others fail: 3 m is below 5 m; 14 m is above 13 m; 0 m needs equal magnitudes.

Concept behind the question. The two extreme resultants occur along a line; every angle in between gives a value inside this range (Section 4).

Question 57

VEC-T2-N-022 · Topic 2 · Medium · Calculation

Three forces of 4 N, 6 N and 9 N act at a point, with directions that can be chosen freely. What is the smallest possible magnitude of their resultant?

(a) 0 N

(b) 1 N

(c) 2 N

(d) 19 N

Answer: (a) 0 N

Explanation

Given: forces 4 N, 6 N, 9 N, directions free.

To find: smallest magnitude of the resultant.

Formula: Three vectors sum to zero if the largest is not more than the sum of the other two (closed triangle).

Working: Largest = 9 N; the other two add to 4 + 6 = 10 N, which is $\geq$ 9 N, so a closed triangle exists.

Result: 0 N $\rightarrow$ option (a)

Why the others fail: The smallest resultant with all three on one line is 1 N, which is not the true minimum; 2 N ignores the third force; 19 N is the maximum.

Concept behind the question. Three vectors can sum to zero exactly when they can close into a triangle (Section 4, polygon law), so the minimum resultant is then zero.

Question 58

VEC-T2-N-023 · Topic 2 · Medium · Calculation

A guard walks 40 m along each side of an equilateral triangular yard and ends where he began. What are his displacement and the distance he walked?

(a) 120 m and 120 m

(b) 40 m and 120 m

(c) 0 m and 0 m

(d) 0 m and 120 m

Answer: (d) 0 m and 120 m

Explanation

Given: three sides of 40 m each, closed path.

To find: displacement and distance walked.

Formula: A closed polygon has vector sum zero; distance walked is the sum of the side lengths.

Working: Displacement = start-to-end = 0 m. Distance = 3 $\times$ 40 = 120 m.

Result: 0 m and 120 m $\rightarrow$ option (d)

Why the others fail: Displacement ignores the route, so it cannot equal the 120 m path; one side (40 m) is a single leg only; the distance walked is not zero.

Concept behind the question. The three equal sides taken head to tail close on themselves, so their vector sum is zero although the path length is not (Sections 1 and 4).

Question 59

VEC-T2-N-024 · Topic 2 · Medium · Calculation

A ship steaming east at 20 km/h changes course to south without changing its speed. Find the magnitude of the change in its velocity (final $-$ initial).

(a) 0 km/h

(b) 20 km/h

(c) 28.3 km/h

(d) 40 km/h

Answer: (c) 28.3 km/h

Explanation

Given: initial velocity 20 km/h east, final velocity 20 km/h south.

To find: $|v_f - v_i|$.

Formula: $v_f - v_i = v_f + (-v_i)$; here $-v_i$ points west and v_f points south, at right angles.

Working: $|\Delta v| = \sqrt{20^2 + 20^2} = 20\sqrt{2} = 28.3$ km/h

Result: 28.3 km/h (directed towards the south-west) $\rightarrow$ option (c)

Why the others fail: Velocity is a vector, so a change of direction is a change of velocity; 40 km/h adds the speeds; 20 km/h forgets the second perpendicular piece.

Concept behind the question. Velocity changes whenever direction changes; the change is found by subtracting the initial velocity vector from the final one (Section 4).

Question 60

VEC-T2-N-025 · Topic 2 · Medium · Calculation

A trawler sails 18 km east, then 9 km north, then 6 km west. Find the magnitude of its displacement from the start.

(a) 15 km

(b) 20.1 km

(c) 25.6 km

(d) 33 km

Answer: (a) 15 km

Explanation

Given: 18 km east, 9 km north, 6 km west.

To find: magnitude of the displacement.

Formula: Combine the collinear east-west legs first; then use Pythagoras with the north leg.

Working: East-west: $18 - 6 = 12$ km east. North: 9 km. R = $\sqrt{12^2 + 9^2} = \sqrt{225} = 15$ km

Result: 15 km $\rightarrow$ option (a)

Why the others fail: 33 km is the path length; 20.1 km leaves out the west leg; 25.6 km adds the west leg to the east leg instead of subtracting.

Concept behind the question. By the polygon law the displacement is the closing side; legs along the same line partly cancel before the perpendicular leg is combined (Section 4).

Question 61

VEC-T2-N-026 · Topic 2 · Medium · Calculation

A boat can be rowed at 4 m/s in still water. The river flows at 3 m/s. Over all directions in which the boat may be rowed, what are the largest and smallest speeds relative to the bank?

(a) 5 m/s and 1 m/s

(b) 7 m/s and 0 m/s

(c) 4 m/s and 3 m/s

(d) 7 m/s and 1 m/s

Answer: (d) 7 m/s and 1 m/s

Explanation

Given: boat speed 4 m/s relative to water; river 3 m/s.

To find: largest and smallest speed relative to the bank.

Formula: Velocity relative to bank = velocity relative to water + river velocity (vector sum).

Working: Largest when rowing downstream (same direction): 4 + 3 = 7 m/s. Smallest when rowing upstream (opposite directions): $4 - 3 = 1$ m/s.

Result: 7 m/s and 1 m/s $\rightarrow$ option (d)

Why the others fail: Zero needs equal speeds ($4 \neq 3$); 5 m/s is the cross-stream value, not the maximum; 4 and 3 m/s are the inputs.

Concept behind the question. The extreme sums of two vectors occur when they lie on one line, same sense for the maximum and opposite sense for the minimum (Section 4).

Question 62

VEC-T2-N-027 · Topic 2 · Medium · Calculation

Two vectors of magnitudes 8 units and 15 units are at right angles to each other. Find the ratio |A + B| : |A − B|.

(a) 23 : 7

(b) 17 : 7

(c) 1 : 1

(d) 23 : 17

Answer: (c) 1 : 1

Explanation

Given: perpendicular vectors of 8 and 15 units.

To find: |A + B| : |A − B|.

Formula: A + B and A − B are the two diagonals of the rectangle with sides A and B (A − B uses −B, also perpendicular to A).

Working: |A + B| = $\sqrt{8^2 + 15^2}$ = 17; |A − B| = $\sqrt{8^2 + 15^2}$ = 17

Result: 17 : 17 = 1 : 1 → option (c)

Why the others fail: Only for parallel vectors do the sizes become 23 and 7; here both diagonals of a rectangle have the same length.

Concept behind the question. The diagonals of a rectangle are equal, so for perpendicular vectors the sum and the difference have equal magnitude (Section 4, parallelogram law).

Question 63

VEC-T2-N-028 · Topic 2 · Medium · Calculation

Two velocity vectors of equal magnitude 10 m/s are such that the magnitude of their sum equals the magnitude of their difference. Find the magnitude of the sum.

(a) 0 m/s

(b) 10 m/s

(c) 14.1 m/s

(d) 20 m/s

Answer: (c) 14.1 m/s

Explanation

Given: |A| = |B| = 10 m/s and |A + B| = |A − B|.

To find: |A + B|.

Formula: A + B and A − B are the two diagonals of the parallelogram on A and B; equal diagonals mean a rectangle, so A ⊥ B. Then |A + B| = $\sqrt{A^2 + B^2}$.

Working: |A + B| = $\sqrt{10^2 + 10^2}$ = $10\sqrt{2}$ = 14.1 m/s

Result: 14.1 m/s → option (c)

Why the others fail: Parallel vectors give 20 m/s for the sum but 0 m/s for the difference (not equal); antiparallel gives the reverse; 10 m/s would need the vectors at 120°, where the sum and the difference are not equal.

Concept behind the question. A parallelogram with equal diagonals is a rectangle, so equal |A + B| and |A − B| means the vectors are at right angles (Section 4, parallelogram law).

Question 64

VEC-T2-N-029 · Topic 2 · Hard · Calculation

Three forces of 3 N, 4 N and 12 N act at a point, with directions that can be chosen freely. What is the smallest possible magnitude of their resultant?

(a) 0 N

(b) 5 N

(c) 8 N

(d) 19 N

Answer: (b) 5 N

Explanation

Given: forces 3 N, 4 N, 12 N, directions free.

To find: smallest magnitude of the resultant.

Formula: The two smaller forces can add to any value from $|3 - 4|$ to 3 + 4; the resultant of all three is smallest when this sum opposes the largest force.

Working: Largest possible opposing sum = 3 + 4 = 7 N, which is less than 12 N, so the triangle cannot close.

Working: R_{min} = 12 − (3 + 4) = 5 N

Result: 5 N → option (b)

Why the others fail: Zero needs the largest force to be no more than the sum of the other two (7 < 12); 19 N is the maximum; 8 N leaves out the 3 N force.

Concept behind the question. When one vector is longer than the sum of the others the polygon cannot close, and the minimum resultant is the excess (Section 4).

Question 65

VEC-T2-N-030 · Topic 2 · Hard · Calculation

A survey vessel sails 40 km east, 3 km north, 16 km west and 4 km north, in that order. By how much does the distance it sails exceed the magnitude of its displacement?

(a) 25 km

(b) 38 km

(c) 56 km

(d) 63 km

Answer: (b) 38 km

Explanation

Given: legs 40 km E, 3 km N, 16 km W, 4 km N.

To find: path length − |displacement|.

Formula: Path length = sum of leg lengths. Displacement: add east-west legs and north-south legs separately, then use Pythagoras.

Working: Path = 40 + 3 + 16 + 4 = 63 km. East-west: 40 − 16 = 24 km east. North: 3 + 4 = 7 km.

Working: |displacement| = $\sqrt{24^2 + 7^2}$ = $\sqrt{625}$ = 25 km

Working: Difference = 63 − 25 = 38 km

Result: 38 km → option (b)

Why the others fail: 63 km is the path length alone; 25 km is the displacement alone; 56 km drops the eastward net 24 km when finding the displacement.

Concept behind the question. Path length adds magnitudes; displacement adds vectors, so legs in opposite directions partly cancel and the displacement is never larger than the path (Sections 1 and 4).

Question 66

VEC-T3-C-001 · Topic 3 · Easy · Concept

A vector **A** of magnitude A makes an angle θ with the x-axis. Its component along the x-axis is

(a) $A/\cos\theta$

(b) $A\tan\theta$

(c) $A\sin\theta$

(d) $A\cos\theta$

Answer: (d) $A\cos\theta$

Explanation

Rule: A_x = A cos θ and A_y = A sin θ, with θ measured from the x-axis (Sec 5, N4).

Reasoning: dropping a perpendicular from the head of A on the x-axis gives a right-angled triangle with hypotenuse A; the side next to θ is A cos θ.

Why the others fail: A sin θ is the y-component; A tan θ and A/cos θ have no meaning as a side of this triangle (A/cos θ would be larger than A).

Concept behind the question. Resolution is the right-triangle relation: the side adjacent to the given angle carries cos, the side opposite carries sin (Sec 5, S2).

Question 67

VEC-T3-C-002 · Topic 3 · Easy · Concept

The unit vector along a non-zero vector **A** (of magnitude A) is

(a) $\mathbf{A}/A$

(b) $\mathbf{A}/A^2$

(c) **A**

(d) $A\,\mathbf{A}$

Answer: (a) $\mathbf{A}/A$

Explanation

Rule: $\hat{\mathbf{n}}$ = A/A, so A = A $\hat{\mathbf{n}}$ (Sec 5, N2).

Reasoning: dividing a vector by its own magnitude keeps the direction and makes the magnitude (A/A) = 1.

Why the others fail: the vector multiplied by A has magnitude A^2; A/A^2 has magnitude 1/A, not 1; A has magnitude A, which is 1 only when A = 1.

Concept behind the question. A unit vector only records direction; dividing by the length removes the size and leaves a pure direction (Sec 5, N2).

Question 68

VEC-T3-C-003 · Topic 3 · Easy · Concept

In the resolution $\mathbf{A} = A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}}$, which of the following is a scalar (a number with a unit, not a vector)?

(a) A_x

(b) **A**

(c) $A_y\hat{\mathbf{j}}$

(d) $A_x\hat{\mathbf{i}}$

Answer: (a) A_x

Explanation

Rule: A_x and A_y are numbers (scalars) that may be positive, negative or zero; A_x $\hat{\mathbf{i}}$ and A_y $\hat{\mathbf{j}}$ are vectors (Sec 5, N6).

Reasoning: A_x is multiplied by the unit vector $\hat{\mathbf{i}}$ to give a vector along the x-axis, so A_x by itself carries no direction of its own except its sign.

Why the others fail: $A_x\,\hat{\mathbf{i}}$ and $A_y\,\hat{\mathbf{j}}$ each include a unit vector, and A is the vector sum.

Concept behind the question. The sign of A_x is the only directional information a scalar component keeps; the unit vector supplies the axis (Sec 5, N6).

Question 69

VEC-T3-C-004 · Topic 3 · Easy · Concept

The components A_x and A_y of a vector in a plane are

(a) scalars that are always positive, like a magnitude

(b) vectors lying along the x- and y-axes

(c) unit vectors without any unit

(d) scalars with the unit of the vector, each of which can be positive, negative or zero

Answer: (d) scalars with the unit of the vector, each of which can be positive, negative or zero

Explanation

Rule: A_x and A_y are scalars and can be positive, negative or zero; they carry the unit of A (Sec 5, N6).

Reasoning: A_x = A cos θ is negative when θ is between 90° and 270°, and A_y = A sin θ is negative when θ is between 180° and 360°; both have the unit of A.

Why the others fail: only the magnitude A is non-negative; the vectors are $A_x\,\hat{\mathbf{i}}$ and $A_y\,\hat{\mathbf{j}}$; unit vectors have magnitude 1 and no unit.

Concept behind the question. A component is a signed number: its sign tells which way along the axis the vector points (Sec 5, N6).

Question 70

VEC-T3-C-005 · Topic 3 · Easy · Concept

A vector makes angles α , β , γ with the x-, y- and z-axes. Its direction cosines always satisfy

(a) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$

(b) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 3$

(c) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

(d) $\cos\alpha + \cos\beta + \cos\gamma = 1$

Answer: (c) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Explanation

Rule: A_x = A cos α, A_y = A cos β, A_z = A cos γ and $A^2 = A_x^2 + A_y^2 + A_z^2$ (Sec 5, N7–N8).

Reasoning: substituting gives $A^2(cos^2\alpha + cos^2\beta + cos^2\gamma) = A^2$, so the sum of the squares of the direction cosines is 1.

Why the others fail: the plain sum of cosines is not fixed (for equal acute angles it is $\sqrt{3}$); the sines give a sum of 2, since each $sin^2 = 1 - cos^2$; a total of 3 is impossible for a non-zero vector.

Concept behind the question. The three direction cosines are the components of the unit vector, so their squares must add to 1 (Sec 5, N8).

Question 71

VEC-T3-C-006 · Topic 3 · Medium · Concept

A vector of fixed magnitude A is turned slowly from $\theta = 0°$ to $\theta = 90°$ with the x-axis. Which quantity stays the same throughout?

(a) $A_x^2 + A_y^2$

(b) $A_x A_y$

(c) $A_x + A_y$

(d) $A_x - A_y$

Answer: (a) $A_x^2 + A_y^2$

Explanation

Rule: A = $\sqrt{A_x^2 + A_y^2}$ (Sec 5, N5), so A_x^2 + A_y^2 = A^2 for every orientation.

Reasoning: A_x^2 + A_y^2 = $A^2(cos^2\theta + sin^2\theta)$ = A^2, a constant. The sum A(cos θ + sin θ) goes from A up to $\sqrt{2}$ A and back to A; the difference and the product also change with θ.

Why the others fail: A_x + A_y, $A_x - A_y$ and A_x A_y all vary with θ (for example A_x A_y is 0 at 0° and A^2/2 at 45°).

Concept behind the question. Rotating a vector changes its components but not its length; Pythagoras links the two (Sec 5, N5).

Question 72

VEC-T3-C-007 · Topic 3 · Medium · Concept

The x- and y-components of a vector are both negative. Measured anticlockwise from the +x-axis, the direction of the vector lies between

(a) 270° and 360°

(b) 90° and 180°

(c) 0° and 90°

(d) 180° and 270°

Answer: (d) 180° and 270°

Explanation

Rule: use the signs of A_x and A_y to place the vector in the correct quadrant (Sec 5, S4).

Reasoning: A_x < 0 and A_y < 0 means the vector points to the left of and below the origin, which is the third quadrant, 180° to 270°.

Why the others fail: (−3, −4) and (3, 4) give the same tan θ, but the first lies in the third quadrant and the second in the first; the other two ranges are the second and fourth quadrants.

Concept behind the question. The ratio A_y/A_x loses the signs, so the quadrant must be read from the signs themselves (Sec 5, Mistakes).

Question 73

VEC-T3-C-008 · Topic 3 · Medium · Concept

Which set of numbers can be the direction cosines (cos α, cos β, cos γ) of a vector in space?

(a) 1, 1, 0

(b) 1/3, 1/3, 1/3

(c) 1/$\sqrt{2}$, 1/$\sqrt{2}$, 1/$\sqrt{2}$

(d) 2/3, 1/3, 2/3

Answer: (d) 2/3, 1/3, 2/3

Explanation

Rule: $cos^2\alpha$ + $cos^2\beta$ + $cos^2\gamma$ = 1 (Sec 5, N8).

Reasoning: for (2/3, 1/3, 2/3): 4/9 + 1/9 + 4/9 = 9/9 = 1, so it is allowed.

Why the others fail: (1/3, 1/3, 1/3) has squares adding to 1/3 (only the plain sum is 1); (1/$\sqrt{2}$, 1/$\sqrt{2}$, 1/$\sqrt{2}$) gives 3/2; (1, 1, 0) gives 2.

Concept behind the question. A set of direction cosines is the set of components of a unit vector, so the sum of squares has to be exactly 1 (Sec 5, N8).

Question 74

VEC-T3-C-009 · Topic 3 · Medium · Concept

Which of the following statements about unit vectors is incorrect?

(a) Any non-zero vector can be written as A = A $\hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is the unit vector along it

(b) A unit vector has magnitude 1

(c) A unit vector along a velocity has the unit m/s

(d) The unit vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, $\hat{\mathbf{k}}$ are mutually perpendicular

Answer: (c) A unit vector along a velocity has the unit m/s

Explanation

Rule: a unit vector has magnitude 1, only shows a direction and has no unit (Sec 5, N2, Mistakes).

Reasoning: $\hat{\mathbf{n}}$ = A/A divides the unit of A by the same unit, so it is a pure number with a direction. Attaching m/s to it is the error the chapter warns about.

Why the others fail: a non-zero vector can be written as its magnitude times its unit vector. A unit vector has magnitude 1, and the three coordinate unit vectors are mutually perpendicular.

Concept behind the question. The unit and size of a vector live in the factor A; the unit vector $\hat{\mathbf{n}}$ carries only direction (Sec 5, N2).

Question 75

VEC-T3-C-010 · Topic 3 · Medium · Concept

$\hat{\mathbf{n}}$ is the unit vector along a non-zero vector $\mathbf{A}$. The unit vector along $-3\mathbf{A}$ is

(a) $-3\hat{\mathbf{n}}$

(b) $-\hat{\mathbf{n}}$

(c) $\hat{\mathbf{n}}$

(d) $3\hat{\mathbf{n}}$

Answer: (b) $-\hat{\mathbf{n}}$

Explanation

Rule: multiplying by a negative number reverses the direction, and the size becomes 3A; a unit vector always has magnitude 1 (Sec 5, N2).

Reasoning: the direction of -3A is opposite to A, so the unit vector there is $-\hat{\mathbf{n}}$, which has magnitude 1.

Why the others fail: $-3\hat{\mathbf{n}}$ has magnitude 3; $\hat{\mathbf{n}}$ points along A, not against it; $3\hat{\mathbf{n}}$ has magnitude 3 and points along A.

Concept behind the question. A unit vector remembers direction only, so scaling a vector changes the unit vector at most by a sign (Sec 5, N2).

Question 76

VEC-T3-C-011 · Topic 3 · Easy · Concept

If $\mathbf{R} = \mathbf{A} - \mathbf{B}$, the y-component of $\mathbf{R}$ is

(a) $A_y - B_y$

(b) $B_y - A_y$

(c) $A_y B_y$

(d) $A_y + B_y$

Answer: (a) $A_y - B_y$

Explanation

Rule: components add (or subtract) matching component by matching component (Sec 6, N1).

Reasoning: R = A − B means $R_x = A_x - B_x$, $R_y = A_y - B_y$ and $R_z = A_z - B_z$.

Why the others fail: $B_y - A_y$ is the y-component of B − A, which is the negative of R; the sum and product belong to different operations.

Concept behind the question. Subtraction by components is addition with minus signs on the second vector, in the stated order (Sec 6, N1).

Question 77

VEC-T3-C-012 · Topic 3 · Easy · Concept

The resultant of two vectors of fixed magnitudes is smallest when the angle between their directions is

(a) 0°

(b) 90°

(c) 120°

(d) 180°

Answer: (d) 180°

Explanation

Rule: R = $\sqrt{A^2 + B^2 + 2ABcos\theta}$; R is largest at θ = 0° and smallest at θ = 180°, where R = |A − B| (Sec 6, N4).

Reasoning: cos θ has its least value, -1, at θ = 180°, which makes R^2 = $(A - B)^2$.

Why the others fail: at 0° R = A + B (maximum); at 90° R = $\sqrt{A^2 + B^2}$, which is larger than |A − B|; at 120° the cross term is $-AB$, still larger than $-2AB$, so R is again larger than |A − B|.

Concept behind the question. The 2AB cos θ term is the only part that depends on θ, so the extremes occur where cos θ = ± 1 (Sec 6, N4).

Question 78

VEC-T3-C-013 · Topic 3 · Easy · Concept

Two vectors of equal magnitude F act at an angle of 120° to each other. The magnitude of their resultant is

(a) $\sqrt{2}$ F

(b) 2F

(c) F

(d) F/2

Answer: (c) F

Explanation

Rule: for equal vectors R = 2F cos(θ/2) (Sec 6, N6); at θ = 120°, R = F.

Reasoning: R = 2F cos 60° = 2F $\times$ 0.5 = F.

Why the others fail: 2F needs θ = 0°; $\sqrt{2}$ F needs θ = 90°; F/2 would need cos(θ/2) = 1/4, which is not a standard angle here.

Concept behind the question. At 120° the two equal vectors and their resultant form an equilateral triangle, which is why three equal forces at 120° cancel (Sec 6, N6, TIP).

Question 79

VEC-T3-C-014 · Topic 3 · Medium · Concept

In $\tan\alpha = \dfrac{B\sin\theta}{A + B\cos\theta}$ for $\mathbf{R} = \mathbf{A} + \mathbf{B}$, the angle α is measured between **R** and

(a) the line that bisects A and B

(b) the direction of **A**

(c) the perpendicular to **A**

(d) the direction of **B**

Answer: (b) the direction of **A**

Explanation

Rule: α is the angle of R from A, and tan α = SN/ON with ON = A + B cos θ, SN = B sin θ (Sec 6, S2–S3).

Reasoning: SN is the perpendicular from the head of R on the line of A, and ON is the projection of R along A, so tan α compares them for the angle at the origin O between R and A.

Why the others fail: the angle between R and B is $\theta - \alpha$; the bisector applies only when A = B; the perpendicular to A is the direction of SN, not an angle used here.

Concept behind the question. The same triangle that gives R also gives its direction, always relative to the first vector A (Sec 6, S1–S3).

Question 80

VEC-T3-C-015 · Topic 3 · Medium · Concept

Two vectors are drawn head to tail to form a triangle. The angle inside the triangle at the joint of the two arrows is 70°. In $R^2 = A^2 + B^2 + 2AB\cos\theta$, the angle θ is

(a) 20°

(b) 70°

(c) 110°

(d) 250°

Answer: (c) 110°

Explanation

Rule: θ is the angle between the directions of A and B drawn tail to tail; the head-to-tail interior angle is $180^\circ - \theta$ (Sec 6, Mistakes).

Reasoning: $\theta = 180^\circ - 70^\circ = 110^\circ$, and cos 110° is negative, so R is less than the value found with cos 70°.

Why the others fail: 70° is the interior angle (the chapter's listed mistake); $90^\circ - 70^\circ = 20^\circ$ and $180^\circ + 70^\circ = 250^\circ$ have no connection to the geometry.

Concept behind the question. Moving one arrow head-to-tail turns the angle between the vectors into its supplement inside the triangle (Sec 6, S1, Mistakes).

Question 81

VEC-T3-C-016 · Topic 3 · Medium · Concept

Two vectors of equal magnitude make an angle of 80° with each other. The resultant makes an angle with either vector of

(a) 40°

(b) 50°

(c) 80°

(d) 160°

Answer: (a) 40°

Explanation

Rule: for equal vectors the resultant lies along the bisector, so it makes $\theta/2$ with each (Sec 6, N6).

Reasoning: the parallelogram on two equal sides is a rhombus, whose diagonal bisects the angle; $80^\circ/2 = 40^\circ$.

Why the others fail: 80° is the angle between the vectors, 160° has no meaning (the vectors themselves are only 80° apart), and $50^\circ = 90^\circ - 40^\circ$.

Concept behind the question. Equal sides make the parallelogram a rhombus, which fixes the resultant along the bisector (Sec 6, N6).

Question 82

VEC-T3-C-017 · Topic 3 · Medium · Concept

Two vectors have magnitudes 6 units and 10 units. Which statement about the magnitude of their resultant is incorrect?

(a) It can be 2 units

(b) It can be 12 units

(c) It can be 16 units

(d) It can be 4 units

Answer: (a) It can be 2 units

Explanation

Rule: $|A - B| \le R \le A + B$ (Sec 6, N5).

Reasoning: the smallest resultant is $10 - 6 = 4$ units (opposite directions) and the largest is $6 + 10 = 16$ units (same direction). So R can be anything from 4 to 16 units, and 2 is outside the range.

Why the others fail: 4, 12 and 16 are all inside the range 4 to 16, so those statements are correct.

Concept behind the question. Any value between the difference and the sum is reachable by choosing the angle; nothing outside can occur (Sec 6, N5).

Question 83

VEC-T3-C-018 · Topic 3 · Hard · Concept

For two non-zero vectors, $|\mathbf{A}+\mathbf{B}| = |\mathbf{A}-\mathbf{B}|$. The angle between **A** and **B** is

(a) 0°

(b) 60°

(c) 90°

(d) 180°

Answer: (c) 90°

Explanation

Rule: $|A + B|^2 = A^2 + B^2 + 2AB\cos\theta$ and $|A - B|^2 = A^2 + B^2 - 2AB\cos\theta$ (Sec 6, N2, N7).

Reasoning: equating them gives $2AB\cos\theta = -2AB\cos\theta$, so $4AB\cos\theta = 0$. As A and B are non-zero, $\cos\theta = 0$ and $\theta = 90^\circ$.

Why the others fail: at 0° the sum is A + B but the difference is $|A - B|$, and at 180° it is the other way round; at 60° the two sides differ by 2AB.

Concept behind the question. The sum and difference have equal length exactly when the two vectors are perpendicular (the diagonals of a rectangle are equal) (Sec 6, N2, N7).

Question 84

VEC-T3-C-019 · Topic 3 · Hard · Concept

Two non-zero vectors **A** and **B** satisfy $|\mathbf{A}+\mathbf{B}| = |\mathbf{A}| = |\mathbf{B}|$. Then $|\mathbf{A}-\mathbf{B}|$ is

(a) $2A$

(b) $\sqrt{2}\,A$

(c) A

(d) $\sqrt{3}\,A$

Answer: (d) $\sqrt{3}\,A$

Explanation

Rule: for equal vectors $R = 2A\cos(\theta/2)$ (Sec 6, N6); $|A - B| = \sqrt{A^2 + B^2 - 2AB\cos\theta}$ (N7).

Reasoning: $R = A$ gives $2A\cos(\theta/2) = A$, so $\cos(\theta/2) = 1/2$, $\theta/2 = 60^\circ$ and $\theta = 120^\circ$. Then $|A - B|^2 = A^2 + A^2 - 2A^2\cos 120^\circ = 2A^2 + A^2 = 3A^2$, so $|A - B| = \sqrt{3}\,A$.

Why the others fail: A would need $\theta = 60^\circ$; 2A would need $\theta = 180^\circ$; $\sqrt{2}$ A would need $\theta = 90^\circ$, but then |A + B| would also be $\sqrt{2}$ A.

Concept behind the question. The condition |A + B| = A fixes the angle at 120°, and the same angle then fixes |A − B| (Sec 6, N6–N7).

Question 85

VEC-T3-C-020 · Topic 3 · Hard · Concept

Three non-zero vectors of equal magnitude add up to the zero vector. The angle between any two of them is

(a) 60°

(b) 90°

(c) 120°

(d) 180°

Answer: (c) 120°

Explanation

Rule: three vectors add to zero only if each is equal and opposite to the sum of the other two (Sec 6, N8).

Reasoning: the other two, each of magnitude A, must give a resultant of magnitude A, so 2A $\cos(\theta/2)$ = A, $\cos(\theta/2)$ = 1/2 and $\theta = 120^\circ$.

Why the others fail: 60° is the angle inside the closed head-to-tail triangle, not the angle between the vectors with tails together (it is $180^\circ - 120^\circ$); at 90° the resultant is $\sqrt{2}$ A; at 180° the resultant is 0.

Concept behind the question. Three equal forces at 120° to each other balance; drawn head to tail they close into an equilateral triangle (Sec 6, N8, TIP).

Question 86

VEC-T3-N-021 · Topic 3 · Easy · Calculation

A tug pulls a barge with a rope force of 800 N, the angle between the rope and the canal bank being 60°. The component of the pull along the bank is

(a) 400 N

(b) 693 N

(c) 800 N

(d) 1600 N

Answer: (a) 400 N

Explanation

Given: F = 800 N, $\theta = 60^\circ$ with the bank.

Formula: component along the bank = F cos θ.

Working: F cos 60° = 800 × 0.5 = 400 N.

Result: 400 N. → option (a)

Why the others fail: 693 N is F sin 60°, the component towards the bank; 800 N ignores the angle; 1600 N is F ÷ cos 60°, larger than the pull itself.

Concept behind the question. Only the part of the pull along the canal moves the barge forward; the rest pulls it sideways (Sec 5, Familiar example).

Question 87

VEC-T3-N-022 · Topic 3 · Easy · Calculation

An aircraft climbs at a speed of 250 m/s along a straight path at 30° above the horizontal. The vertical component of its velocity is

(a) 125 m/s

(b) 144.3 m/s

(c) 216.5 m/s

(d) 250 m/s

Answer: (a) 125 m/s

Explanation

Given: v = 250 m/s, θ = 30° above the horizontal.

Formula: v_y = v sin θ (θ measured from the horizontal).

Working: v_y = 250 × sin 30° = 250 × 0.5 = 125 m/s.

Result: 125 m/s. → option (a)

Why the others fail: 216.5 m/s is v cos 30°, the horizontal component; 250 m/s ignores the angle; 144.3 m/s is v tan 30°.

Concept behind the question. A velocity vector resolves exactly like any other vector: sin gives the part perpendicular to the reference line (Sec 5, N4).

Question 88

VEC-T3-N-023 · Topic 3 · Easy · Calculation

A vector in the xy-plane has an x-component of 5 units and a y-component of −12 units. Its magnitude is

(a) 7 units

(b) 13 units

(c) 17 units

(d) 169 units

Answer: (b) 13 units

Explanation

Given: A_x = 5, A_y = −12.

Formula: A = $\sqrt{A_x^2 + A_y^2}$.

Working: A = $\sqrt{5^2 + (-12)^2}$ = $\sqrt{25 + 144}$ = $\sqrt{169}$ = 13 units.

Result: 13 units. → option (b)

Why the others fail: 7 is the magnitude of the incorrect scalar sum 5 + (−12) = −7; 17 adds the sizes (a straight-line shortcut is shorter than two legs); 169 is A^2 without the root.

Concept behind the question. The two components are the perpendicular sides of a right triangle and the magnitude is its hypotenuse; the minus sign vanishes on squaring (Sec 5, N5).

Question 89

VEC-T3-N-024 · Topic 3 · Easy · Calculation

Find the unit vector along $6\hat{\mathbf{i}} - 8\hat{\mathbf{j}}$.

(a) $0.6\hat{\mathbf{i}} + 0.8\hat{\mathbf{j}}$

(b) $0.8\hat{\mathbf{i}} - 0.6\hat{\mathbf{j}}$

(c) $0.06\hat{\mathbf{i}} - 0.08\hat{\mathbf{j}}$

(d) $0.6\hat{\mathbf{i}} - 0.8\hat{\mathbf{j}}$

Answer: (d) $0.6\hat{\mathbf{i}} - 0.8\hat{\mathbf{j}}$

Explanation

Given: A = $6\hat{\mathbf{i}} - 8\hat{\mathbf{j}}$.

Formula: $\hat{\mathbf{n}}$ = A/A with A = $\sqrt{A_x^2 + A_y^2}$.

Working: A = $\sqrt{6^2 + (-8)^2} = \sqrt{100}$ = 10. $\hat{\mathbf{n}}$ = $(6\hat{\mathbf{i}} - 8\hat{\mathbf{j}})/10 = 0.6\hat{\mathbf{i}} - 0.8\hat{\mathbf{j}}$.

Result: $0.6\hat{\mathbf{i}} - 0.8\hat{\mathbf{j}}$ (check: 0.6^2 + $(-0.8)^2$ = 1). → option (d)

Why the others fail: swapping components gives a vector at a different angle; dropping the minus sign points it into the first quadrant; dividing by A^2 = 100 gives magnitude 0.1, not 1.

Concept behind the question. Dividing by the magnitude makes the length 1 while keeping the direction; the squares of the new components must add to 1 (Sec 5, TIP).

Question 90

VEC-T3-N-025 · Topic 3 · Easy · Calculation

For **A** = $2\hat{\mathbf{i}} + 5\hat{\mathbf{j}}$ and **B** = $3\hat{\mathbf{i}} - 2\hat{\mathbf{j}}$, the sum **A + B** is

(a) $5\hat{\mathbf{i}} + 7\hat{\mathbf{j}}$

(b) $5\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$

(c) $-\hat{\mathbf{i}} + 7\hat{\mathbf{j}}$

(d) $6\hat{\mathbf{i}} - 10\hat{\mathbf{j}}$

Answer: (b) $5\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$

Explanation

Given: A = (2, 5), B = (3, −2).

Formula: $R_x = A_x + B_x$, $R_y = A_y + B_y$.

Working: R_x = 2 + 3 = 5; R_y = 5 + (−2) = 3.

Result: $5\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$. → option (b)

Why the others fail: $-\hat{\mathbf{i}} + 7\hat{\mathbf{j}}$ is A − B; adding 5 + 2 = 7 drops the sign of B_y; multiplying components is not a vector operation here.

Concept behind the question. Adding by components turns a geometric construction into two ordinary additions (Sec 6, N1).

Question 91

VEC-T3-N-026 · Topic 3 · Easy · Calculation

Given **A** = $4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$ and **B** = $\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$, find **A** − **B** .

(a) $5\hat{\mathbf{i}} + 8\hat{\mathbf{j}} - \hat{\mathbf{k}}$

(b) $-3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$

(c) $3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$

(d) $3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$

Answer: (d) $3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$

Explanation

Given: A = (4, 3, 1), B = (1, 5, −2).

Formula: $(\text{A} - \text{B})_x = A_x - B_x$, and similarly for y and z.

Working: x: 4 − 1 = 3; y: 3 − 5 = −2; z: 1 − (−2) = 3.

Result: $3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$. → option (d)

Why the others fail: $-3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$ is B − A; $5\hat{\mathbf{i}} + 8\hat{\mathbf{j}} - \hat{\mathbf{k}}$ is A + B; the third option gets z = 1 − 2 = −1, but subtracting −2 adds 2.

Concept behind the question. Subtraction works on each component in the stated order; the minus sign must be applied carefully to negative components (Sec 6, N1).

Question 92

VEC-T3-N-027 · Topic 3 · Easy · Calculation

Two forces of 7 N and 24 N act at a point at right angles to each other. The magnitude of the resultant is

(a) 17 N

(b) 25 N

(c) 31 N

(d) 625 N

Answer: (b) 25 N

Explanation

Given: A = 7 N, B = 24 N, $\theta = 90^\circ$.

Formula: R = $\sqrt{A^2 + B^2 + 2AB\cos\theta}$; cos 90° = 0, so R = $\sqrt{A^2 + B^2}$.

Working: R = $\sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625}$ = 25 N.

Result: 25 N. → option (b)

Why the others fail: 31 N is the sum, valid only for $\theta = 0^\circ$; 17 N is the difference, valid only for $\theta = 180^\circ$; 625 is R^2 without the root.

Concept behind the question. At 90° the cross term vanishes and the law of cosines reduces to Pythagoras (Sec 6, N4).

Question 93

VEC-T3-N-028 · Topic 3 · Easy · Calculation

Two forces of 15 N each act at a point with an angle of 60° between them. The resultant has magnitude

(a) $15\sqrt{2}$ N

(b) 15 N

(c) $15\sqrt{3}$ N

(d) 30 N

Answer: (c) $15\sqrt{3}$ N

Explanation

Given: A = B = 15 N, $\theta = 60^\circ$.

Formula: for equal vectors R = 2A cos(θ/2).

Working: R = 2 × 15 × cos 30° = 30 × 0.866 ≈ 26.0 N, and cos 30° = $\sqrt{3}$/2 exactly, so R = $15\sqrt{3}$ N.

Result: $15\sqrt{3}$ N (about 26.0 N). → option (c)

Why the others fail: 15 N uses the full angle, 2A cos 60°; 30 N needs $\theta = 0^\circ$; $15\sqrt{2}$ N (about 21.2 N) is the right-angle result.

Concept behind the question. The half-angle form is a shortcut to the law of cosines for equal vectors, and the resultant lies along the bisector (Sec 6, N6).

Question 94

VEC-T3-N-029 · Topic 3 · Easy · Calculation

Two forces of 12 N and 5 N act at a point. The smallest possible magnitude of their resultant is

(a) 5 N

(b) 7 N

(c) 13 N

(d) 17 N

Answer: (b) 7 N

Explanation

Given: A = 12 N, B = 5 N.

Formula: $|A - B| \le R \le A + B$; the minimum occurs at $\theta = 180^\circ$.

Working: R_{min} = 12 − 5 = 7 N.

Result: 7 N. → option (b)

Why the others fail: 17 N is the maximum ($\theta = 0^\circ$); 13 N is the value at $\theta = 90^\circ$; 5 N is not a limit of the range.

Concept behind the question. The resultant ranges from the difference (opposite directions) to the sum (same direction) of the magnitudes (Sec 6, N4–N5).

Question 95

VEC-T3-N-030 · Topic 3 · Easy · Calculation

A ship's velocity has components $6\sqrt{3}$ m/s towards the east and 6 m/s towards the north. Its direction is

(a) 30° north of east

(b) 60° north of east

(c) 45° north of east

(d) 30° north of west

Answer: (a) 30° north of east

Explanation

Given: $v_x = 6\sqrt{3}$ m/s (east), v_y = 6 m/s (north).

Formula: $\tan\theta = v_y / v_x$, θ measured from the east (x) axis.

Working: $\tan\theta = 6 / (6\sqrt{3}) = 1/\sqrt{3} = 0.577$, so $\theta = 30^\circ$. Both components are positive, so the direction is north of east.

Result: 30° north of east. → option (a)

Why the others fail: 60° comes from $\tan\theta = \sqrt{3}$ (ratio upside down); 45° needs equal components; "north of west" would need a negative x-component.

Concept behind the question. The angle from the x-axis is $tan^{-1}(A_y/A_x)$; the signs of the components then fix the quadrant (Sec 5, S4).

Question 96

VEC-T3-N-031 · Topic 3 · Easy · Calculation

A vector is $4\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ (in metres). If it makes an angle β with the y-axis, then cos β equals

(a) 1/3

(b) 3/2

(c) 1/9

(d) 2/3

Answer: (d) 2/3

Explanation

Given: A_x = 4, A_y = 4, A_z = 2.

Formula: $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$, $\cos\beta = A_y / A$.

Working: $A = \sqrt{16 + 16 + 4} = \sqrt{36}$ = 6 m; $\cos\beta$ = 4/6 = 2/3.

Result: 2/3. → option (d)

Why the others fail: 3/2 is greater than 1, impossible for a cosine; 1/3 is $\cos\gamma = A_z/A$; 1/9 is 4/36 with A^2 in place of A.

Concept behind the question. Each direction cosine is a component divided by the magnitude; a value above 1 is the quick sign of an inverted ratio (Sec 5, N7).

Question 97

VEC-T3-N-032 · Topic 3 · Easy · Calculation

Raindrops have a vertical velocity component of 8 m/s and a horizontal velocity component of 6 m/s due to a steady wind. The resultant velocity of the raindrops has magnitude

(a) 2 m/s

(b) 10 m/s

(c) 14 m/s

(d) 100 m/s

Answer: (b) 10 m/s

Explanation

Given: vertical 8 m/s, horizontal 6 m/s (perpendicular).

Formula: R = $\sqrt{A^2 + B^2}$ for perpendicular vectors.

Working: R = $\sqrt{8^2 + 6^2}$ = $\sqrt{100}$ = 10 m/s.

Result: 10 m/s. → option (b)

Why the others fail: 14 m/s is the sum for parallel vectors; 2 m/s is the difference for opposite vectors; 100 is R^2 without the root.

Concept behind the question. Perpendicular vectors combine by Pythagoras; this is the chapter's rain-and-wind example (Sec 6, S5).

Question 98

VEC-T3-N-033 · Topic 3 · Medium · Calculation

A vector in the xy-plane has x-component -4 units and y-component $-4\sqrt{3}$ units. Its direction, measured anticlockwise from the +x-axis, is

(a) 60°

(b) 120°

(c) 240°

(d) 300°

Answer: (c) 240°

Explanation

Given: $A_x = -4$, $A_y = -4\sqrt{3}$.

Formula: tan $\phi = |A_y/A_x|$ for the reference angle; then place the vector using the signs.

Working: tan $\phi = 4\sqrt{3}/4 = \sqrt{3}$, so $\phi = 60^\circ$. Both components are negative, so the vector is in the third quadrant: $\theta = 180^\circ + 60^\circ = 240^\circ$.

Result: 240°. → option (c)

Why the others fail: 60° is the calculator value for $(4, 4\sqrt{3})$ in the first quadrant; 120° would need $A_x < 0$, $A_y > 0$; 300° would need $A_x > 0$, $A_y < 0$.

Concept behind the question. Equal ratios hide opposite quadrants; the signs of both components decide the final angle (Sec 5, S4, Mistakes).

Question 99

VEC-T3-N-034 · Topic 3 · Medium · Calculation

A vector of magnitude 12 units makes an angle of 120° with the +x-axis, measured anticlockwise. Written in component form it is

(a) $-6\hat{\mathbf{i}} + 10.39\hat{\mathbf{j}}$

(b) $10.39\hat{\mathbf{i}} - 6\hat{\mathbf{j}}$

(c) $6\hat{\mathbf{i}} + 10.39\hat{\mathbf{j}}$

(d) $-6\hat{\mathbf{i}} - 10.39\hat{\mathbf{j}}$

Answer: (a) $-6\hat{\mathbf{i}} + 10.39\hat{\mathbf{j}}$

Explanation

Given: A = 12, θ = 120° from +x.

Formula: A_x = A cos θ, A_y = A sin θ.

Working: A_x = 12 cos 120° = 12 × (−0.5) = −6; A_y = 12 sin 120° = 12 × 0.866 = 10.39.

Result: $-6\hat{\mathbf{i}} + 10.39\hat{\mathbf{j}}$. → option (a)

Why the others fail: the second option has A_x positive although cos 120° is negative; the third uses A_x = A sin θ; the fourth makes A_y negative although sin 120° is positive.

Concept behind the question. Obtuse angles from the +x-axis give a negative cosine, so the x-component automatically comes out negative (Sec 5, N4).

Question 100

VEC-T3-N-035 · Topic 3 · Medium · Calculation

Find the vector of magnitude 15 units that points in the direction of $8\hat{\mathbf{i}} - 6\hat{\mathbf{j}}$.

(a) $\frac{16}{3}\hat{\mathbf{i}} - 4\hat{\mathbf{j}}$

(b) $120\hat{\mathbf{i}} - 90\hat{\mathbf{j}}$

(c) $9\hat{\mathbf{i}} - 12\hat{\mathbf{j}}$

(d) $12\hat{\mathbf{i}} - 9\hat{\mathbf{j}}$

Answer: (d) $12\hat{\mathbf{i}} - 9\hat{\mathbf{j}}$

Explanation

Given: direction of (8, −6); required magnitude 15.

Formula: B = B $\hat{\mathbf{n}}$, with $\hat{\mathbf{n}}$ = A/A.

Working: A = $\sqrt{8^2 + (-6)^2}$ = 10; $\hat{\mathbf{n}} = 0.8\hat{\mathbf{i}} - 0.6\hat{\mathbf{j}}$; B = 15 × $\hat{\mathbf{n}}$ = (12, −9).

Result: $12\hat{\mathbf{i}} - 9\hat{\mathbf{j}}$ (check: $\sqrt{12^2 + 9^2}$ = 15). → option (d)

Why the others fail: (120, −90) has magnitude 150; multiplying by 10/15 instead of 15/10 gives a vector of magnitude 10 × 10/15 ≈ 6.7; the swapped version has the right size but a different direction.

Concept behind the question. “Magnitude M along a given direction” always means M times the unit vector in that direction (Sec 5, N2).

Question 101

VEC-T3-N-036 · Topic 3 · Medium · Calculation

Three forces (in newton) act at a point: $\mathbf{F}_1 = 4\hat{\mathbf{i}} + 9\hat{\mathbf{j}}$, $\mathbf{F}_2 = 5\hat{\mathbf{i}} - 3\hat{\mathbf{j}}$ and $\mathbf{F}_3 = -3\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$. The magnitude of the resultant force is

(a) 10 N

(b) 10.8 N

(c) 18.4 N

(d) 19.3 N

Answer: (a) 10 N

Explanation

Given: F_1 = (4, 9), F_2 = (5, −3), F_3 = (−3, 2).

Formula: $R_x = \Sigma F_x$, $R_y = \Sigma F_y$, R = $\sqrt{R_x^2 + R_y^2}$.

Working: R_x = 4 + 5 + (−3) = 6; R_y = 9 + (−3) + 2 = 8; R = $\sqrt{6^2 + 8^2} = \sqrt{100}$ = 10 N.

Result: 10 N. → option (a)

Why the others fail: 19.3 N adds the magnitudes 9.85 + 5.83 + 3.61, valid only if all three were parallel; 10.8 N uses F_1 + F_2 only; 18.4 N treats every component as positive.

Concept behind the question. With many vectors, add components first and apply Pythagoras only once at the end (Sec 6, N1, S4).

Question 102

VEC-T3-N-037 · Topic 3 · Medium · Calculation

Given $\mathbf{A} = 3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\mathbf{B} = -\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$, the vector **C** for which **A** + **B** + **C** = 0 is

(a) $-2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$

(b) $2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$

(c) $-2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$

(d) $-4\hat{\mathbf{i}} + 6\hat{\mathbf{j}} + \hat{\mathbf{k}}$

Answer: (c) $-2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$

Explanation

Given: A = (3, −2, 1), B = (−1, 4, 2).

Formula: C = −(A + B), component by component.

Working: A + B = (2, 2, 3); so C = (−2, −2, −3).

Result: $-2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$ (check: sum of all three = 0). → option (c)

Why the others fail: option (a) has the wrong sign on z; option (b) is A + B rather than −(A + B); option (d) is B − A.

Concept behind the question. For three vectors to sum to zero, the third must be equal and opposite to the sum of the other two (Sec 6, N8).

Question 103

VEC-T3-N-038 · Topic 3 · Medium · Calculation

Two displacements of 7 km and 8 km are at 60° to each other (tails together). The magnitude of their resultant is

(a) 7.5 km

(b) 10.6 km

(c) 13 km

(d) 15 km

Answer: (c) 13 km

Explanation

Given: A = 7 km, B = 8 km, θ = 60°.

Formula: R = $\sqrt{A^2 + B^2 + 2ABcos\theta}$.

Working: R^2 = 7^2 + 8^2 + 2 × 7 × 8 × cos 60° = 49 + 64 + 56 = 169, so R = 13 km.

Result: 13 km. → option (c)

Why the others fail: 15 km is the sum (θ = 0°); 10.6 km drops the cross term; 7.5 km has the sign of the cross term wrong.

Concept behind the question. The cross term 2AB cos θ is positive for acute angles, so R exceeds $\sqrt{A^2 + B^2}$ (Sec 6, N2).

Question 104

VEC-T3-N-039 · Topic 3 · Medium · Calculation

Forces of 2 N and 4 N act at a point with 120° between them. The resultant makes what angle with the 2 N force?

(a) 30°

(b) 60°

(c) 90°

(d) 120°

Answer: (c) 90°

Explanation

Given: A = 2 N, B = 4 N, θ = 120°.

Formula: tan α = B sin θ / (A + B cos θ), α measured from A.

Working: A + B cos 120° = 2 + 4(−0.5) = 0; B sin 120° = 4 × 0.866 = 3.464. The denominator is 0, so tan α is infinite and α = 90°.

Result: 90° (R = 3.46 N = $2\sqrt{3}$ N; check: $2^2 + R^2 = 4^2$). → option (c)

Why the others fail: 60° would be the bisector direction; 30° = 120° − 90° is the angle from the 4 N force; 120° is the angle between the forces themselves.

Concept behind the question. A zero denominator in tan α is not an error: it simply means the resultant is perpendicular to A (Sec 6, N3).

Question 105

VEC-T3-N-040 · Topic 3 · Medium · Calculation

Two vectors of equal magnitude have a resultant of magnitude $\sqrt{3}$ times the magnitude of each. The angle between them is

(a) 30°

(b) 60°

(c) 120°

(d) 150°

Answer: (b) 60°

Explanation

Given: R = $\sqrt{3}$ A, with both magnitudes equal to A.

Formula: R = 2A cos(θ/2).

Working: $\sqrt{3}$ A = 2A cos(θ/2), so cos(θ/2) = $\sqrt{3}$/2 = 0.866, θ/2 = 30° and θ = 60°.

Result: 60° (check: $R^2 = 2A^2(1 + \cos 60°) = 3A^2$). → option (b)

Why the others fail: 30° is the half-angle; 120° comes from $R^2 = 2A^2 - 2A^2 \cos\theta = 3A^2$ (wrong sign) giving cos θ = −1/2; 150° = 180° − 30°.

Concept behind the question. For equal vectors, the factor 2cos(θ/2) = $\sqrt{3}$ pins the angle; the answer is the full angle between the vectors, not the angle each makes with the resultant (Sec 6, N6).

Question 106

VEC-T3-N-041 · Topic 3 · Medium · Calculation

Raindrops have a vertical velocity component of 20 m/s and a horizontal velocity component of 15 m/s due to a steady wind. Taking tan 37° = 3/4, the raindrops' resultant velocity makes with the vertical an angle of about

(a) 37°

(b) 41°

(c) 49°

(d) 53°

Answer: (a) 37°

Explanation

Given: vertical component 20 m/s, horizontal component 15 m/s.

Formula: angle with the vertical θ: tan θ = (horizontal)/(vertical).

Working: tan θ = 15/20 = 0.75 = 3/4, so $\theta \approx 37°$.

Result: 37° (the umbrella must be tilted by this angle from the vertical). → option (a)

Why the others fail: 53° has tan = 4/3 and is the angle from the horizontal; the resultant is $\sqrt{20^2 + 15^2}$ = 25 m/s, not 20, so sin^{-1}(15/20) and cos^{-1}(15/20) (49° and 41°) are wrong.

Concept behind the question. Tilt from the vertical uses the horizontal-to-vertical ratio, as in the chapter's umbrella example (Sec 6, S5).

Question 107

VEC-T3-N-042 · Topic 3 · Medium · Calculation

A motorboat heads due north at 20 km/h in a current of 10 km/h that flows 60° east of south. Adding the two velocities, the boat's speed over the ground is

(a) 17.3 km/h

(b) 22.4 km/h

(c) 26.5 km/h

(d) 30 km/h

Answer: (a) 17.3 km/h

Explanation

Given: boat 20 km/h north; current 10 km/h at 60° east of south.

Formula: R = $\sqrt{A^2 + B^2 + 2ABcos\theta}$, θ = angle between the two velocity directions = 180° − 60° = 120°.

Working: R^2 = 20^2 + 10^2 + 2 × 20 × 10 × cos 120° = 400 + 100 − 200 = 300, so R = 17.3 km/h (= $10\sqrt{3}$ km/h).

Result: 17.3 km/h. → option (a)

Why the others fail: 30 km/h needs the same direction; 26.5 km/h uses cos 60° (+) and so overstates R; 22.4 km/h ignores the cross term.

Concept behind the question. The angle in the law of cosines is between the directions of the two vectors; "60° east of south" is 120° from north (Sec 6, S6).

Question 108

VEC-T3-N-043 · Topic 3 · Medium · Calculation

For $\mathbf{A} = 4\hat{\mathbf{i}} + \hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ and $\mathbf{B} = 2\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - \hat{\mathbf{k}}$, the magnitude of $\mathbf{A} - \mathbf{B}$ is

(a) 1.9 units

(b) 7 units

(c) 8.8 units

(d) 11 units

Answer: (b) 7 units

Explanation

Given: A = (4, 1, 5), B = (2, 4, −1).

Formula: subtract components, then magnitude = $\sqrt{x^2 + y^2 + z^2}$.

Working: A − B = (2, −3, 6); magnitude = $\sqrt{4 + 9 + 36}$ = $\sqrt{49}$ = 7.

Result: 7 units. → option (b)

Why the others fail: 8.8 = $\sqrt{6^2 + 5^2 + 4^2}$ belongs to A + B; 1.9 = $\sqrt{42} - \sqrt{21}$ treats magnitudes as simple numbers to subtract; 11 = 2 + 3 + 6 skips the squares and the root.

Concept behind the question. |A − B| is not |A| − |B|: subtract the vectors by components first, then take the magnitude (Sec 6, N1, N7).

Question 109

VEC-T3-N-044 · Topic 3 · Medium · Calculation

A vector is $\hat{\mathbf{i}} + \hat{\mathbf{j}} + \sqrt{2}\,\hat{\mathbf{k}}$. The angle it makes with the z-axis is

(a) 35.3°

(b) 45°

(c) 54.7°

(d) 60°

Answer: (b) 45°

Explanation

Given: A_x = 1, A_y = 1, A_z = $\sqrt{2}$.

Formula: A = $\sqrt{A_x^2 + A_y^2 + A_z^2}$; cos γ = A_z / A.

Working: A = $\sqrt{1+1+2}$ = 2; cos γ = $\sqrt{2}$/2 = 0.707, so γ = 45°.

Result: 45°. → option (b)

Why the others fail: cos α = 1/2 gives 60° (the angle with the x-axis); using a ratio of only two components omits the third component and does not give the angle with the z-axis here (54.7° and 35.3° are incorrect).

Concept behind the question. The angle with an axis always comes from cos^{-1}(component/magnitude), because the magnitude is the hypotenuse (Sec 5, N7).

Question 110

VEC-T3-N-045 · Topic 3 · Medium · Calculation

A vector in the first octant makes equal angles with the x-, y- and z-axes. Each of these angles is about

(a) 35.3°

(b) 45°

(c) 54.7°

(d) 70.5°

Answer: (c) 54.7°

Explanation

Given: α = β = γ, with all three components positive.

Formula: $cos^2\alpha$ + $cos^2\beta$ + $cos^2\gamma$ = 1.

Working: 3 $cos^2\alpha$ = 1, so cos α = 1/$\sqrt{3}$ = 0.577 and α = 54.7°.

Result: 54.7°. → option (c)

Why the others fail: 70.5° comes from 3 cos α = 1; 45° belongs to a vector in a plane ($cos^2\alpha$ + $cos^2\beta$ = 1); 35.3° is the angle whose cosine is $\sqrt{2/3}$, i.e. 1/$\sqrt{3}$ is its sine.

Concept behind the question. Equal angles with all three axes describe the cube diagonal; the identity on direction cosines gives its angle (Sec 5, N8).

Question 111

VEC-T3-N-046 · Topic 3 · Medium · Calculation

A davit cable pulls a lifeboat with a force of 120 N, the cable making 30° with the vertical. The horizontal component of the pull is

(a) 60 N

(b) 103.9 N

(c) 120 N

(d) 240 N

Answer: (a) 60 N

Explanation

Given: F = 120 N, θ = 30° from the vertical.

Formula: the component next to the angle gets cos, so $F_{vertical}$ = F cos θ and $F_{horizontal}$ = F sin θ.

Working: $F_{horizontal}$ = 120 × sin 30° = 120 × 0.5 = 60 N.

Result: 60 N. → option (a)

Why the others fail: 103.9 N = 120 cos 30° is the vertical component; 120 N ignores the angle; 240 N is 120 ÷ sin 30°.

Concept behind the question. When the angle is given from the y-axis rather than the x-axis, the roles of sin and cos swap; always ask which side is adjacent to the given angle (Sec 5, S2, Mistakes).

Question 112

VEC-T3-N-047 · Topic 3 · Hard · Calculation

Two forces have a largest resultant of 21 N and a smallest resultant of 3 N. When they act at right angles to each other, the resultant is

(a) 12 N

(b) 15 N

(c) 18 N

(d) 21.2 N

Answer: (b) 15 N

Explanation

Given: A + B = 21 N and A − B = 3 N (A the larger force).

Formula: R_{max} = A + B, R_{min} = A − B; at 90°, R = $\sqrt{A^2 + B^2}$.

Working: A = (21 + 3)/2 = 12 N; B = (21 − 3)/2 = 9 N; R = $\sqrt{12^2 + 9^2}$ = $\sqrt{225}$ = 15 N.

Result: 15 N. → option (b)

Why the others fail: 21.2 N = $\sqrt{21^2 + 3^2}$ treats the limits as the forces; 12 N is just the larger force; 18 N is 2B.

Concept behind the question. The extreme resultants are the sum and the difference of the magnitudes, so they can be solved back for the two forces (Sec 6, N4–N5).

Question 113

VEC-T3-N-048 · Topic 3 · Hard · Calculation

A boat's own velocity (relative to the water) has magnitude 5 m/s. The current is 3 m/s due east. The boat's velocity over the ground is the vector sum of the two and it is due north. If the river is 120 m wide, the time to cross is

(a) 15 s

(b) 24 s

(c) 30 s

(d) 40 s

Answer: (c) 30 s

Explanation

Given: own velocity 5 m/s (direction to be found), current 3 m/s east, resultant due north, width 120 m.

Formula: resultant = own + current; with the resultant due north, the east-west components cancel.

Working: the own velocity has an x-component of −3 m/s (west) to cancel the current. Its northward component is $\sqrt{5^2 - 3^2}$ = $\sqrt{16}$ = 4 m/s, which is the resultant speed. Time = 120 ÷ 4 = 30 s.

Result: 30 s. → option (c)

Why the others fail: 24 s uses 5 m/s although part of it is spent cancelling the current; 15 s uses 8 m/s, which would need the two velocities to be parallel; 40 s uses the current speed.

Concept behind the question. "Resultant due north" is a component condition: the east-west parts of the two vectors must cancel, and what remains is the northward speed (Sec 6, N1).

Question 114

VEC-T3-N-049 · Topic 3 · Hard · Calculation

Three forces of 10 N each act at a point in a plane, at 0°, 60° and 120° to the x-axis. The magnitude of the resultant is

(a) 10 N

(b) 17.3 N

(c) 20 N

(d) 30 N

Answer: (c) 20 N

Explanation

Given: three forces of 10 N at 0°, 60°, 120°.

Formula: $R_x = \Sigma F \cos\theta$, $R_y = \Sigma F \sin\theta$, $R = \sqrt{R_x^2 + R_y^2}$.

Working: $R_x = 10(1 + 0.5 - 0.5) = 10$ N; $R_y = 10(0 + 0.866 + 0.866) = 17.32$ N; $R = \sqrt{10^2 + 17.32^2} = \sqrt{400} = 20$ N.

Result: 20 N, at 60° to the x-axis. → option (c)

Why the others fail: 30 N needs all three parallel; 17.3 N is only R_y; 10 N is only R_x. Both components are needed.

Concept behind the question. When several vectors point in different directions, resolve every one along the same axes, add each axis separately and combine at the end (Sec 5 and Sec 6, S4).

Question 115

VEC-T3-N-050 · Topic 3 · Hard · Calculation

A force of 13 N acts along the direction $5\hat{\mathbf{i}}+12\hat{\mathbf{j}}$ and a force of 15 N acts along $3\hat{\mathbf{i}}-4\hat{\mathbf{j}}$. The magnitude of their resultant is

(a) 2 N

(b) 11.3 N

(c) 14 N

(d) 28 N

Answer: (c) 14 N

Explanation

Given: F_1 = 13 N along (5, 12); F_2 = 15 N along (3, −4).

Formula: $\mathbf{F} = F\,\hat{\mathbf{n}}$ with $\hat{\mathbf{n}} = \mathbf{d}/|\mathbf{d}|$; then add components.

Working: $|d_1| = \sqrt{5^2 + 12^2} = 13$, so F_1 = (5, 12) N. $|d_2| = 5$, so $F_2 = 15 \times (3, -4)/5 = (9, -12)$ N. Sum = (14, 0) N.

Result: 14 N, along the x-axis. → option (c)

Why the others fail: 28 N requires parallel forces; 11.3 N adds the raw direction vectors (the direction vectors have lengths 13 and 5, not the forces); 2 N requires opposite forces.

Concept behind the question. A direction given as a vector only fixes the unit vector; the force vector is its magnitude times that unit vector (Sec 5, N2, and Sec 6, N1).

Question 116

VEC-T4-C-001 · Topic 4 · Easy · Concept

For two vectors **A** and **B** , which of the following is a scalar?

(a) $3\mathbf{A}$

(b) $\mathbf{A} \times \mathbf{B}$

(c) $\mathbf{A} + \mathbf{B}$

(d) $\mathbf{A} \cdot \mathbf{B}$

Answer: (d) $\mathbf{A} \cdot \mathbf{B}$

Explanation

Rule: The scalar (dot) product is A·B = AB cos θ, a plain number (Sec 7, N1).

Reasoning: AB cos θ is built from two magnitudes and a cosine, so it has no direction.

Reasoning: A × B, A + B and 3A each carry a direction, so each is a vector.

Why the others fail: A × B is the vector product; A + B is the resultant vector; 3A is a stretched copy of A. All three have direction.

Concept behind the question. Sec 7: the dot product keeps only the "overlap" of the two directions and so loses direction altogether, which is why its result is called a scalar.

Question 117

VEC-T4-C-002 · Topic 4 · Easy · Concept

For the unit vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, $\hat{\mathbf{k}}$ along the three axes, which pair of results is correct?

(a) $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 1$ and $\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{k}}$

(b) $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 0$ and $\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = 0$

(c) $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 0$ and $\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = 1$

(d) $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 1$ and $\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = 0$

Answer: (d) $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 1$ and $\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = 0$

Explanation

Rule: i·i = j·j = k·k = 1 and i·j = j·k = k·i = 0 (Sec 7, N4).

Reasoning: i·i = (1)(1)cos 0° = 1, and i·j = (1)(1)cos 90° = 0.

Why the others fail: i·j is a scalar, so it cannot be k; the pair (0, 1) is the right pair upside down; and i·i is 1, not 0 (zero is the cross product i × i).

Concept behind the question. Sec 7: the dot product of a unit vector with itself is 1 because the angle is 0°; with a perpendicular unit vector it is 0 because cos 90° = 0.

Question 118

VEC-T4-C-003 · Topic 4 · Easy · Concept

For two vectors, $\mathbf{A} \cdot \mathbf{B} = -12$. Then $\mathbf{B} \cdot \mathbf{A}$ equals

(a) -12

(b) 0

(c) 12

(d) 144

Answer: (a) -12

Explanation

Rule: The dot product is commutative: A·B = B·A (Sec 7, N5).

Reasoning: AB cos θ is unchanged when A and B swap places, so B·A = A·B = -12.

Why the others fail: 12 incorrectly reverses the sign; unlike the cross product, the dot product is commutative; 0 and 144 have no basis.

Concept behind the question. Sec 7: the product AB cos θ contains no sense of order, unlike the cross product, which changes sign when the order is reversed (Sec 8).

Question 119

VEC-T4-C-004 · Topic 4 · Easy · Concept

Two non-zero vectors make an angle of 150° with each other. Their scalar product is

(a) zero

(b) impossible to tell without the magnitudes

(c) positive

(d) negative

Answer: (d) negative

Explanation

Rule: A·B = AB cos θ; it is positive for θ < 90°, zero at 90° and negative for θ > 90° (Sec 7, N6).

Reasoning: cos 150° = -0.866 is negative, while A and B (magnitudes) are positive, so the product is negative.

Why the others fail: the magnitudes A and B are always positive, so only cos θ fixes the sign; zero needs exactly 90°.

Concept behind the question. Sec 7: the sign of the dot product tells at a glance whether the angle between the vectors is acute (positive), right (zero) or obtuse (negative).

Question 120

VEC-T4-C-005 · Topic 4 · Easy · Concept

For two non-zero vectors **A** and **B** , $\mathbf{A} \cdot \mathbf{B} = 0$. The vectors are

(a) antiparallel (opposite in direction)

(b) parallel to each other

(c) equal in magnitude

(d) perpendicular to each other

Answer: (d) perpendicular to each other

Explanation

Rule: Two non-zero vectors are perpendicular if and only if A·B = 0 (Sec 7, N6).

Reasoning: AB cos θ = 0 with A $\neq$ 0 and B $\neq$ 0 forces cos θ = 0, so θ = 90°.

Why the others fail: parallel or antiparallel vectors make cos $\theta = \pm 1$ (largest dot product in size); zero is the cross-product test for them. Magnitudes play no part.

Concept behind the question. Sec 7 (Mistakes): ”dot product zero” means perpendicular. ”Cross product zero” means parallel or antiparallel.

Question 121

VEC-T4-C-006 · Topic 4 · Easy · Concept

A vector **a** is measured in m/s^2 and a vector **d** in m. The scalar product $\mathbf{a} \cdot \mathbf{d}$ has unit

(a) m^2/s^4

(b) m^2/s^2

(c) s^{-2}

(d) m/s^2

Answer: (b) m^2/s^2

Explanation

Rule: The dot product is a number times the product of the two units (Sec 7, S2).

Reasoning: (m/s^2) $\times$ (m) = m^2/s^2.

Why the others fail: s^{-2} comes from dividing the units; m/s^2 forgets the second vector; m^2/s^4 squares the unit of a.

Concept behind the question. Sec 7: multiplying two physical quantities multiplies their units, and cos θ is a pure number, so it adds no unit.

Question 122

VEC-T4-C-007 · Topic 4 · Easy · Concept

For any vector **A** of magnitude A, the product $\mathbf{A} \cdot \mathbf{A}$ equals

(a) 2A

(b) 0

(c) A

(d) A^2

Answer: (d) A^2

Explanation

Rule: A·A = AA cos 0° = A^2 (Sec 7, N7).

Reasoning: a vector makes an angle 0° with itself and cos 0° = 1, so A·A = A·A·1 = A^2.

Why the others fail: A has the unit of A, not of A^2; 0 belongs to the cross product A × A; 2A is a sum A + A, not a product.

Concept behind the question. Sec 7: the dot product of a vector with itself gives the square of its magnitude, which is how a magnitude is recovered from components: $A^2 = A_x^2 + A_y^2 + A_z^2$.

Question 123

VEC-T4-C-008 · Topic 4 · Easy · Concept

Three vectors satisfy $\mathbf{A} \cdot \mathbf{B} = 6$ and $\mathbf{A} \cdot \mathbf{C} = -2$. Then $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C})$ equals

(a) −12

(b) −3

(c) 4

(d) 8

Answer: (c) 4

Explanation

Rule: The dot product is distributive: A·(B + C) = A·B + A·C (Sec 7, N5).

Reasoning: A·(B + C) = 6 + (−2) = 4.

Why the others fail: 8 subtracts the second term; −12 and −3 multiply or divide the products, which the distributive law never does.

Concept behind the question. Sec 7: the dot product distributes over vector addition exactly like ordinary multiplication distributes over addition of numbers.

Question 124

VEC-T4-C-009 · Topic 4 · Medium · Concept

Two non-zero vectors **A** and **B** point in exactly opposite directions (antiparallel). Their scalar product is

(a) −AB

(b) +AB

(c) 0

(d) −(A + B)

Answer: (a) −AB

Explanation

Rule: A·B = AB cos θ, with θ the angle between the vectors (Sec 7, N1).

Reasoning: for opposite directions θ = 180° and cos 180° = −1, so A·B = −AB, the most negative value possible.

Why the others fail: +AB is for parallel vectors (θ = 0°); 0 is for θ = 90°; a dot product multiplies magnitudes, it never adds them.

Concept behind the question. Sec 7: the dot product runs from +AB (parallel) through 0 (perpendicular) to −AB (antiparallel).

Question 125

VEC-T4-C-010 · Topic 4 · Medium · Concept

Which of the following statements about the scalar product is incorrect?

(a) $\mathbf{A}\cdot\mathbf{A} = A^2$ for every vector $\mathbf{A}$.

(b) $\mathbf{A}\cdot\mathbf{B} = \mathbf{B}\cdot\mathbf{A}$ for all vectors.

(c) $\hat{\mathbf{j}}\cdot\hat{\mathbf{k}} = 0$ and $\hat{\mathbf{k}}\cdot\hat{\mathbf{k}} = 1$.

(d) If $\mathbf{A}\cdot\mathbf{B} = 0$, then $\mathbf{A}$ or $\mathbf{B}$ must be the null vector.

Answer: (d) If $\mathbf{A}\cdot\mathbf{B} = 0$, then $\mathbf{A}$ or $\mathbf{B}$ must be the null vector.

Explanation

Rule: A·B = 0 also happens for non-zero perpendicular vectors (Sec 7, N6 and Clearing the confusions).

Reasoning: for example i·j = 0 although neither i nor j is the null vector, so the statement is false.

Why the others fail: the other three are exactly Sec 7, N5, N4 and N7.

Concept behind the question. Sec 7 and Clearing the confusions: a zero dot product means one vector is null OR the two are perpendicular, never only the first.

Question 126

VEC-T4-C-011 · Topic 4 · Medium · Concept

For $\mathbf{A} = A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}}$, the scalar product $\mathbf{A}\cdot\hat{\mathbf{j}}$ equals

(a) 0

(b) A_y

(c) $A_y\hat{\mathbf{j}}$

(d) $A_x + A_y + A_z$

Answer: (b) A_y

Explanation

Rule: A·B = A_xB_x + A_yB_y + A_zB_z (Sec 7, N3).

Reasoning: j has components (0, 1, 0), so A·j = A_x(0) + A_y(1) + A_z(0) = A_y.

Why the others fail: A_y j is a vector, but a dot product is a scalar; 0 would need A_y = 0; the sum uses components that j does not have.

Concept behind the question. Sec 7 and Sec 5: dotting with a unit vector picks out the component along that unit vector.

Question 127

VEC-T4-C-012 · Topic 4 · Medium · Concept

For non-zero **B**, the scalar component (projection) of **A** along the direction of **B** is

(a) $\dfrac{\mathbf{A}\cdot\mathbf{B}}{AB}$

(b) $\mathbf{A}\cdot\mathbf{B}$

(c) $\dfrac{\mathbf{A}\cdot\mathbf{B}}{A}$

(d) $\dfrac{\mathbf{A}\cdot\mathbf{B}}{B}$

Answer: (d) $\dfrac{\mathbf{A}\cdot\mathbf{B}}{B}$

Explanation

Rule: The component of B along A is A·B/A (Sec 7, S5); swapping the roles of A and B gives the component of A along B.

Reasoning: A·B = B(A cos θ), so dividing by B leaves A cos θ, the part of A lying along B.

Why the others fail: dividing by A gives B cos θ (the other projection); no division leaves AB cos θ; dividing by AB leaves a pure number cos θ.

Concept behind the question. Sec 7 (N2): the dot product is B times the projection of A on B, so dividing it by B recovers that projection.

Question 128

VEC-T4-C-013 · Topic 4 · Medium · Concept

For two non-zero vectors **A** and **B** , the magnitudes | **A** + **B** | and | **A** − **B** | are equal. The vectors are

(a) perpendicular to each other

(b) parallel to each other

(c) antiparallel to each other

(d) equal in magnitude

Answer: (a) perpendicular to each other

Explanation

Rule: |A + B|^2 = A^2 + B^2 + 2A·B and |A − B|^2 = A^2 + B^2 − 2A·B.

Reasoning: equal magnitudes give 2A·B = −2A·B, so A·B = 0 and the vectors are perpendicular (Sec 7, N6).

Why the others fail: for parallel vectors |A+B| = A + B but |A−B| is only the difference of the sizes, so they are unequal; for antiparallel vectors |A+B| is the difference of the sizes and |A−B| = A + B, again unequal. Equal sizes A = B do not force |A+B| = |A−B|.

Concept behind the question. Sec 7 with Sec 6 (N7): a rectangle has equal diagonals, which is the geometric face of A·B = 0.

Question 129

VEC-T4-C-014 · Topic 4 · Medium · Concept

For any two vectors **A** and **B** , the square of the magnitude of their sum, | **A** + **B** |^2, equals

(a) $(A + B)^2$ with A and B the magnitudes

(b) $A^2 + B^2 + 2\,\mathbf{A} \cdot \mathbf{B}$

(c) $A^2 + B^2 - 2\,\mathbf{A} \cdot \mathbf{B}$

(d) $A^2 + B^2$

Answer: (b) $A^2 + B^2 + 2\,\mathbf{A} \cdot \mathbf{B}$

Explanation

Rule: A·A = A^2 and the dot product is distributive and commutative (Sec 7, N5, N7).

Reasoning: |A + B|^2 = (A + B)·(A + B) = A·A + A·B + B·A + B·B = A^2 + B^2 + 2A·B.

Why the others fail: −2A·B belongs to |A − B|^2; A^2 + B^2 holds only if A·B = 0; (A + B)^2 holds only if A·B = AB, i.e. parallel vectors.

Concept behind the question. Sec 7 with Sec 6 (N2): since A·B = AB cos θ, this is exactly the law of cosines R^2 = A^2 + B^2 + 2AB cos θ.

Question 130

VEC-T4-C-015 · Topic 4 · Hard · Concept

Three non-zero vectors satisfy **A** · **B** = **A** · **C** . Which conclusion is always true?

(a) **B** − **C** is perpendicular to **A** , or is the null vector

(b) **B** and **C** have equal magnitudes

(c) **B** = **C**

(d) **B** is parallel to **C**

Answer: (a) $\mathbf{B} - \mathbf{C}$ is perpendicular to $\mathbf{A}$, or is the null vector

Explanation

Rule: The dot product is distributive, so A·B − A·C = A·(B − C) (Sec 7, N5), and zero means perpendicular (N6).

Reasoning: A·B = A·C gives A·(B − C) = 0, so B − C is perpendicular to A (or null).

Reasoning: counter-example with A = i, B = i + j, C = i + 2j: A·B = A·C = 1, but $B \neq C$, $|B| \neq |C|$ and B is not parallel to C.

Why the others fail: a dot product cannot be ”divided out” the way a common number can.

Concept behind the question. Sec 7: unlike ordinary numbers, a dot product has no cancellation law, because many different vectors share the same overlap with A.

Question 131

VEC-T4-C-016 · Topic 4 · Hard · Concept

$\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$ are four vectors. Which expression is NOT meaningful?

(a) $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C})$

(b) $(\mathbf{A} \cdot \mathbf{B})\,\mathbf{C}$

(c) $(\mathbf{A} \cdot \mathbf{B})(\mathbf{C} \cdot \mathbf{D})$

(d) $\mathbf{A} \cdot (\mathbf{B} \cdot \mathbf{C})$

Answer: (d) $\mathbf{A} \cdot (\mathbf{B} \cdot \mathbf{C})$

Explanation

Rule: A·B is a scalar, and the dot product is defined only between two vectors (Sec 7, N1).

Reasoning: B·C is a scalar, so A·(B·C) would be a dot product of a vector with a number, which is not defined.

Why the others fail: (A·B)C multiplies a vector by a number (Sec 3); (A·B)(C·D) multiplies two numbers; B + C is a vector, so A·(B + C) is a proper dot product.

Concept behind the question. Sec 7: the result of a dot product is a scalar, so it cannot be fed into another dot product as if it were a vector.

Question 132

VEC-T4-N-017 · Topic 4 · Easy · Calculation

Two vectors have magnitudes 15 units and 4 units and the angle between them is 37° (cos 37° = 0.8, sin 37° = 0.6). Find their scalar product.

(a) 15.2

(b) 36

(c) 48

(d) 60

Answer: (c) 48

Explanation

Given: A = 15, B = 4, $\theta = 37^\circ$, cos 37° = 0.8.

Formula: A·B = AB cos θ.

Working: A·B = $15 \times 4 \times 0.8 = 48$.

Result: 48 (a scalar). → option (c)

Why the others fail: 36 uses sin 37°; 60 forgets cos θ; 15.2 adds the magnitudes first.

Concept behind the question. Sec 7 (N1): the dot product is the product of the two magnitudes with the cosine of the angle between them.

Question 133

VEC-T4-N-018 · Topic 4 · Easy · Calculation

Find $\mathbf{A} \cdot \mathbf{B}$ for $\mathbf{A} = 5\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ and $\mathbf{B} = 3\hat{\mathbf{i}} - 4\hat{\mathbf{j}}$.

(a) -14

(b) 7

(c) 15

(d) 23

Answer: (b) 7

Explanation

Given: A = (5, 2), B = (3, -4).

Formula: A·B = A_xB_x + A_yB_y.

Working: A_xB_x = 5 $\times$ 3 = 15.

Working: A_yB_y = 2 $\times$ (-4) = -8.

Working: A·B = 15 + (-8) = 7.

Result: 7. $\rightarrow$ option (b)

Why the others fail: 23 adds the sizes and loses the sign; -14 multiplies unlike components (i·j = 0 kills them); 15 ignores the j part.

Concept behind the question. Sec 7 (N3, N4): only matching components multiply, because i·j = 0, and the answer is a signed number.

Question 134

VEC-T4-N-019 · Topic 4 · Easy · Calculation

Work is defined as W = F·d = Fd cos θ. A tug pulls a barge with a steady force of 800 N while the rope makes 60° with the barge's displacement of 25 m. Find W.

(a) 20000 J

(b) 1.73 $\times$ 10^4 J

(c) -1.9 $\times$ 10^4 J

(d) 10000 J

Answer: (d) 10000 J

Explanation

Given: F = 800 N, d = 25 m, θ = 60°, and W = F·d = Fd cos θ (given).

Formula: W = Fd cos θ.

Working: cos 60° = 0.5.

Working: W = 800 $\times$ 25 $\times$ 0.5 = 10000 J.

Result: 10000 J. $\rightarrow$ option (d)

Why the others fail: 20000 J forgets the cosine; 1.73 $\times$ 10^4 J uses sin 60°; -1.9 $\times$ 10^4 J treats 60 as radians.

Concept behind the question. Sec 7 (N8): the cosine keeps only the part of the pull along the displacement; the rest of the rope force does not count in F·d.

Question 135

VEC-T4-N-020 · Topic 4 · Easy · Calculation

For $\mathbf{A} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}$, find $\mathbf{A} \cdot \mathbf{A}$.

(a) 5

(b) 7

(c) 31

(d) 49

Answer: (d) 49

Explanation

Given: A = (2, −3, 6).

Formula: A·A = A_x^2 + A_y^2 + A_z^2 (= A^2).

Working: 4 + 9 + 36 = 49.

Result: 49. → option (d)

Why the others fail: 7 is the magnitude $\sqrt{49}$; 5 just adds the components; 31 makes $(-3)^2$ negative.

Concept behind the question. Sec 7 (N7): A·A = A^2. The squares of components are all positive, so A·A is never negative.

Question 136

VEC-T4-N-021 · Topic 4 · Easy · Calculation

Find b so that $3\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ is perpendicular to $4\hat{\mathbf{i}} + b\hat{\mathbf{j}}$.

(a) 6

(b) −6

(c) 8/3

(d) −8/3

Answer: (b) −6

Explanation

Given: A = (3, 2), B = (4, b), A ⊥ B.

Formula: perpendicular means A·B = 0, with A·B = A_xB_x + A_yB_y.

Working: 3(4) + 2(b) = 0, so 12 + 2b = 0.

Working: b = −12/2 = −6.

Result: b = −6. → option (b)

Why the others fail: b = 6 gives 12 + 2(6) = 24, not 0; 8/3 comes from the parallel condition; −8/3 from multiplying unlike components.

Concept behind the question. Sec 7 (TIP): to find an unknown that makes two vectors perpendicular, set the component dot product to zero and solve.

Question 137

VEC-T4-N-022 · Topic 4 · Easy · Calculation

Two vectors have magnitudes 6 and 3, and their scalar product is −9. Find the angle between them.

(a) 30°

(b) 60°

(c) 120°

(d) 180°

Answer: (c) 120°

Explanation

Given: A = 6, B = 3, A·B = −9.

Formula: cos θ = A·B/(AB).

Working: cos θ = −9/(6 × 3) = −9/18 = −0.5.

Working: θ = cos^{-1}(−0.5) = 120°.

Result: 120°. → option (c)

Why the others fail: 60° has cos θ = +0.5; 180° needs cos θ = −1 (−9/9); 30° uses the wrong trig function.

Concept behind the question. Sec 7 (N7): a negative dot product always means an angle greater than 90° and at most 180°. Here it is 120°.

Question 138

VEC-T4-N-023 · Topic 4 · Easy · Calculation

A wind has velocity $10\hat{\mathbf{i}} + 5\hat{\mathbf{j}}$ km/h. A runway lies along $4\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$ (east and north axes). Find the component of the wind along the runway direction.

(a) 2.2 km/h

(b) 4.92 km/h

(c) 11 km/h

(d) 55 km/h

Answer: (c) 11 km/h

Explanation

Given: A = (10, 5) km/h, B = (4, 3).

Formula: component of A along B = A·B/B.

Working: A·B = 10×4 + 5×3 = 55.

Working: B = $\sqrt{4^2 + 3^2}$ = 5.

Working: component = 55/5 = 11 km/h.

Result: 11 km/h. → option (c)

Why the others fail: 55 is A·B undivided; 4.92 divides by A = $\sqrt{125}$; 2.2 divides by B^2 = 25.

Concept behind the question. Sec 7 (S5): a projection needs a division by the length of the vector you project onto, so the answer keeps the unit of the projected vector.

Question 139

VEC-T4-N-024 · Topic 4 · Easy · Calculation

Find the angle between $\mathbf{A} = 2\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ and $\mathbf{B} = -3\hat{\mathbf{i}} - 3\hat{\mathbf{j}}$.

(a) 0°

(b) 45°

(c) 90°

(d) 180°

Answer: (d) 180°

Explanation

Given: A = (2, 2), B = (−3, −3).

Formula: cos θ = A·B/(AB).

Working: A·B = 2(−3) + 2(−3) = −12; A = $2\sqrt{2}$, B = $3\sqrt{2}$, so AB = 12.

Working: cos θ = −12/12 = −1, so θ = 180°.

Result: 180°. → option (d)

Why the others fail: 0° needs cos θ = +1; 90° needs A·B = 0; 45° is the direction angle of A alone, not the angle between the two vectors.

Concept behind the question. Sec 7 (N6, N7): cos θ = −1 means the vectors lie along one line but point opposite ways (antiparallel). B here is −1.5 A.

Question 140

VEC-T4-N-025 · Topic 4 · Medium · Calculation

Two perpendicular vectors **A** and **B** have magnitudes 5 and 12. Find $(\mathbf{A}+\mathbf{B})\cdot(\mathbf{A}-\mathbf{B})$.

(a) -119

(b) 0

(c) 119

(d) 169

Answer: (a) -119

Explanation

Given: A = 5, B = 12, A·B = 0 (perpendicular).

Formula: (A + B)·(A − B) = A·A − A·B + B·A − B·B = $A^2 - B^2$ (distributive and commutative laws).

Working: $A^2 - B^2$ = 25 − 144 = -119.

Result: -119. → option (a)

Why the others fail: 119 has the order reversed; 169 is A^2 + B^2; 0 would need A = B.

Concept behind the question. Sec 7 (N5, N7): expanding with the distributive law, the cross terms cancel because the dot product is commutative, leaving $A^2 - B^2$.

Question 141

VEC-T4-N-026 · Topic 4 · Medium · Calculation

Find λ so that $\lambda\hat{\mathbf{i}}+2\hat{\mathbf{j}}-3\hat{\mathbf{k}}$ is perpendicular to $2\hat{\mathbf{i}}-5\hat{\mathbf{j}}+4\hat{\mathbf{k}}$.

(a) -1

(b) 1

(c) 5

(d) 11

Answer: (d) 11

Explanation

Given: A = (λ, 2, -3), B = (2, -5, 4), A $\perp$ B.

Formula: A·B = A_xB_x + A_yB_y + A_zB_z = 0.

Working: 2λ + (2)(-5) + (-3)(4) = 0, i.e. $2\lambda - 10 - 12 = 0$.

Working: 2λ = 22, so λ = 11.

Result: λ = 11. → option (d)

Why the others fail: -1 turns -12 into +12; 5 drops the k-term; 1 turns -10 into +10.

Concept behind the question. Sec 7 (TIP): perpendicularity is a single linear equation in the unknown, so every constant of this kind is a one-line solve once the dot product is written out.

Question 142

VEC-T4-N-027 · Topic 4 · Medium · Calculation

Find the angle between $\mathbf{A}=\hat{\mathbf{i}}+2\hat{\mathbf{j}}+\hat{\mathbf{k}}$ and $\mathbf{B}=-2\hat{\mathbf{i}}-\hat{\mathbf{j}}+\hat{\mathbf{k}}$.

(a) 30°

(b) 60°

(c) 120°

(d) 150°

Answer: (c) 120°

Explanation

Given: A = (1, 2, 1), B = (-2, -1, 1).

Formula: cos θ = A·B/(AB).

Working: A·B = (1)(-2) + (2)(-1) + (1)(1) = -3.

Working: A = $\sqrt{1+4+1} = \sqrt{6}$ and B = $\sqrt{4+1+1} = \sqrt{6}$, so AB = 6.

Working: $\cos\theta = -3/6 = -0.5$, so $\theta = 120^\circ$.

Result: 120°. $\rightarrow$ option (c)

Why the others fail: 60° has $\cos\theta = +0.5$; 30° and 150° have $\cos\theta = \pm 0.866$, not ± 0.5.

Concept behind the question. Sec 7 (N7): the two lengths are equal here, so the dot product is half of AB in size; its negative sign puts the angle in the obtuse range.

Question 143

VEC-T4-N-028 · Topic 4 · Medium · Calculation

Two vectors have magnitudes 8 and 9, and $\mathbf{A}\cdot\mathbf{B} = 12$. Find the magnitude of their sum $\mathbf{A}+\mathbf{B}$.

(a) 11

(b) 12.04

(c) 13

(d) 17

Answer: (c) 13

Explanation

Given: A = 8, B = 9, A·B = 12.

Formula: $|A + B|^2 = A^2 + B^2 + 2A\cdot B$.

Working: $|A + B|^2 = 64 + 81 + 2(12) = 169$.

Working: $|A + B| = \sqrt{169} = 13$.

Result: 13. $\rightarrow$ option (c)

Why the others fail: 17 assumes parallel vectors; 12.04 assumes perpendicular vectors; 11 is $|A - B|$.

Concept behind the question. Sec 7 with Sec 6 (N2, N7): the sum and difference magnitudes differ only in the sign of the 2A·B term.

Question 144

VEC-T4-N-029 · Topic 4 · Medium · Calculation

Work is defined as W = F·d. A rope pulls a sledge with a force of 60 N whose direction makes 120° with the sledge's displacement of 8 m. Find W.

(a) -240 J

(b) 240 J

(c) 390.8 J

(d) 415.7 J

Answer: (a) -240 J

Explanation

Given: F = 60 N, d = 8 m, $\theta = 120^\circ$, W = F·d = Fd $\cos\theta$ (given).

Formula: W = Fd $\cos\theta$.

Working: $\cos 120^\circ = -0.5$.

Working: $W = 60 \times 8 \times (-0.5) = -240$ J.

Result: -240 J. $\rightarrow$ option (a)

Why the others fail: 240 J loses the negative sign of $\cos 120^\circ$; 415.7 J uses sin; 390.8 J reads 120 as radians.

Concept behind the question. Sec 7 (N6): an angle above 90° means the force component along the displacement is backwards, so the dot product is negative.

Question 145

VEC-T4-N-030 · Topic 4 · Medium · Calculation

A buoy lies at position $2\hat{\mathbf{i}}+\hat{\mathbf{j}}+2\hat{\mathbf{k}}$ km from a ship (x east, y north, z up). Using the scalar product with $\hat{\mathbf{i}}$, find the cosine of the angle between the position vector and the x-axis.

(a) 2

(b) 2/3

(c) 2/9

(d) 1/3

Answer: (b) 2/3

Explanation

Given: r = (2, 1, 2) km.

Formula: cos α = (r·$\hat{\mathbf{i}}$)/(r × 1) = r_x/r, because r·$\hat{\mathbf{i}}$ = r_x and |$\hat{\mathbf{i}}$| = 1.

Working: r = $\sqrt{2^2+1^2+2^2}$ = $\sqrt{9}$ = 3.

Working: r·i = 2; cos α = 2/3.

Result: 2/3. → option (b)

Why the others fail: 2 is r·i undivided (cos cannot exceed 1); 1/3 is r_y/r; 2/9 divides by r^2 = 9.

Concept behind the question. Sec 7 with Sec 5 (N7): dotting with a unit vector gives the component, and dividing by the magnitude gives the direction cosine.

Question 146

VEC-T4-N-031 · Topic 4 · Medium · Calculation

For $\mathbf{A}=\lambda\hat{\mathbf{i}}+3\hat{\mathbf{j}}+2\hat{\mathbf{k}}$ and $\mathbf{B}=4\hat{\mathbf{i}}-2\hat{\mathbf{j}}+5\hat{\mathbf{k}}$, the magnitudes of $\mathbf{A}+\mathbf{B}$ and $\mathbf{A}-\mathbf{B}$ are equal. Find λ.

(a) -4

(b) -1

(c) 1

(d) 1.5

Answer: (b) -1

Explanation

Given: A = (λ, 3, 2), B = (4, -2, 5), |A + B| = |A $-$ B|.

Formula: |A + B|^2 $-$ |A $-$ B|^2 = 4A·B, so equal magnitudes need A·B = 0.

Working: A·B = 4λ + 3(-2) + 2(5) = 4λ $-$ 6 + 10 = 4λ + 4 = 0.

Working: λ = $-4/4$ = -1.

Result: λ = -1. → option (b)

Why the others fail: 1 gets the sign of λ wrong; 1.5 drops the 2 × 5 term; -4 turns -6 into +6.

Concept behind the question. Sec 7: equal sum and difference magnitudes is a disguised way of saying the vectors are perpendicular, so the condition is again A·B = 0.

Question 147

VEC-T4-N-032 · Topic 4 · Medium · Calculation

Two vectors have magnitudes 5 and 7, and the magnitude of their difference is $|\mathbf{A}-\mathbf{B}|=8$. Find $\mathbf{A}\cdot\mathbf{B}$.

(a) -5

(b) 5

(c) 10

(d) 69

Answer: (b) 5

Explanation

Given: A = 5, B = 7, |A − B| = 8.

Formula: |A − B|^2 = A^2 + B^2 − 2A·B, so A·B = (A^2 + B^2 − |A − B|^2)/2.

Working: A·B = (25 + 49 − 64)/2 = (10)/2 = 5.

Result: 5. → option (b)

Why the others fail: 10 is twice the answer; −5 is the negative; 69 adds the |A − B|^2 term.

Concept behind the question. Sec 7 (N5, N7): expanding |A − B|^2 = (A − B)·(A − B) lets you find the dot product from three lengths alone, with no angle needed.

Question 148

VEC-T4-N-033 · Topic 4 · Medium · Calculation

Vector **A** has magnitude 10 and points along +x. Vector **B** has magnitude 8 and makes 60° with the negative x-axis. Find **A** · **B** .

(a) −80

(b) 69.3

(c) 40

(d) −40

Answer: (d) −40

Explanation

Given: A = 10 along +x; B = 8 at 60° to −x.

Formula: A·B = AB cos θ, where θ is measured from A (+x) to B.

Working: the angle between +x and a direction 60° from −x is 180° − 60° = 120°.

Working: A·B = 10 × 8 × cos 120° = 80 × (−0.5) = −40.

Result: −40. → option (d)

Why the others fail: 40 uses 60°; 69.3 uses sin; −80 would need exactly 180°.

Concept behind the question. Sec 7 (N1): the angle in A·B = AB cos θ is the angle between the two vectors tail to tail, not the angle B makes with an axis.

Question 149

VEC-T4-N-034 · Topic 4 · Medium · Calculation

Work is defined as W = F·d. A winch cable exerts a force $\mathbf{F} = 2\hat{\mathbf{i}} + 7\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$ N on a crate that moves through $\mathbf{d} = 5\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$ m. Find W.

(a) −16 J

(b) 8 J

(c) 16 J

(d) 36 J

Answer: (a) −16 J

Explanation

Given: F = (2, 7, −3) N, d = (5, −2, 4) m, W = F·d (given).

Formula: W = $F_x d_x$ + $F_y d_y$ + $F_z d_z$.

Working: $F_x d_x$ = 10, $F_y d_y$ = −14, $F_z d_z$ = −12.

Working: W = 10 + (−14) + (−12) = −16 J.

Result: −16 J. → option (a)

Why the others fail: 16 J flips the final sign; 36 J adds absolute values; 8 J uses +12 for the z-term.

Concept behind the question. Sec 7 (N3, N8): in three dimensions the dot product simply adds one more matching-component product; a negative total means the force opposes the motion overall.

Question 150

VEC-T4-N-035 · Topic 4 · Medium · Calculation

For two vectors, $\mathbf{A}\cdot\mathbf{B} = 5$. Find $(3\mathbf{A})\cdot(-2\mathbf{B})$.

(a) -30

(b) -10

(c) 6

(d) 30

Answer: (a) -30

Explanation

Given: A·B = 5.

Formula: (mA)·(nB) = mn (A·B), because the numbers m and n come out of the dot product (distributive law).

Working: (3A)·(−2B) = (3)(−2)(A·B) = $-6 \times 5 = -30$.

Result: -30. → option (a)

Why the others fail: 30 loses the sign; 6 adds; -10 uses only -2.

Concept behind the question. Sec 7 (N5): scaling a vector scales the dot product by the same factor; reversing a vector (negative factor) reverses the sign of the dot product.

Question 151

VEC-T4-N-036 · Topic 4 · Medium · Calculation

A ship has velocity $8\hat{\mathbf{i}}+6\hat{\mathbf{j}}$ m/s. A channel runs along the unit vector $\hat{\mathbf{n}} = 0.6\hat{\mathbf{i}}+0.8\hat{\mathbf{j}}$. Find the ship's speed component along the channel.

(a) 4.8 m/s

(b) 9.6 m/s

(c) 10 m/s

(d) 14 m/s

Answer: (b) 9.6 m/s

Explanation

Given: v = (8, 6) m/s; $\hat{\mathbf{n}}$ = (0.6, 0.8), with $|\hat{\mathbf{n}}|^2 = 1$.

Formula: component of v along $\hat{\mathbf{n}}$ = v·$\hat{\mathbf{n}}$ ($\hat{\mathbf{n}}$ has magnitude 1, so no division is needed).

Working: v·$\hat{\mathbf{n}}$ = 8(0.6) + 6(0.8) = 4.8 + 4.8 = 9.6 m/s.

Result: 9.6 m/s. → option (b)

Why the others fail: 10 is |v|, the full speed, not the part along the channel; 14 just adds 8 + 6; 4.8 counts only the x-part.

Concept behind the question. Sec 7 (S5): for a unit vector, the scalar projection (v·$\hat{\mathbf{n}}$)/|$\hat{\mathbf{n}}$| is just v·$\hat{\mathbf{n}}$, so a dot product with a unit vector directly gives a component.

Question 152

VEC-T4-N-037 · Topic 4 · Hard · Calculation

Vectors **A** and **B** have magnitudes 4 and 3, and the angle between them is 60°. Find $(2\mathbf{A}-\mathbf{B})\cdot(\mathbf{A}+3\mathbf{B})$.

(a) 5

(b) 35

(c) 47

(d) 57

Answer: (b) 35

Explanation

Given: A = 4, B = 3, θ = 60°.

Formula: expand with the distributive law and use A·B = B·A.

Working: (2A − B)·(A + 3B) = 2A·A + 6A·B − B·A − 3B·B = $2A^2$ + 5A·B − $3B^2$.

Working: A·B = 4 × 3 × cos 60° = 6.

Working: 2(16) + 5(6) − 3(9) = 32 + 30 − 27 = 35.

Result: 35. → option (b)

Why the others fail: 5 omits 5A·B; 47 uses 7A·B; 57 takes A·B = AB sin 60°.

Concept behind the question. Sec 7 (N5): the distributive and commutative laws let you multiply brackets like ordinary algebra, with A·A = A^2 and B·A = A·B collecting the cross terms.

Question 153

VEC-T4-N-038 · Topic 4 · Hard · Calculation

Let $\mathbf{A} = 5\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$ and $\mathbf{B} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$. Find the number λ for which $\mathbf{A} + \lambda\mathbf{B}$ is perpendicular to $\mathbf{B}$.

(a) −6

(b) −2

(c) −0.5

(d) 2

Answer: (b) −2

Explanation

Given: A = (5, 2, 3), B = (2, 1, 2); (A + λB)·B = 0.

Formula: (A + λB)·B = A·B + λ(B·B) = 0, so λ = −A·B/B^2.

Working: A·B = 10 + 2 + 6 = 18.

Working: B^2 = 4 + 1 + 4 = 9.

Working: λ = −18/9 = −2.

Result: λ = −2. → option (b)

Why the others fail: 2 has the wrong sign; −6 divides by B = 3; −0.5 inverts the ratio.

Concept behind the question. Sec 7 (N5, N7): A + λB is A with its component along B removed when λ = −A·B/B^2, which is why the result ends up perpendicular to B.

Question 154

VEC-T4-N-039 · Topic 4 · Hard · Calculation

A vector $\mathbf{A} = 9\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ m is resolved into a part along $\mathbf{B} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ and a part perpendicular to $\mathbf{B}$. Find the magnitude of the perpendicular part.

(a) 6 m

(b) 2.22 m

(c) 7 m

(d) 36 m

Answer: (a) 6 m

Explanation

Given: A = (9, 2) m, B = (3, 4).

Formula: parallel part = A·B/B; perpendicular part = $\sqrt{A^2 - (parallel part)^2}$.

Working: A·B = 27 + 8 = 35; B = 5; parallel part = 35/5 = 7 m.

Working: A^2 = 81 + 4 = 85.

Working: perpendicular part = $\sqrt{85 - 7^2}$ = $\sqrt{85 - 49}$ = $\sqrt{36}$ = 6 m.

Result: 6 m. → option (a)

Why the others fail: 7 m is the parallel part; 36 m is the square of the answer; 2.22 m subtracts $|A| - 7$.

Concept behind the question. Sec 7 (S5) with Sec 5: A, its part along B and its part perpendicular to B form a right-angled triangle, so Pythagoras finishes the job once the dot product has given the projection.

Question 155

VEC-T4-N-040 · Topic 4 · Hard · Calculation

Two velocity vectors **A** and **B** are each of magnitude 6 m/s, and the magnitude of their sum is also 6 m/s. Find the angle between **A** and **B** .

(a) 60°

(b) 90°

(c) 120°

(d) 180°

Answer: (c) 120°

Explanation

Given: A = B = 6 m/s, |A + B| = 6 m/s.

Formula: $|A + B|^2 = A^2 + B^2 + 2A\cdot B = A^2 + B^2 + 2AB\cos\theta$.

Working: $36 = 36 + 36 + 2(6)(6)\cos\theta$, so $36 = 72 + 72\cos\theta$.

Working: $\cos\theta = (36 - 72)/72 = -0.5$, so $\theta = 120^\circ$.

Result: 120°. $\rightarrow$ option (c)

Why the others fail: 60° has $\cos\theta = +0.5$ (that makes $|A - B| = A$); 90° gives $|A + B| = \sqrt{2}\,A$; 180° gives $|A + B| = 0$.

Concept behind the question. Sec 7 with Sec 6 (N6): for two equal vectors the sum equals each of them only at 120°, which is why three equal forces at 120° to each other balance.

Question 156

VEC-T5-C-001 · Topic 5 · Easy · Concept

The cross product $\mathbf{A} \times \mathbf{B}$ of two non-parallel vectors is

(a) a scalar equal to $AB\cos\theta$

(b) a vector lying in the plane of A and B

(c) a scalar equal to $AB\sin\theta$

(d) a vector perpendicular to the plane of A and B

Answer: (d) a vector perpendicular to the plane of A and B

Explanation

Rule: $A \times B$ is a vector of magnitude $AB\sin\theta$, perpendicular to the plane of A and B (Section 8, N1).

Reasoning: The definition gives a magnitude AND a direction, so the result is a vector, not a scalar.

Reasoning: The direction is fixed by the right-hand rule and is at right angles to both A and B, so it cannot lie in their plane.

Why the others fail: ”A scalar” options describe the dot product ($AB\cos\theta$) or only the size of the cross product; ”in the plane” ignores the perpendicular direction.

Concept behind the question. A cross product has both size and direction, so it belongs to the vector family; its direction points out of the plane that A and B span (Section 8).

Question 157

VEC-T5-C-002 · Topic 5 · Easy · Concept

For two non-zero, non-parallel vectors A and B, compared with $\mathbf{A} \times \mathbf{B}$, the vector $\mathbf{B} \times \mathbf{A}$ has

(a) the same magnitude and the opposite direction

(b) the same magnitude and the same direction

(c) twice the magnitude and the opposite direction

(d) zero magnitude

Answer: (a) the same magnitude and the opposite direction

Explanation

Rule: $\mathrm{B} \times \mathrm{A} = -(\mathrm{A} \times \mathrm{B})$ (anticommutative), Section 8, N3.

Reasoning: Both products have magnitude AB sin θ, because sin θ does not depend on the order.

Reasoning: Reversing the order reverses the curl of the fingers in the right-hand rule, so the thumb points the opposite way.

Why the others fail: ”Same direction” is the dot-product habit of commutativity; ”twice” is not what the minus sign means. The product cannot be zero because the vectors are non-parallel.

Concept behind the question. The cross product is anticommutative: swapping the order flips only the direction, never the size (Section 8).

Question 158

VEC-T5-C-003 · Topic 5 · Easy · Concept

For any vector A, the cross product $\mathbf{A} \times \mathbf{A}$ equals

(a) the null vector 0

(b) a vector of magnitude A along A

(c) A^2

(d) a vector of magnitude A^2, perpendicular to A

Answer: (a) the null vector 0

Explanation

Rule: $\mathrm{A} \times \mathrm{A} = 0$, because the angle between A and itself is 0° and $\sin 0^\circ = 0$ (Section 8, N4).

Reasoning: $|\mathrm{A} \times \mathrm{A}| = \mathrm{A} \cdot \mathrm{A} \cdot \sin 0^\circ = 0$, so the product is the null vector.

Why the others fail: A^2 is the value of the dot product A·A ($\cos 0^\circ = 1$); the other options need $\sin \theta$ to be non-zero.

Concept behind the question. Any two parallel vectors, including a vector with itself, have a zero cross product because $\sin 0^\circ = 0$ (Section 8).

Question 159

VEC-T5-C-004 · Topic 5 · Easy · Concept

With right-handed axes, $\hat{\mathbf{i}} \times \hat{\mathbf{k}}$ equals

(a) $\hat{\mathbf{k}}$

(b) 0

(c) $\hat{\mathbf{j}}$

(d) $-\hat{\mathbf{j}}$

Answer: (d) $-\hat{\mathbf{j}}$

Explanation

Rule: The cyclic order $\mathrm{i} \to \mathrm{j} \to \mathrm{k} \to \mathrm{i}$ gives plus: $\mathrm{i} \times \mathrm{j} = \mathrm{k}$, $\mathrm{j} \times \mathrm{k} = \mathrm{i}$, $\mathrm{k} \times \mathrm{i} = \mathrm{j}$; the reverse order gives minus (Section 8, N5).

Reasoning: $\mathrm{i} \times \mathrm{k}$ is the reverse of $\mathrm{k} \times \mathrm{i} = \mathrm{j}$.

Reasoning: Therefore $\mathrm{i} \times \mathrm{k} = -(\mathrm{k} \times \mathrm{i}) = -\mathrm{j}$.

Why the others fail: +j ignores the reversal; k belongs to $\mathrm{i} \times \mathrm{j}$; zero is the dot product i·k.

Concept behind the question. The unit-vector table is easy to remember as a circle i $\to$ j $\to$ k $\to$ i: going round the circle is positive, going against it is negative.

Question 160

VEC-T5-C-005 · Topic 5 · Easy · Concept

Two non-zero vectors satisfy $\mathbf{A} \times \mathbf{B} = 0$. They must be

(a) perpendicular to each other

(b) at 45° to each other

(c) equal in magnitude

(d) parallel or antiparallel

Answer: (d) parallel or antiparallel

Explanation

Rule: $|A \times B| = AB \sin\theta$, which is zero for non-zero vectors only when $\sin\theta = 0$ (Section 8, N4).

Reasoning: $\sin\theta = 0$ means $\theta = 0°$ or 180°, i.e. the vectors are parallel or antiparallel.

Why the others fail: Perpendicular vectors make the DOT product zero ($\cos 90° = 0$), not the cross product.

Concept behind the question. Zero dot product means perpendicular; zero cross product means parallel or antiparallel. The two tests are opposites (Section 8, comparison table).

Question 161

VEC-T5-C-006 · Topic 5 · Easy · Concept

To find the direction of $\mathbf{A} \times \mathbf{B}$ by the right-hand rule, you should

(a) curl the fingers of the left hand from A towards B; the thumb gives the direction

(b) point the fingers along A and the thumb along B; the direction is along the thumb

(c) curl the fingers of the right hand from A towards B; the thumb gives the direction

(d) curl the fingers of the right hand from B towards A; the thumb gives the direction

Answer: (c) curl the fingers of the right hand from A towards B; the thumb gives the direction

Explanation

Rule: Curl the fingers of the right hand from A towards B through the smaller angle; the thumb points along $A \times B$ (Section 8, N2).

Reasoning: The order matters: from the first vector to the second vector.

Reasoning: The right hand is used because the axes are right-handed ($i \times j = k$).

Why the others fail: Curling from B to A gives $B \times A$ (the opposite direction); the left hand also reverses the thumb; the thumb is the ANSWER, not a vector of the product.

Concept behind the question. The right-hand rule fixes only the sense of the perpendicular line; the curl from the first vector to the second vector decides which way along that line the result points.

Question 162

VEC-T5-C-007 · Topic 5 · Easy · Concept

Take east as +x, north as +y and vertically up as +z (a right-handed set). A ship's displacement A is due east and a second displacement B is due north. The vector $A \times B$ points

(a) vertically upward

(b) vertically downward

(c) due east

(d) due north

Answer: (a) vertically upward

Explanation

Rule: $i \times j = k$ for right-handed axes; $A \times B$ is perpendicular to both A and B (Section 8, N2, N5).

Reasoning: A is along +x (east) and B is along +y (north), so A $\times$ B is along i $\times$ j = +k.

Reasoning: +z is vertically up, so the product points upward.

Why the others fail: Downward is B $\times$ A; east or north lies in the plane of A and B, but the cross product is perpendicular to that plane.

Concept behind the question. East, north, up form a right-handed set, so east $\times$ north = up; this matches the screw rule: turn a right-handed screw from east to north and it advances upward.

Question 163

VEC-T5-C-008 · Topic 5 · Easy · Concept

For two vectors of fixed magnitudes, the magnitude of $\mathbf{A} \times \mathbf{B}$ is greatest when the angle between them is

(a) 0°

(b) 45°

(c) 90°

(d) 180°

Answer: (c) 90°

Explanation

Rule: $|A \times B| = AB \sin\theta$ (Section 8, N1).

Reasoning: For fixed A and B the factor $\sin\theta$ is largest (equal to 1) at $\theta = 90^\circ$.

Why the others fail: At 0° and 180° $\sin\theta = 0$, so the cross product is zero; 0° maximises $\cos\theta$ (the dot product), not $\sin\theta$.

Concept behind the question. Cross product measures how much the vectors are ”across” each other, so perpendicular vectors give the largest value, AB.

Question 164

VEC-T5-C-009 · Topic 5 · Medium · Concept

Which one of the following statements about the dot and cross products is incorrect?

(a) $\mathbf{A} \times \mathbf{B} = \mathbf{B} \times \mathbf{A}$

(b) $\mathbf{A} \times \mathbf{B}$ is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$

(c) $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$

(d) $\mathbf{A} \times \mathbf{A} = \mathbf{0}$

Answer: (a) $\mathbf{A} \times \mathbf{B} = \mathbf{B} \times \mathbf{A}$

Explanation

Rule: B $\times$ A = $-$(A $\times$ B); A $\times$ A = 0; A·B = B·A; A $\times$ B is perpendicular to A and to B (Section 8, N1, N3, N4; Section 7, N5).

Reasoning: The first option claims the cross product is commutative. It is anticommutative, so it is incorrect.

Reasoning: The other three options are all standard true results.

Why the others fail: They are correct statements, so they are not the one asked for.

Concept behind the question. The dot product is commutative but the cross product is not: this single difference is a favourite entrance-paper trap.

Question 165

VEC-T5-C-010 · Topic 5 · Medium · Concept

Two non-parallel vectors A and B form a parallelogram and a triangle as adjacent sides from a common point. The ratio (area of triangle) : (area of parallelogram) is

(a) 1 : 4

(b) 1 : 1

(c) 2 : 1

(d) 1 : 2

Answer: (d) 1 : 2

Explanation

Rule: Area of parallelogram = $|A \times B|$; area of the triangle = $\frac{1}{2}|A \times B|$ (Section 8, N7).

Reasoning: The triangle is half of the parallelogram, because the diagonal splits the parallelogram into two equal triangles.

Reasoning: Ratio = $\frac{1}{2}|A \times B| : |A \times B| = 1 : 2$.

Why the others fail: 1 : 1 forgets the half; 2 : 1 is the reverse; 1 : 4 applies the half twice.

Concept behind the question. The diagonal of a parallelogram cuts it into two congruent triangles, which is why the triangle area is $\frac{1}{2}|A \times B|$.

Question 166

VEC-T5-C-011 · Topic 5 · Medium · Concept

A and B are two non-zero, non-parallel vectors. The number of unit vectors perpendicular to both A and B is

(a) 2

(b) infinitely many

(c) 1

(d) 4

Answer: (a) 2

Explanation

Rule: A unit vector perpendicular to both is $\hat{\mathbf{n}} = \pm(A \times B)/|A \times B|$ (Section 8, N8).

Reasoning: The sign $\pm$ gives exactly two unit vectors, one on each side of the plane of A and B.

Reasoning: $B \times A = -(A \times B)$ gives the same two vectors, not two new ones.

Why the others fail: 1 forgets the minus sign; 4 double-counts; infinitely many is true for the vectors perpendicular to ONE vector, not to two non-parallel vectors.

Concept behind the question. Two non-parallel vectors fix a plane, and only one line is perpendicular to that plane, so there are just two directions along it.

Question 167

VEC-T5-C-012 · Topic 5 · Medium · Concept

For non-zero, non-parallel vectors A and B, the value of $(\mathbf{A} \times \mathbf{B}) \cdot A$ is

(a) AB sin θ

(b) A^2B sin θ

(c) 0

(d) A^2B cos θ

Answer: (c) 0

Explanation

Rule: $A \times B$ is perpendicular to both A and B, so its dot product with either is zero (Section 8, S4).

Reasoning: The angle between $A \times B$ and A is 90°, so $(A \times B) \cdot A = |A \times B| A \cos 90^\circ = 0$.

Why the others fail: The three non-zero options assume the cross product has a component along A, which it does not.

Concept behind the question. This is the quick check for any cross-product calculation: dot the answer with A or B and you must get zero.

Question 168

VEC-T5-C-013 · Topic 5 · Medium · Concept

For any two vectors A and B (with magnitudes A and B and angle θ), which relation always holds?

(a) $|\mathbf{A}\times\mathbf{B}|^2+(\mathbf{A}\cdot\mathbf{B})^2=AB$

(b) $|\mathbf{A}\times\mathbf{B}|^2-(\mathbf{A}\cdot\mathbf{B})^2=A^2B^2$

(c) $|\mathbf{A}\times\mathbf{B}|+\mathbf{A}\cdot\mathbf{B}=AB$

(d) $|\mathbf{A}\times\mathbf{B}|^2+(\mathbf{A}\cdot\mathbf{B})^2=A^2B^2$

Answer: (d) $|\mathbf{A}\times\mathbf{B}|^2+(\mathbf{A}\cdot\mathbf{B})^2=A^2B^2$

Explanation

Rule: $|A\times B|^2 + (A\cdot B)^2 = A^2B^2$ (Section 8, Tip).

Reasoning: $|A\times B|^2 = A^2B^2\,sin^2\theta$ and $(A\cdot B)^2 = A^2B^2\,cos^2\theta$.

Reasoning: Adding gives $A^2B^2(sin^2\theta + cos^2\theta) = A^2B^2$.

Why the others fail: $\sin\theta + \cos\theta \neq 1$ and $sin^2\theta - cos^2\theta \neq 1$ in general; the right side must have dimension $(AB)^2$, not AB.

Concept behind the question. This identity is just $sin^2\theta + cos^2\theta = 1$ written for the two products; it lets you find one product when the other is known.

Question 169

VEC-T5-C-014 · Topic 5 · Medium · Concept

Can two non-zero vectors have both $\mathbf{A}\cdot\mathbf{B}=0$ and $\mathbf{A}\times\mathbf{B}=0$ at the same time?

(a) No, because sin θ and cos θ are never zero together

(b) Yes, when $\theta = 0^\circ$

(c) Yes, when $\theta = 180^\circ$

(d) Yes, when $\theta = 90^\circ$

Answer: (a) No, because sin θ and cos θ are never zero together

Explanation

Rule: $A\cdot B = AB\cos\theta$ and $|A\times B| = AB\sin\theta$ (Sections 7 and 8).

Reasoning: For non-zero A and B, both are zero only if $\cos\theta = 0$ and $\sin\theta = 0$ together.

Reasoning: No angle does that, since $sin^2\theta + cos^2\theta = 1$.

Why the others fail: Each angle listed makes just one of the products vanish (90°: dot; 0° and 180°: cross).

Concept behind the question. If both products vanish, at least one of the vectors must be the zero vector; for non-zero vectors the dot and cross products can never both be zero.

Question 170

VEC-T5-C-015 · Topic 5 · Medium · Concept

The vector $\mathbf{A}\times(-2\mathbf{B})$ is equal to

(a) $2(\mathbf{A}\times\mathbf{B})$

(b) $-\frac{1}{2}(\mathbf{A}\times\mathbf{B})$

(c) $-(\mathbf{A}\times\mathbf{B})$

(d) $-2(\mathbf{A}\times\mathbf{B})$

Answer: (d) $-2(\mathbf{A} \times \mathbf{B})$

Explanation

Rule: A scalar factor can be taken outside a cross product: A $\times$ (λB) = λ(A $\times$ B) (Section 8, distributive/linear property).

Reasoning: With $\lambda = -2$: A $\times$ ($-$2B) = $-$2(A $\times$ B).

Why the others fail: +2 ignores the minus sign; -1 ignores the 2; $-\frac{1}{2}$ divides instead of multiplying.

Concept behind the question. The cross product is linear in each vector: a factor on one vector simply multiplies the result, with its sign.

Question 171

VEC-T5-C-016 · Topic 5 · Medium · Concept

With right-handed axes, A lies along +z and B lies along +x. The direction of A $\times$ B is

(a) $-$y

(b) +y

(c) +z

(d) +x

Answer: (b) +y

Explanation

Rule: k $\times$ i = j (cyclic order), Section 8, N5.

Reasoning: A is along k and B along i, so A $\times$ B is along k $\times$ i = +j, i.e. the +y direction.

Why the others fail: $-$y is i $\times$ k (B $\times$ A); +x and +z lie along the original vectors, while the cross product is perpendicular to both.

Concept behind the question. The cyclic order i $\to$ j $\to$ k $\to$ i tells you the answer at once: the third axis follows when you go forward round the circle.

Question 172

VEC-T5-C-017 · Topic 5 · Hard · Concept

For any vectors A and B, the vector $\mathbf{A} \times (\mathbf{A} + \mathbf{B})$ equals

(a) the null vector 0

(b) $2(\mathbf{A} \times \mathbf{B})$

(c) $\mathbf{A} \times \mathbf{B}$

(d) $\mathbf{B} \times \mathbf{A}$

Answer: (c) $\mathbf{A} \times \mathbf{B}$

Explanation

Rule: The cross product is distributive: A $\times$ (A + B) = A $\times$ A + A $\times$ B (Section 8, N3).

Reasoning: A $\times$ A = 0 (Section 8, N4).

Reasoning: Hence A $\times$ (A + B) = 0 + A $\times$ B = A $\times$ B.

Why the others fail: B $\times$ A has the opposite sign; 2(A $\times$ B) wrongly uses A $\times$ A = A $\times$ B; the null vector forgets that A $\times$ B remains.

Concept behind the question. Adding a multiple of A to B does not change A $\times$ B, since the extra part is parallel to A: the same parallelogram, sheared along A, keeps the same area.

Question 173

VEC-T5-C-018 · Topic 5 · Hard · Concept

A is a non-zero vector and $\mathbf{A} \times \mathbf{B} = \mathbf{A} \times \mathbf{C}$. Which conclusion is always correct?

(a) $\mathbf{B} = \mathbf{C}$

(b) $\mathbf{B} - \mathbf{C}$ is collinear with $\mathbf{A}$ (or is zero)

(c) $\mathbf{B} + \mathbf{C}$ is parallel to $\mathbf{A}$

(d) $\mathbf{B} - \mathbf{C}$ is perpendicular to $\mathbf{A}$

Answer: (b) $\mathbf{B} - \mathbf{C}$ is collinear with $\mathbf{A}$ (or is zero)

Explanation

Rule: A × B − A × C = A × (B − C) (distributive property, Section 8, N3).

Reasoning: The given equality means A × (B − C) = 0.

Reasoning: With A non-zero, this holds only if B − C is zero or collinear with A (either parallel or antiparallel) (Section 8, N4).

Why the others fail: B = C is only one possibility (B and C can differ by any multiple of A); perpendicular is the dot-product condition; B + C is not what appears.

Concept behind the question. Division by a vector is not defined, so A × B = A × C cannot be cancelled; it only tells us B and C differ by a vector along A.

Question 174

VEC-T5-N-019 · Topic 5 · Easy · Calculation

Two displacement vectors of magnitudes 6 m and 8 m make an angle of 30° with each other. Find the magnitude of their vector product.

(a) 12 m^2

(b) 24 m^2

(c) 41.6 m^2

(d) 48 m^2

Answer: (b) 24 m^2

Explanation

Given: A = 6 m, B = 8 m, θ = 30°.

To find: |A × B|.

Formula: |A × B| = AB sin θ.

Working: sin 30° = 0.5.

Working: |A × B| = 6 × 8 × 0.5 = 24 m^2.

Result: 24 m^2. → option (b)

Why the others fail: 41.6 uses cos 30°; 48 leaves out sin θ; 12 is half the correct value.

Concept behind the question. The magnitude of the cross product uses the sine of the angle between the vectors, tails together (Section 8, N1).

Question 175

VEC-T5-N-020 · Topic 5 · Easy · Calculation

From a lighthouse, a buoy is 3 km due east (vector A) and another buoy is 4 km due north (vector B). Find |A × B|.

(a) 0 km^2

(b) 5 km^2

(c) 7 km^2

(d) 12 km^2

Answer: (d) 12 km^2

Explanation

Given: A = 3 km east, B = 4 km north, so $\theta = 90^\circ$.

To find: |A × B|.

Formula: |A × B| = AB sin θ.

Working: sin 90° = 1.

Working: |A × B| = 3 × 4 × 1 = 12 km^2.

Result: 12 km^2. → option (d)

Why the others fail: 0 is A·B (cos 90° = 0); 7 adds A + B; 5 is |A + B|.

Concept behind the question. For perpendicular vectors the cross product has its largest value AB, the area of the rectangle they form.

Question 176

VEC-T5-N-021 · Topic 5 · Easy · Calculation

A field is a parallelogram with adjacent sides 8 m and 5 m, and the angle between those sides is 150°. Find its area.

(a) 10 m^2

(b) 20 m^2

(c) 34.6 m^2

(d) 40 m^2

Answer: (b) 20 m^2

Explanation

Given: sides 8 m and 5 m, angle 150° between them.

To find: area of the parallelogram.

Formula: Area = |A × B| = AB sin θ.

Working: sin 150° = sin $(180^\circ - 150^\circ)$ = sin 30° = 0.5.

Working: Area = 8 × 5 × 0.5 = 20 m^2.

Result: 20 m^2. → option (b)

Why the others fail: 40 omits sin θ; 34.6 uses |cos 150°| instead of sin 150°; 10 is the area of half the parallelogram.

Concept behind the question. sin θ = sin $(180^\circ - \theta)$, so a parallelogram has the same area whether its angle is taken as 30° or 150° (Section 8, N7).

Question 177

VEC-T5-N-022 · Topic 5 · Easy · Calculation

Find $\mathbf{A} \times \mathbf{B}$ for $\mathbf{A} = 3\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ and $\mathbf{B} = \hat{\mathbf{i}} + 5\hat{\mathbf{j}}$.

(a) $13\hat{\mathbf{k}}$

(b) $3\hat{\mathbf{i}} + 10\hat{\mathbf{j}}$

(c) $-13\hat{\mathbf{k}}$

(d) $17\hat{\mathbf{k}}$

Answer: (a) $13\hat{\mathbf{k}}$

Explanation

Given: A = (3, 2, 0), B = (1, 5, 0).

To find: A × B.

Formula: A × B = $(A_yB_z - A_zB_y)$ i + $(A_zB_x - A_xB_z)$ j + $(A_xB_y - A_yB_x)$ k.

Working: i-component = $A_yB_z - A_zB_y$ = 2×0 − 0×5 = 0

Working: j-component = $A_zB_x - A_xB_z$ = 0×1 − 3×0 = 0

Working: k-component = $A_xB_y - A_yB_x$ = 3×5 − 2×1 = 13

Result: A × B = 13 k, i.e. $13\hat{\mathbf{k}}$. → option (a)

Why the others fail: −13 k is B × A; 17 k adds 3×5 and 2×1; 3i + 10j multiplies matching components.

Concept behind the question. For vectors in the xy-plane only the k component survives: $A_xB_y - A_yB_x$, a vector perpendicular to the plane (Section 8, N6).

Question 178

VEC-T5-N-023 · Topic 5 · Easy · Calculation

If $\mathbf{A} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 3\hat{\mathbf{k}}$ and $\mathbf{B} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$, find $\mathbf{A} \times \mathbf{B}$.

(a) $-5\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$

(b) $5\hat{\mathbf{i}} - 5\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$

(c) $-5\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$

(d) $-5\hat{\mathbf{i}} - 5\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$

Answer: (c) $-5\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$

Explanation

Given: A = (2, −1, 3), B = (1, 2, −1).

To find: A × B.

Formula: A × B = $(A_yB_z - A_zB_y)$ i + $(A_zB_x - A_xB_z)$ j + $(A_xB_y - A_yB_x)$ k.

Working: i-component = $A_yB_z - A_zB_y$ = (−1)×(−1) − 3×2 = −5

Working: j-component = $A_zB_x - A_xB_z$ = 3×1 − 2×(−1) = 5

Working: k-component = $A_xB_y - A_yB_x$ = 2×2 − (−1)×1 = 5

Result: A × B = $-5\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$. → option (c)

Why the others fail: $5\hat{\mathbf{i}} - 5\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$ is B × A; $-5\hat{\mathbf{i}} - 5\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ reverses the j term; $-5\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$ gets the k component by adding (2×2 + (−1)×1).

Concept behind the question. Check by dotting with A: (−5)(2) + 5(−1) + 5(3) = 0, as it must be for a vector perpendicular to A (Section 8, S4).

Question 179

VEC-T5-N-024 · Topic 5 · Easy · Calculation

Simplify $(4\hat{\mathbf{j}}) \times (5\hat{\mathbf{k}}) + (3\hat{\mathbf{k}}) \times (2\hat{\mathbf{j}})$.

(a) 14 $\hat{\mathbf{i}}$

(b) 26 $\hat{\mathbf{i}}$

(c) −14 $\hat{\mathbf{i}}$

(d) 14 $\hat{\mathbf{k}}$

Answer: (a) 14 $\hat{\mathbf{i}}$

Explanation

Given: (4j) × (5k) + (3k) × (2j).

Formula: j × k = i, and k × j = −i; numbers come out of the product.

Working: (4j) × (5k) = 4 × 5 (j × k) = 20 i.

Working: (3k) × (2j) = 3 × 2 (k × j) = 6 × (−i) = −6 i.

Working: Sum = 20 i − 6 i = 14 i.

Result: 14 i. → option (a)

Why the others fail: 26 i treats k × j as +i; −14 i reverses everything; k is the wrong unit vector.

Concept behind the question. The numerical factors multiply as ordinary numbers, but the order of the unit vectors decides the sign (Section 8, N5).

Question 180

VEC-T5-N-025 · Topic 5 · Easy · Calculation

For two vectors, $\mathbf{A} \times \mathbf{B} = 4\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$. Find $\mathbf{B} \times \mathbf{A}$.

(a) $-4\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$

(b) $-4\hat{\mathbf{i}} - 3\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$

(c) $-4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$

(d) $4\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$

Answer: (c) $-4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$

Explanation

Given: A $\times$ B = $4\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$.

To find: B $\times$ A.

Formula: B $\times$ A = $-$(A $\times$ B).

Working: Reverse the sign of every component: $4 \to -4$, $-3 \to +3$, $2 \to -2$.

Result: B $\times$ A = $-4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$. $\to$ option (c)

Why the others fail: Each wrong option leaves some component with its original sign; the minus sign applies to all three components.

Concept behind the question. Anticommutativity reverses the whole vector, not selected components (Section 8, N3).

Question 181

VEC-T5-N-026 · Topic 5 · Easy · Calculation

A parallelogram is formed by the vectors $5\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ m and $3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ m. Find its area.

(a) -14 m^2

(b) 14 m^2

(c) 23 m^2

(d) 26 m^2

Answer: (b) 14 m^2

Explanation

Given: A = (5, 2) m, B = (3, 4) m.

To find: area of the parallelogram.

Formula: Area = |A $\times$ B|, and for plane vectors A $\times$ B = ($A_xB_y - A_yB_x$) k.

Working: $A_xB_y - A_yB_x$ = 5$\times$4 $-$ 2$\times$3 = 20 $-$ 6 = 14.

Working: |A $\times$ B| = 14 m^2.

Result: 14 m^2. $\to$ option (b)

Why the others fail: 26 = 5$\times$4 + 2$\times$3; 23 = 5$\times$3 + 2$\times$4 is the dot product; -14 is B $\times$ A, and area is the magnitude.

Concept behind the question. The area of the parallelogram is the size of the cross product, so it is always positive (Section 8, N7).

Question 182

VEC-T5-N-027 · Topic 5 · Easy · Calculation

Two sides of a triangle, drawn from the same corner, are $6\hat{\mathbf{i}}$ m and $2\hat{\mathbf{i}} + 5\hat{\mathbf{j}}$ m. Find the area of the triangle.

(a) 6 m^2

(b) 15 m^2

(c) 16.2 m^2

(d) 30 m^2

Answer: (b) 15 m^2

Explanation

Given: A = (6, 0) m, B = (2, 5) m (sides from one corner).

To find: area of the triangle.

Formula: Area = $\frac{1}{2}$|A × B|, with A × B = ($A_xB_y - A_yB_x$) k.

Working: $A_xB_y - A_yB_x$ = 6×5 − 0×2 = 30.

Working: Area = $\frac{1}{2}$ × 30 = 15 m^2.

Result: 15 m^2. → option (b)

Why the others fail: 30 is the parallelogram area; 6 is $\frac{1}{2}$(A·B); 16.2 = $\frac{1}{2}$ × 6 × $\sqrt{29}$ ignores the angle.

Concept behind the question. A triangle is half of the parallelogram on the same two sides, so its area is $\frac{1}{2}$|A × B| (Section 8, N7).

Question 183

VEC-T5-N-028 · Topic 5 · Easy · Calculation

The turning effect (torque) of a force about a point is defined as τ = r × F, where r runs from the point to where the force acts. A spanner is 0.5 m long along +x and a force of 6 N along +y acts at its end. Find τ about the bolt (in N m).

(a) $12\hat{\mathbf{k}}$

(b) 0

(c) $-3\hat{\mathbf{k}}$

(d) $3\hat{\mathbf{k}}$

Answer: (d) $3\hat{\mathbf{k}}$

Explanation

Given: r = 0.5 i m, F = 6 j N, τ = r × F (defined).

To find: τ.

Formula: i × j = k, so (r_x i) × (F_y j) = r_xF_y k.

Working: τ = (0.5)(6) (i × j) = 3 k.

Result: τ = 3 k N m (along +z, perpendicular to the spanner and the force). → option (d)

Why the others fail: −3 k is F × r; 0 is the dot product r·F; 12 k divides instead of multiplying.

Concept behind the question. A push at right angles to the handle gives the largest turning effect, and the cross product points along the bolt axis (Section 8, familiar example).

Question 184

VEC-T5-N-029 · Topic 5 · Medium · Calculation

For $\mathbf{A} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$ and $\mathbf{B} = \hat{\mathbf{j}} + \hat{\mathbf{k}}$, use $\mathbf{A} \cdot \mathbf{B}$ and $|\mathbf{A} \times \mathbf{B}|$ to find the angle between them.

(a) 30°

(b) 45°

(c) 60°

(d) 90°

Answer: (c) 60°

Explanation

Given: A = (1, 1, 0), B = (0, 1, 1).

To find: θ.

Formula: A·B = AB cos θ and |A × B| = AB sin θ, so tan θ = |A × B|/(A·B).

Working: A·B = 1×0 + 1×1 + 0×1 = 1.

Working: i-component = $A_yB_z - A_zB_y$ = 1×1 − 0×1 = 1

Working: j-component = $A_zB_x - A_xB_z$ = 0×0 − 1×1 = −1

Working: k-component = $A_xB_y - A_yB_x$ = 1×1 − 1×0 = 1

Working: A × B = $\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$, so |A × B| = $\sqrt{1+1+1} = \sqrt{3}$.

Working: tan $\theta = \sqrt{3}/1 = \sqrt{3}$, so $\theta = 60^\circ$.

Result: 60°. → option (c)

Why the others fail: tan $\theta = 1/\sqrt{3}$ would give 30°; tan $\theta = 1$ would give 45°; 90° would need A·B = 0.

Concept behind the question. Dividing the two products removes the product AB, so tan θ = |A × B|/(A·B) determines the angle when the quadrant is fixed using the sign of the dot product (Section 8, Tip).

Question 185

VEC-T5-N-030 · Topic 5 · Medium · Calculation

Find the unit vector in the direction of $\mathbf{A} \times \mathbf{B}$ for $\mathbf{A} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ and $\mathbf{B} = 2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$.

(a) $2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$

(b) $\frac{1}{3}(2\hat{\mathbf{i}} - \hat{\mathbf{j}} - 2\hat{\mathbf{k}})$

(c) $\frac{1}{3}(-2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}})$

(d) $\frac{1}{3}(2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}})$

Answer: (d) $\frac{1}{3}(2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}})$

Explanation

Given: A = (1, 2, 2), B = (2, 2, 3).

To find: $\hat{\mathbf{n}}$ = (A × B)/|A × B|.

Formula: A × B = ($A_yB_z - A_zB_y$) i + ($A_zB_x - A_xB_z$) j + ($A_xB_y - A_yB_x$) k; $\hat{\mathbf{n}}$ = (A × B)/|A × B|.

Working: i-component = $A_yB_z - A_zB_y$ = 2×3 − 2×2 = 2

Working: j-component = $A_zB_x - A_xB_z$ = 2×2 − 1×3 = 1

Working: k-component = $A_xB_y - A_yB_x$ = 1×2 − 2×2 = −2

Working: |A × B| = $\sqrt{4+1+4}$ = 3.

Working: $\hat{\mathbf{n}}$ = (2 i + j − 2 k)/3.

Result: $\hat{\mathbf{n}} = \frac{1}{3}(2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}})$. → option (d)

Why the others fail: the second option has the wrong j sign; the third points the opposite way (B × A); option (a) has magnitude 3, not 1.

Concept behind the question. A unit vector must have magnitude 1, so the cross product is divided by its own size (Section 8, N8).

Question 186

VEC-T5-N-031 · Topic 5 · Medium · Calculation

Find the magnitude of $\mathbf{A} \times \mathbf{B}$ for $\mathbf{A} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ and $\mathbf{B} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$.

(a) 0

(b) 3

(c) 9

(d) 81

Answer: (c) 9

Explanation

Given: A = (2, 1, 2), B = (1, 2, −2).

To find: |A × B|.

Formula: A × B from the determinant, then |A × B| = $\sqrt{x^2+y^2+z^2}$.

Working: i-component = $A_yB_z - A_zB_y$ = 1×(−2) − 2×2 = −6

Working: j-component = $A_zB_x - A_xB_z$ = 2×1 − 2×(−2) = 6

Working: k-component = $A_xB_y - A_yB_x$ = 2×2 − 1×1 = 3

Working: |A × B| = $\sqrt{36+36+9}$ = $\sqrt{81}$ = 9.

Result: 9. → option (c)

Why the others fail: 0 is A·B; 3 = −6 + 6 + 3; 81 is the sum of squares without the root.

Concept behind the question. Here A·B = 0, so the vectors are perpendicular and |A × B| = AB = 3 × 3 = 9, which agrees with the component method.

Question 187

VEC-T5-N-032 · Topic 5 · Medium · Calculation

For what value of λ are the vectors $2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$ and $4\hat{\mathbf{i}} + \lambda\hat{\mathbf{j}}$ parallel?

(a) 6

(b) −6

(c) 8/3

(d) −8/3

Answer: (a) 6

Explanation

Given: A = 2 i + 3 j, B = 4 i + λ j.

To find: λ so that A ∥ B, i.e. A × B = 0.

Formula: A × B = ($A_xB_y - A_yB_x$) k = 0.

Working: $A_xB_y - A_yB_x$ = 2λ − 3×4 = 2λ − 12.

Working: 2λ − 12 = 0, so λ = 6.

Result: λ = 6 (check: B = 2A). → option (a)

Why the others fail: −6 comes from 2λ + 12 = 0; 8/3 inverts the ratio 2/3 = 4/λ; −8/3 solves 8 + 3λ = 0, which makes the vectors perpendicular.

Concept behind the question. Parallel vectors have a zero cross product (Section 8, N4), whereas a zero dot product would make them perpendicular.

Question 188

VEC-T5-N-033 · Topic 5 · Medium · Calculation

Evaluate $(\hat{\mathbf{i}} \times \hat{\mathbf{j}}) \times \hat{\mathbf{j}}$.

(a) $\hat{\mathbf{k}}$

(b) 0

(c) $\hat{\mathbf{i}}$

(d) $-\hat{\mathbf{i}}$

Answer: (d) $-\hat{\mathbf{i}}$

Explanation

Given: (i × j) × j.

Formula: i × j = k; j × k = i, so k × j = −i.

Working: i × j = k.

Working: k × j = −(j × k) = −i.

Result: −i. → option (d)

Why the others fail: +i forgets the reversal; 0 would need parallel vectors, but k and j are perpendicular; k is only the first step.

Concept behind the question. Work from the bracket outwards, and use the cyclic table with the minus sign for the reversed order (Section 8, N5).

Question 189

VEC-T5-N-034 · Topic 5 · Medium · Calculation

A triangular plate has its corners at (1, 0, 0), (0, 2, 0) and (0, 0, 3), with coordinates in metres. Find its area.

(a) 3 m^2

(b) 3.5 m^2

(c) 7 m^2

(d) 24.5 m^2

Answer: (b) 3.5 m^2

Explanation

Given: P(1, 0, 0), Q(0, 2, 0), R(0, 0, 3) in metres.

To find: area of triangle PQR.

Formula: Area = $\frac{1}{2}|\text{PQ} \times \text{PR}|$, with PQ = Q − P and PR = R − P.

Working: PQ = (−1, 2, 0), PR = (−1, 0, 3).

Working: i: 2×3 − 0×0 = 6; j: 0×(−1) − (−1)×3 = 3; k: (−1)×0 − 2×(−1) = 2, so PQ × PR = (6, 3, 2).

Working: $|\text{PQ} \times \text{PR}| = \sqrt{36 + 9 + 4} = 7$.

Working: Area = $\frac{1}{2} \times 7 = 3.5\ m^2$.

Result: 3.5 m^2. → option (b)

Why the others fail: 7 is the parallelogram area; 24.5 = 49/2 skips the root; 3 is $\frac{1}{2}|\text{Q} \times \text{R}|$, which uses position vectors from the origin rather than sides from one corner.

Concept behind the question. The sides must be vectors drawn from the same corner; subtracting position vectors gives them (Section 8, N7).

Question 190

VEC-T5-N-035 · Topic 5 · Medium · Calculation

Let $\mathbf{C} = \mathbf{A} \times \mathbf{B}$ with $\mathbf{A} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\mathbf{B} = \hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$. Find $\mathbf{C} \cdot \mathbf{B}$.

(a) −3

(b) 0

(c) 8.66

(d) 11

Answer: (b) 0

Explanation

Given: A = (2, −1, 1), B = (1, 3, −2), C = A × B.

To find: C·B.

Formula: C = A × B from the determinant; C·B = C_xB_x + C_yB_y + C_zB_z.

Working: i-component = $A_yB_z - A_zB_y$ = (−1)×(−2) − 1×3 = −1

Working: j-component = $A_zB_x - A_xB_z$ = 1×1 − 2×(−2) = 5

Working: k-component = $A_xB_y - A_yB_x$ = 2×3 − (−1)×1 = 7

Working: C = $-\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + 7\hat{\mathbf{k}}$.

Working: C·B = (−1)(1) + (5)(3) + (7)(−2) = −1 + 15 − 14 = 0.

Result: 0. → option (b)

Why the others fail: −3 is A·B; 8.66 is |C|; 11 is −1 + 5 + 7.

Concept behind the question. A × B is perpendicular to B, so C·B = 0 whatever the numbers are; this is the standard check on a cross product (Section 8, S4).

Question 191

VEC-T5-N-036 · Topic 5 · Medium · Calculation

Two vectors of magnitudes 5 and 4 have a scalar product equal to 12. Find the magnitude of their vector product.

(a) 8

(b) 12

(c) 16

(d) 19.7

Answer: (c) 16

Explanation

Given: A = 5, B = 4, A·B = 12.

To find: |A × B|.

Formula: $|A \times B|^2 + (A \cdot B)^2 = A^2B^2$.

Working: $A^2B^2 = (5\times4)^2 = 400$.

Working: $|A \times B|^2 = 400 - 12^2 = 400 - 144 = 256$.

Working: |A × B| = 16.

Result: 16. → option (c)

Why the others fail: 8 = 20 − 12; 19.7 = $\sqrt{400-12}$; 12 is the dot product itself (cos θ = 0.6 instead of sin θ = 0.8).

Concept behind the question. Equivalent check: cos θ = 12/20 = 0.6, so sin θ = 0.8 and |A × B| = 20 × 0.8 = 16 (Section 8, Tip).

Question 192

VEC-T5-N-037 · Topic 5 · Medium · Calculation

A parallelogram has adjacent sides 9 m and 8 m and an area of 36 m^2. Find the acute angle between the sides.

(a) 30°

(b) 60°

(c) 90°

(d) 150°

Answer: (a) 30°

Explanation

Given: sides 9 m and 8 m, area 36 m^2.

To find: acute angle θ.

Formula: Area = AB sin θ.

Working: sin θ = 36/(9×8) = 36/72 = 0.5.

Working: sin θ = 0.5 gives θ = 30° (acute) or 150° (obtuse).

Result: 30°. → option (a)

Why the others fail: 60° has sin 60° = 0.866; 90° would need $\frac{1}{2}$AB = 36 (ignores that the area is the full parallelogram); 150° is the obtuse partner of 30°.

Concept behind the question. Since sin θ = sin (180° − θ), the area alone cannot tell the acute angle from the obtuse one, which is why the question asks for the acute angle (Section 8, N1).

Question 193

VEC-T5-N-038 · Topic 5 · Medium · Calculation

For $\mathbf{A} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 4\hat{\mathbf{k}}$ and $\mathbf{B} = \hat{\mathbf{i}} + 3\hat{\mathbf{j}} + \lambda\hat{\mathbf{k}}$, for what value of λ does $\mathbf{A} \times \mathbf{B}$ have no $\hat{\mathbf{j}}$ component?

(a) -2

(b) -1.25

(c) 2

(d) 12

Answer: (c) 2

Explanation

Given: A = (2, 1, 4), B = (1, 3, λ).

To find: λ such that the j component of A $\times$ B is zero.

Formula: j-component = $A_z B_x - A_x B_z$.

Working: j-component = $4\times1 - 2\times\lambda = 4 - 2\lambda$.

Working: $4 - 2\lambda = 0$, so $\lambda = 2$.

Result: λ = 2 (check: A $\times$ B = $(1\times2 - 4\times3)$ i + 0 j + $(6 - 1)$ k = -10 i + 5 k). $\rightarrow$ option (c)

Why the others fail: 12 solves $\lambda - 12 = 0$ (the i component); -2 comes from $4 + 2\lambda = 0$; -1.25 solves A·B = $5 + 4\lambda = 0$.

Concept behind the question. Each component of the cross product is a 2×2 determinant; the middle one is $A_z B_x - A_x B_z$, the term the chapter warns about (Section 8, S3).

Question 194

VEC-T5-N-039 · Topic 5 · Medium · Calculation

Let $\mathbf{A} = 4\hat{\mathbf{i}} + \hat{\mathbf{j}}$ and $\mathbf{B} = 2\hat{\mathbf{i}} + b\hat{\mathbf{j}}$. If $\mathbf{A} \times \mathbf{B} = 18\hat{\mathbf{k}}$, find b.

(a) 4

(b) 4.5

(c) 5

(d) 10

Answer: (c) 5

Explanation

Given: A = 4 i + j, B = 2 i + b j, A $\times$ B = 18 k.

To find: b.

Formula: A $\times$ B = $(A_x B_y - A_y B_x)$ k for plane vectors.

Working: $A_x B_y - A_y B_x$ = $4b - 1\times2 = 4b - 2$.

Working: $4b - 2 = 18$, so $4b = 20$ and $b = 5$.

Result: b = 5. $\rightarrow$ option (c)

Why the others fail: 4 solves $4b + 2 = 18$; 4.5 solves $4b = 18$; 10 solves $2b = 20$.

Concept behind the question. Only the k component is non-zero for vectors in the xy-plane, so one equation fixes the unknown (Section 8, N6).

Question 195

VEC-T5-N-040 · Topic 5 · Medium · Calculation

Find $\sin\theta$, where θ is the angle between $3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ and $4\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$.

(a) -0.28

(b) 0.28

(c) 0.96

(d) 1.4

Answer: (b) 0.28

Explanation

Given: A = (3, 4), B = (4, 3).

To find: sin θ.

Formula: sin θ = |A × B|/(AB).

Working: A × B = (3×3 − 4×4) k = −7 k, so |A × B| = 7.

Working: A = $\sqrt{9+16}$ = 5 and B = $\sqrt{16+9}$ = 5.

Working: sin θ = 7/(5×5) = 7/25 = 0.28.

Result: 0.28. → option (b)

Why the others fail: 0.96 = 24/25 is cos θ; −0.28 keeps the sign of the k component, but |A × B| is a magnitude; 1.4 = 7/5 divides by only one magnitude (a sine cannot exceed 1).

Concept behind the question. The angle between two vectors lies between 0° and 180°, where sin θ is never negative, so only the magnitude |A × B| is used (Section 8, N1).

Question 196

VEC-T5-N-041 · Topic 5 · Medium · Calculation

For two vectors, A·B = −5 and |A × B| = $5\sqrt{3}$. Find the angle between them.

(a) 30°

(b) 60°

(c) 120°

(d) 150°

Answer: (c) 120°

Explanation

Given: A·B = −5, |A × B| = $5\sqrt{3}$.

To find: θ.

Formula: tan θ = |A × B|/(A·B), with the sign of A·B telling whether θ is acute or obtuse.

Working: A·B is negative, so cos θ < 0 and θ is obtuse (90° to 180°).

Working: |tan θ| = $5\sqrt{3}/5 = \sqrt{3}$, so the reference angle is 60°.

Working: θ = 180° − 60° = 120°.

Result: 120°. → option (c)

Why the others fail: 60° is the acute reference angle, which needs a positive dot product; 150° and 30° come from tan $\theta = 1/\sqrt{3}$.

Concept behind the question. sin θ alone cannot tell an acute angle from its obtuse partner; the sign of the dot product settles it (Sections 7 and 8).

Question 197

VEC-T5-N-042 · Topic 5 · Hard · Calculation

Given $\mathbf{A} \times \mathbf{B} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$, find $(\mathbf{A}+\mathbf{B}) \times (\mathbf{A}-\mathbf{B})$.

(a) $-4\hat{\mathbf{i}} + 6\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$

(b) 0

(c) $4\hat{\mathbf{i}} - 6\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$

(d) $-2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}}$

Answer: (a) $-4\hat{\mathbf{i}} + 6\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$

Explanation

Given: A × B = $2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$.

To find: (A + B) × (A − B).

Formula: The cross product is distributive; A × A = 0 and B × B = 0; B × A = −(A × B).

Working: (A + B) × (A − B) = A × A − A × B + B × A − B × B.

Working: = 0 − A × B − A × B − 0 = −2(A × B), since B × A = −(A × B).

Working: −2 × (2, −3, 1) = (−4, 6, −2).

Result: $-4\hat{\mathbf{i}} + 6\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$. → option (a)

Why the others fail: +2(A × B) treats B × A as +A × B; 0 forgets the cross terms; −(A × B) counts them once.

Concept behind the question. This is the cross-product version of (a + b)(a − b), but it does not cancel: the anticommutative cross terms add instead of cancelling (Section 8, N3).

Question 198

VEC-T5-N-043 · Topic 5 · Hard · Calculation

The corners of a triangular sail are P(1, 2, 1), Q(2, 0, 3) and R(3, 4, 2), with coordinates in metres. Find the area of the sail.

(a) 3.64 m^2

(b) 4.5 m^2

(c) 9 m^2

(d) 40.5 m^2

Answer: (b) 4.5 m^2

Explanation

Given: P(1, 2, 1), Q(2, 0, 3), R(3, 4, 2) in metres.

To find: area of triangle PQR.

Formula: Area = $\frac{1}{2}$|PQ × PR|, with PQ = Q − P, PR = R − P.

Working: PQ = (2−1, 0−2, 3−1) = (1, −2, 2); PR = (3−1, 4−2, 2−1) = (2, 2, 1).

Working: i-component = $PQ_yPR_z - PQ_zPR_y$ = (−2)×1 − 2×2 = −6

Working: j-component = $PQ_zPR_x - PQ_xPR_z$ = 2×2 − 1×1 = 3

Working: k-component = $PQ_xPR_y - PQ_yPR_x$ = 1×2 − (−2)×2 = 6

Working: PQ × PR = (−6, 3, 6), so |PQ × PR| = $\sqrt{36+9+36}$ = 9.

Working: Area = $\frac{1}{2}$ × 9 = 4.5 m^2.

Result: 4.5 m^2. → option (b)

Why the others fail: 9 is the parallelogram area; 40.5 = 81/2 skips the root; 3.64 uses $\frac{1}{2}$|P × Q|, which is not the area of this triangle.

Concept behind the question. Always form the two side vectors from one corner first; the cross product of position vectors from the origin has no connection with the triangle (Section 8, N7).

Question 199

VEC-T5-N-044 · Topic 5 · Hard · Calculation

For **A** = $3\hat{\mathbf{i}} - 2\hat{\mathbf{j}}$ and **B** = $2\hat{\mathbf{j}} - \hat{\mathbf{k}}$, find the vector of magnitude 14 in the direction of **A** × **B** .

(a) $2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}$

(b) $\frac{1}{7}(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}})$

(c) $-4\hat{\mathbf{i}} - 6\hat{\mathbf{j}} - 12\hat{\mathbf{k}}$

(d) $4\hat{\mathbf{i}} + 6\hat{\mathbf{j}} + 12\hat{\mathbf{k}}$

Answer: (d) $4\hat{\mathbf{i}} + 6\hat{\mathbf{j}} + 12\hat{\mathbf{k}}$

Explanation

Given: A = (3, −2, 0), B = (0, 2, −1), required magnitude 14.

To find: 14 $\hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is the unit vector along A × B.

Formula: $\hat{\mathbf{n}}$ = (A × B)/|A × B|, required vector = 14 $\hat{\mathbf{n}}$.

Working: i-component = $A_yB_z - A_zB_y$ = (−2)×(−1) − 0×2 = 2

Working: j-component = $A_zB_x - A_xB_z$ = 0×0 − 3×(−1) = 3

Working: k-component = $A_xB_y - A_yB_x$ = 3×2 − (−2)×0 = 6

Working: |A × B| = $\sqrt{4+9+36}$ = 7.

Working: $\hat{\mathbf{n}}$ = (2 i + 3 j + 6 k)/7, so 14 $\hat{\mathbf{n}}$ = 2 × (2 i + 3 j + 6 k) = 4 i + 6 j + 12 k.

Result: $4\hat{\mathbf{i}} + 6\hat{\mathbf{j}} + 12\hat{\mathbf{k}}$. → option (d)

Why the others fail: 2i + 3j + 6k has magnitude 7; (2i + 3j + 6k)/7 has magnitude 1; the third option points along B × A.

Concept behind the question. Direction comes from the unit vector $\hat{\mathbf{n}}$ and size from the multiplier: vector = (required magnitude) × $\hat{\mathbf{n}}$ (Section 8, N8, with Section 5).

Question 200

VEC-T5-N-045 · Topic 5 · Hard · Calculation

The torque about a point is defined as τ = r × F, where r is drawn from that point to the point of application of the force. A force F = $\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$ N acts at the point (3, 2, 2) m. Find |τ| about the point (1, 1, 0) m.

(a) 0 N m

(b) 9 N m

(c) 12 N m

(d) 81 N m

Answer: (b) 9 N m

Explanation

Given: F = (1, 2, −2) N at (3, 2, 2) m; the point of reference is (1, 1, 0) m; τ = r × F (defined).

To find: |τ|.

Formula: r = (point of application) − (reference point); τ = r × F; |τ| = $\sqrt{x^2+y^2+z^2}$.

Working: r = (3−1, 2−1, 2−0) = (2, 1, 2) m.

Working: i-component = $r_yF_z - r_zF_y$ = 1×(−2) − 2×2 = −6

Working: j-component = $r_zF_x - r_xF_z$ = 2×1 − 2×(−2) = 6

Working: k-component = $r_xF_y - r_yF_x$ = 2×2 − 1×1 = 3

Working: τ = (−6, 6, 3) N m, so |τ| = $\sqrt{36+36+9}$ = 9 N m.

Result: 9 N m. → option (b)

Why the others fail: 12 uses r = (3, 2, 2) from the origin; 0 is r·F (r and F happen to be perpendicular, so the dot product is zero but the torque is not); 81 skips the root.

Concept behind the question. Torque depends on the position vector measured from the chosen reference point; shifting the point changes r and so changes τ.